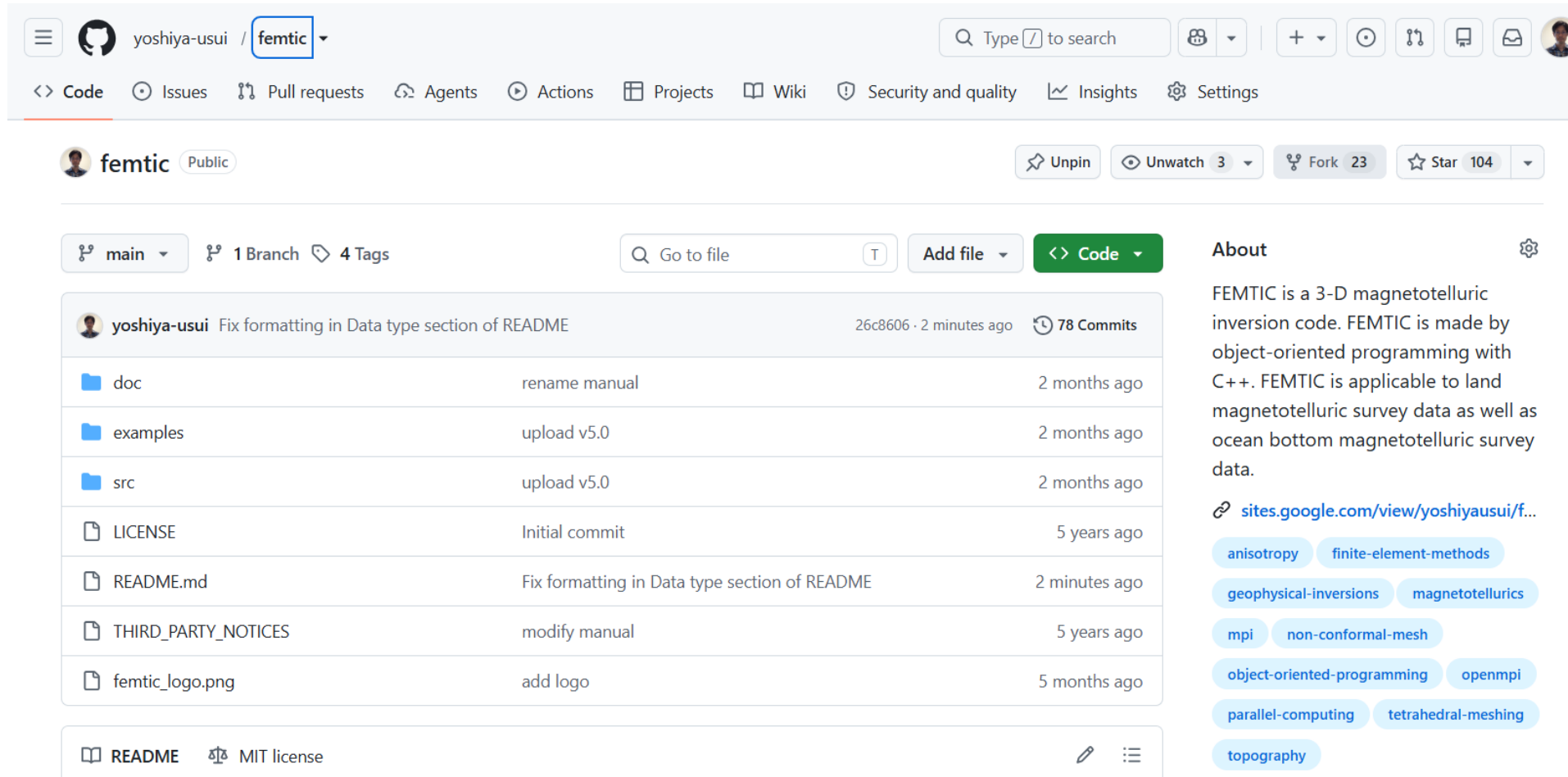


FEMTIC - A user's guide

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- FEMTIC is available on GitHub.
- This code was made using object-oriented programming with C++.



The screenshot displays the GitHub repository page for `yoshiya-usui/femtic`. The repository is public and has 3 unpins, 3 unwatchers, 23 forks, and 104 stars. The file list shows the following files and their commit history:

File	Commit Message	Time Ago
doc	rename manual	2 months ago
examples	upload v5.0	2 months ago
src	upload v5.0	2 months ago
LICENSE	Initial commit	5 years ago
README.md	Fix formatting in Data type section of README	2 minutes ago
THIRD_PARTY_NOTICES	modify manual	5 years ago
femtic_logo.png	add logo	5 months ago

The README section is visible at the bottom, showing the MIT license and a link to the repository.

<https://github.com/yoshiya-usui/femtic>

Mesh type: Tetrahedral mesh / Non-conforming deformed hexahedral mesh / Hexahedral brick mesh

Model parameter: Subsurface electrical resistivity, including anisotropic resistivity / Galvanic distortion parameters

Data type: Impedance tensor / Vertical magnetic transfer function / Inter-station horizontal magnetic transfer function / Phase tensor / Apparent resistivity & Phase / Network-MT response function

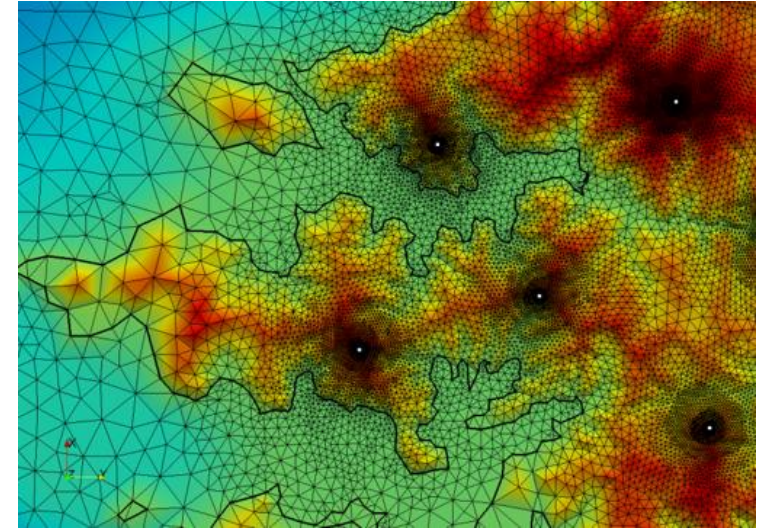
Inversion algorithm: Model-space Gauss-Newton method / Data-space Gauss-Newton method

Parallel computation: Multiple processes parallel computation with MPI / Multiple threads parallel computation with OpenMP / MPI & OpenMP hybrid parallel computation



1. Accurate representation of topography and bathymetry in the computational mesh

Tetrahedral and deformed hexahedral elements allow us to incorporate topography and bathymetry into the model with high accuracy.



2. Stable inversion algorithm

Because FEMTIC uses a direct solver, the forward computation can be performed robustly.

The Gauss-Newton method requires a relatively small number of iterations to obtain the resistivity structure.

3. No dependence on the prior model

Because FEMTIC does not use the prior model in the regularization, it does not depend on it.

(However, if a reliable prior model is available, the inability to use the prior model is a drawback.)

Hardware requirements:

Preferably more than one hundred GB RAM and more than several hundreds GB disk sizes.

Required RAM and disk sizes depend on the numbers of data and elements.

For a small model, FEMTIC requires less than several tens GB.

However, for a large model with fine elements, several hundreds GB of RAM may be required.

Software requirements:

Intel® Math Kernel Library should be installed.

Input files

File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
distortion_iter0.dat	Initial values of galvanic distortion parameters
mesh.dat	Computational mesh
resistivity_block_iter0.dat	Parameter cells and initial resistivity values

Output files

File name	Content
femtic_#.log	Log information
femtic.cnv	Convergence of Gauss-Newton iteration
result_#_iter*.csv	Calculated and observed response functions
resistivity_block_iter*.dat	Parameter cells and resistivity values obtained by the inversion
distortion_iter*.dat	Galvanic distortion parameters obtained by the inversion

‘#’ and ‘*’ indicate process number and iteration number, respectively.

In addition to those files, files for ParaView visualization are output.

File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
distortion_iter0.dat	Initial values of galvanic distortion parameters
mesh.dat	Computational mesh
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Input files

File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
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resistivity_block_iter0.dat	Parameter cells and initial resistivity values

- This file has many parameters.
- However, some parameters are only for minor functions.
- I introduce only important parameters in next several slides.
- More detailed information is available in the manual of FEMTIC.

https://github.com/yoshiya-usui/femtic/blob/main/doc/Manual_Of_FEMTIC.pdf

Example

```

MESH_TYPE
2
OUTPUT_PARAM
1
0
DIV_NUM_RHS_FWD
10
INV_METHOD
1
SMALL_VALUE
1.0e-4
DISTORTION
0
TRADE_OFF_PARAM
10.0
ITERATION
0 20
STEP_LENGTH
0.5 0.1 0.8
2
0.6 1.2
END

```

control.dat (1) Mesh type

1: Tetrahedral mesh

2: Deformed hexahedral mesh

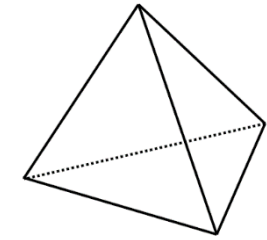
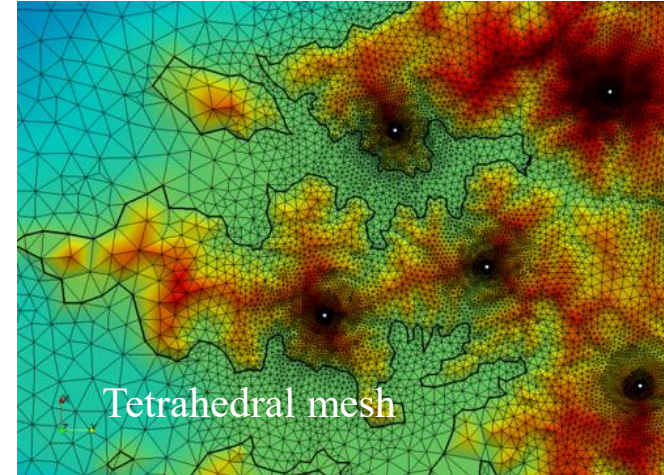
Discretization method of non-conforming deformed hexahedral mesh is based on that of Grayver & Bürg (2014).

Grayver, A. V., & Bürg, M. (2014). Robust and scalable 3-D geo-electromagnetic modelling approach using the finite element method. Geophysical Journal International.
<https://doi.org/10.1093/gji/ggu119>

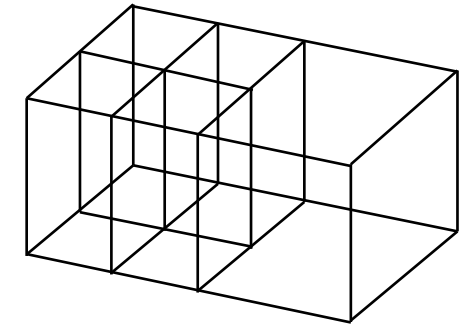
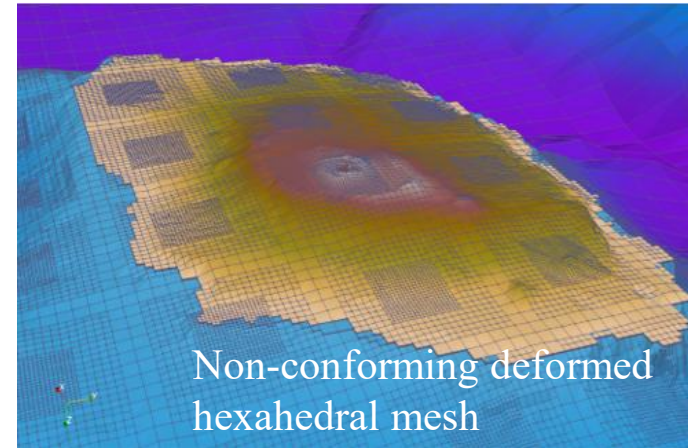
In FEMTIC, the division number of one side can be doubled only in the horizontal direction.

Example

```
MESH_TYPE
2
```



Tetrahedral
element

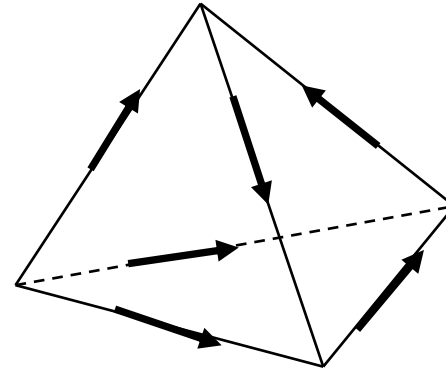
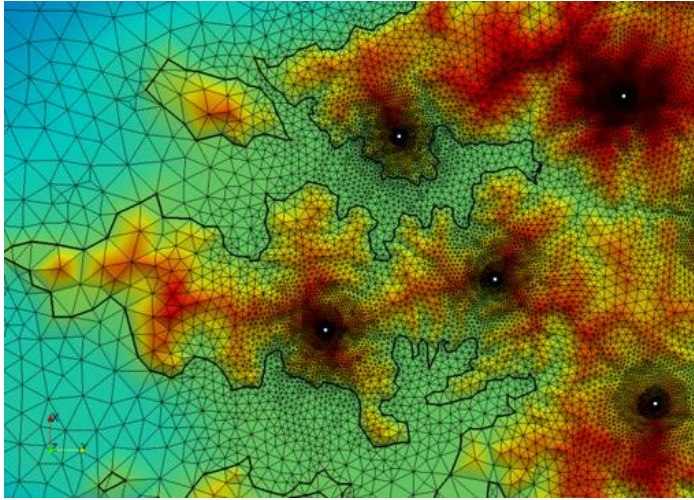


Non-conforming deformed
hexahedral element

By default, or when 0 is written as the mesh type, the brick hexahedral mesh is selected.

However, the brick element is unnecessary because the same mesh can be made with the non-conforming hexahedral mesh (type 2).

Tetrahedral mesh VS Hexahedral mesh



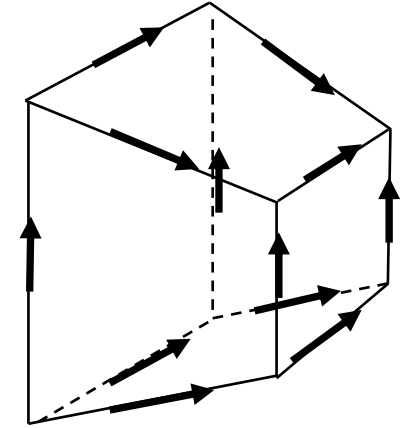
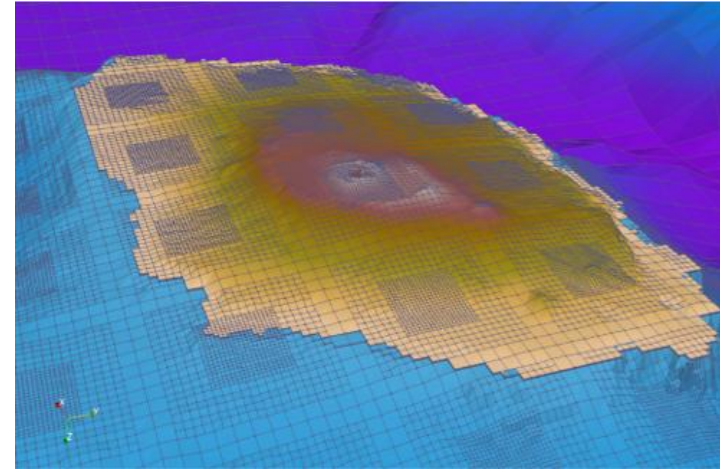
Tetrahedral element

Advantages

- Accurate modeling of coastlines.
- Flexible adjustment of element sizes.

Disadvantages

- Mesh generation is not easy and requires several tries and errors, at least.
- Many elements are required to represent thin layered structures.



Hexahedral element

Advantages

- Easier mesh generation.
- Shallow layered structures can be imaged more easily.

Disadvantages

- Step-like representation of coastlines is inevitable.
- Fineness of the mesh changes discontinuously.

I recommend using a hexahedral mesh (type=2) for the initial trial of FEMTIC.

Under the keyword ‘DISTORTION’, the distortion type should be selected.

Example

DISTORTION
1

0: Distortion matrix is NOT estimated

1: Distortion matrix components are estimated

$$\mathbf{C}' = \mathbf{C} - \mathbf{I} = \begin{pmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} c'_{xx} & c'_{xy} \\ c'_{yx} & c'_{yy} \end{pmatrix}$$

Unknowns

3: Only gains are estimated

$$\mathbf{C} = \begin{pmatrix} 10^{G_x} & 0 \\ 0 & 10^{G_y} \end{pmatrix} \quad \text{Unknowns: } G_x \text{ and } G_y$$

$$\mathbf{Z}_{dist} = \mathbf{C}\mathbf{Z}$$

$\mathbf{Z}$: (undistorted) impedance tensor

$\mathbf{Z}_{dist}$: impedance tensor affected by the galvanic distortion

$\mathbf{C} = \begin{pmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{pmatrix}$: distortion matrix

When 2 is written under ‘DISTORTION’, the gains and rotations of the distortion matrix are estimated.

However, I recommend not using type 2 because the convergence of the inversion becomes worse.

control.dat (3) Trade-off parameters

The trade-off parameters should be written under the keyword 'TRADE_OFF_PARAM'.

The number of the trade-off parameters depends on the distortion parameter type.

Example

```
TRADE_OFF_PARAM
10.0 0.1
```

No distortion matrix

```
DISTORTION
0
```



$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha^2 \phi_m(\mathbf{m})$$

α, β : trade-off parameters

I recommend using a small value for β (e.g., $\beta=0.1$) when DISTORTION=1 (Usui et al., 2017; Usui and Uyeshima, 2025).

Distortion matrix components

```
DISTORTION
1
```



$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha^2 \phi_m(\mathbf{m}) + \beta^2 \phi_c(\mathbf{m})$$

$$\phi_c(\mathbf{m}) = \sum_{i=1}^{\#sites} (|c'_{xx}|^2 + |c'_{xy}|^2 + |c'_{yx}|^2 + |c'_{yy}|^2)$$

Gains

```
DISTORTION
3
```



$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha^2 \phi_m(\mathbf{m}) + \beta^2 \phi_c(\mathbf{m})$$

$$\phi_c(\mathbf{m}) = \sum_{i=1}^{\#sites} (G_x^2 + G_y^2)$$

$$\phi_d(\mathbf{m}) = \|\mathbf{W}(\mathbf{d} - \mathbf{F}(\mathbf{m}))\|^2: \text{data misfit}$$

$$\phi_m(\mathbf{m}) = \|\mathbf{R}\mathbf{m}\|^2 = \|(L_1 \cdots L_i \cdots L_M)^T\|^2: \text{model roughness}$$

$\mathbf{W}$: diagonal matrix composed of the inverses of the data errors

$\mathbf{d}$: observed data vector

$\mathbf{F}(\mathbf{m})$: calculated data vector

$$L_i = N_i m_i - \sum_{j=1}^{N_i} m_{j,i}$$

Laplacian

M : number of unknown resistivity

N_i : number of faces

m_i : log-resistivity

$m_{j,i}$: log-resistivity of adjacent element

control.dat (4) Starting and ending iteration numbers

Example

```
ITERATION  
0 20
```

The starting and ending iteration numbers should be written under the keyword 'ITERATION'.

When only the forward computation is performed, please write '0 0'.

```
ITERATION  
0 0
```



Only the forward computation is performed using the initial model.

The inversion can be restarted from an intermediate result by setting the starting iteration number to a value greater than zero.

→ Even if the inversion stops accidentally, it can restart from the last iteration.

```
ITERATION  
5 20
```



The inversion restarts from the result of the 5th iteration.

control.dat (5) Parameters for damping factor adjustment

When the model update computed by the Gauss-Newton method is directly used, the inversion often diverges.

To stabilize the inversion, the model update is multiplied by a damping factor γ in FEMTIC.

Model at the N-1th iteration:

Model at the Nth iteration: $\mathbf{m}_N = \mathbf{m}_{N-1} + \gamma \delta \mathbf{m}_{N-1}$

Damping factor
 $0 \leq \gamma \leq 1$

Model update obtained by the Gauss-Newton method

FEMTIC automatically adjusts the damping factor γ during the inversion.

Parameters for the damping factor adjustment can be specified under the keywords ‘STEP_LENGTH’ and ‘RETRIAL’.

STEP_LENGTH

γ_{init} γ_{min} γ_{max}
 N_1

c_1 c_2

RETRIAL

N_2

γ_{init} : Initial damping factor

γ_{min} : Lower limit of damping factor

γ_{max} : Upper limit of damping factor

N_1 : Number of iteration for increasing damping factor

c_1 : Factor for decreasing damping factor

c_2 : Factor for increasing damping factor

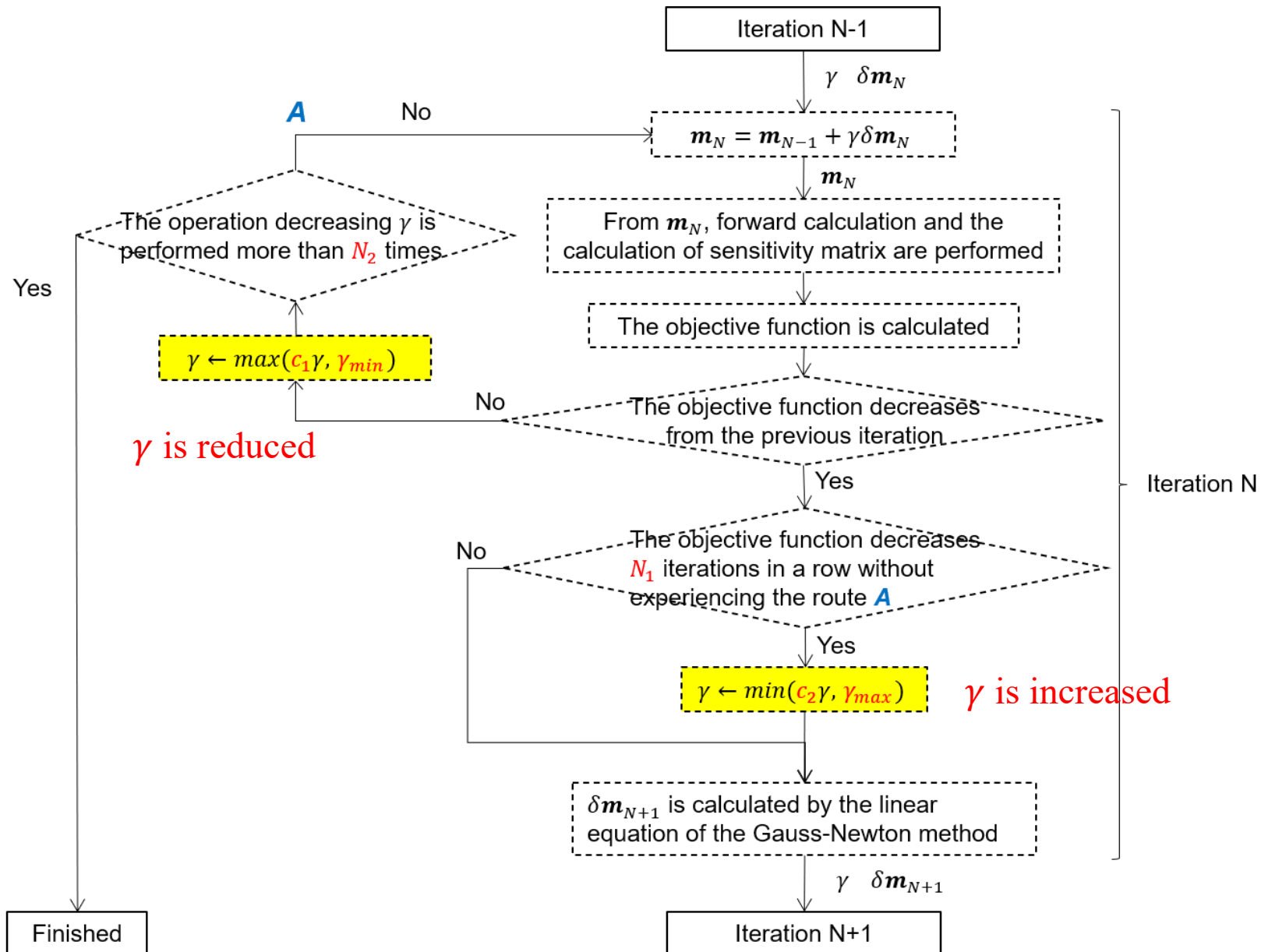
N_2 : Maximum number of the decrease in the damping factor

Example

```
STEP_LENGTH
0.5 0.1 0.8
3
0.5 1.2
RETRIAL
3
```

control.dat (5) Parameters for damping factor adjustment

7 October EMinars Season 4



STEP_LENGTH

γ_{init} γ_{min} γ_{max}
 N_1

c_1 c_2

RETRIAL

N_2

γ_{init} : Initial value of γ

γ_{min} : Lower limit of γ

γ_{max} : Upper limit of γ

N_1 : Number of iteration for increasing γ

c_1 : Factor for decreasing γ

c_2 : Factor for increasing γ

N_2 : Maximum number of the decrease in γ

control.dat (6) Recommended setting for initial trial

```
MESH_TYPE
2
OUTPUT_PARAM
1
0
DIV_NUM_RHS_FWD
10
INV_METHOD
1
SMALL_VALUE
1.0e-4
DISTORTION
0
TRADE_OFF_PARAM
10.0
ITERATION
0 20
STEP_LENGTH
0.5 0.1 0.8
2
0.6 1.2
END
```

Deformed non-conforming hexahedral mesh

Output estimated resistivity values to binary files for visualization

To save memory, use a large division number for the right-hand-side vectors in the sensitivity matrix calculation.

Data space method

No galvanic distortion parameter estimation

A large trade-off parameter α is recommended for the first trial

Ending flag is necessary at the end of the file

File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
distortion_iter0.dat	Initial values of galvanic distortion parameters
mesh.dat	Computational mesh
resistivity_block_iter0.dat	Parameter cells and initial resistivity values

This input file contains the observed response functions and their errors.

Impedance tensor

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

Vertical magnetic transfer function (Tipper)

$$H_z = \begin{pmatrix} T_{zx} & T_{zy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

Inter-station horizontal magnetic transfer function

$$\begin{pmatrix} H_x^{Site1} \\ H_y^{Site1} \end{pmatrix} = \begin{pmatrix} T_{xx} & T_{xy} \\ T_{yx} & T_{yy} \end{pmatrix} \begin{pmatrix} H_x^{Site2} \\ H_y^{Site2} \end{pmatrix}$$

Phase tensor (Caldwell et al. 2004)

$$\begin{pmatrix} \Phi_{11} & \Phi_{12} \\ \Phi_{21} & \Phi_{22} \end{pmatrix} = \begin{pmatrix} Re(Z_{xx}) & Re(Z_{xy}) \\ Re(Z_{yx}) & Re(Z_{yy}) \end{pmatrix}^{-1} \begin{pmatrix} Im(Z_{xx}) & Im(Z_{xy}) \\ Im(Z_{yx}) & Im(Z_{yy}) \end{pmatrix}$$

Apparent resistivity

$$\rho_{a,ij} = \frac{1}{\omega\mu_0} |Z_{ij}|^2$$

Phase

$$\theta_{ij} = \arg(Z_{ij})$$

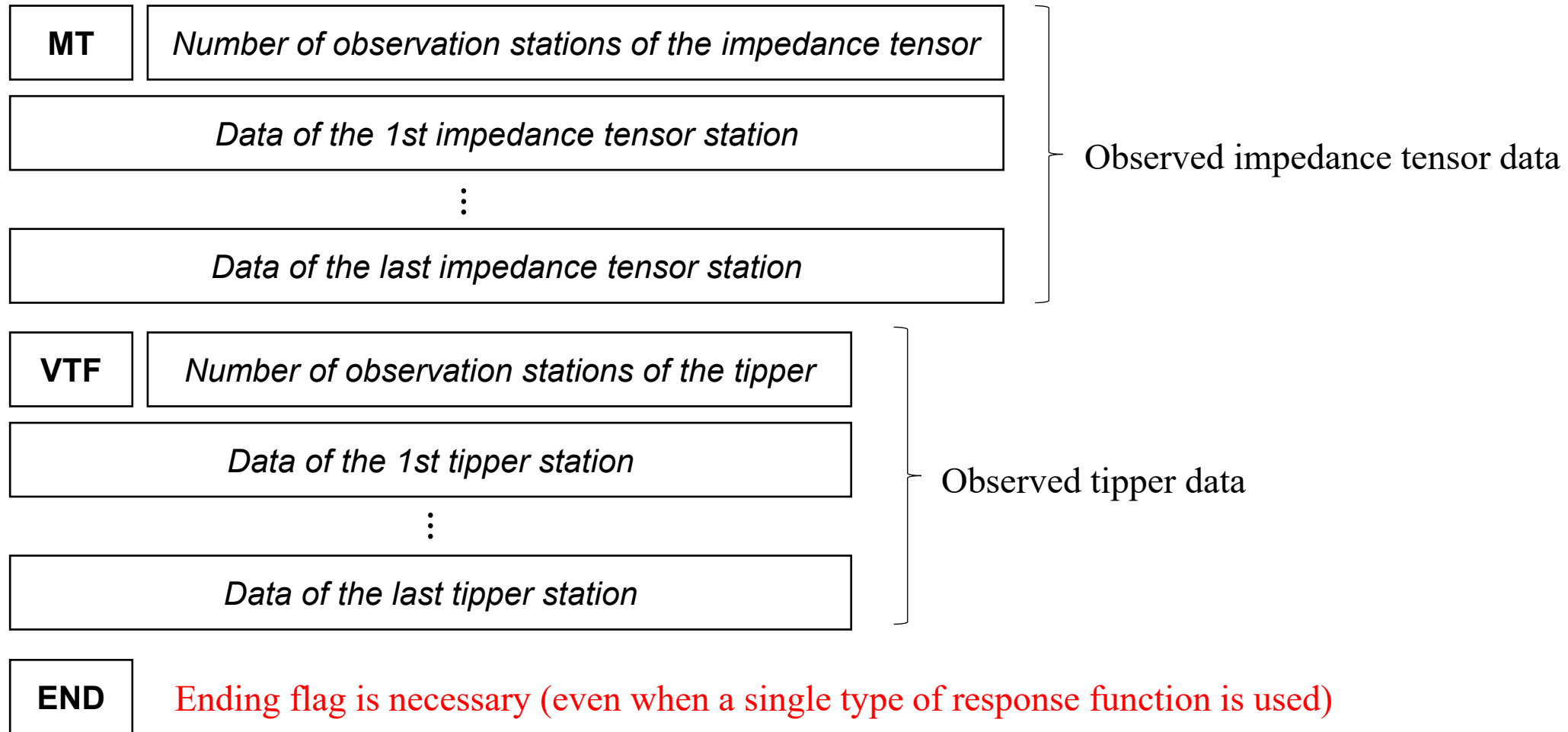
The use of the apparent resistivity and phase is NOT recommended. The quadrant of the phase changes, particularly for the diagonal components, worsening the convergence of the inversion.

Network-MT response function (Uyeshima et al. 2001)

$$V = \begin{pmatrix} Y_x & Y_y \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

Response functions must be in SI units with $e^{-i\omega t}$ time dependence.

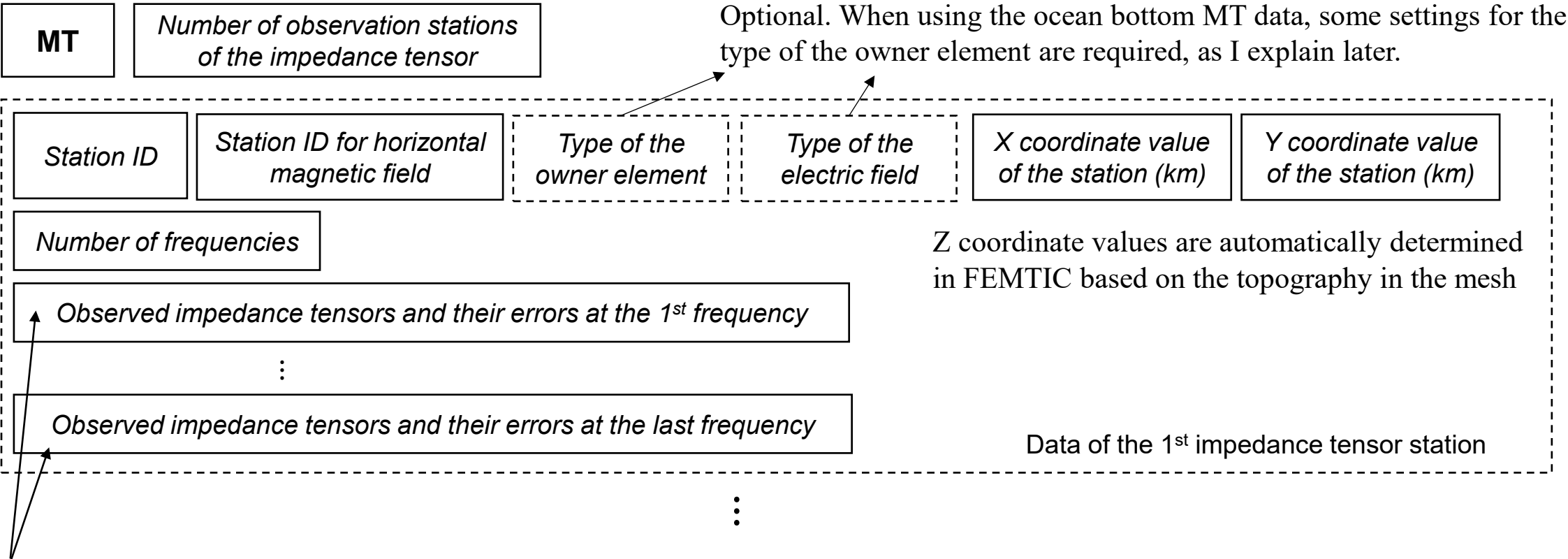
observe.dat (2) Structure of the data file



Please refer to the sample data files of FEMTIC on GitHub.

<https://github.com/yoshiya-usui/femtic/tree/main/examples>

observe.dat (3) Data format for the impedance tensor

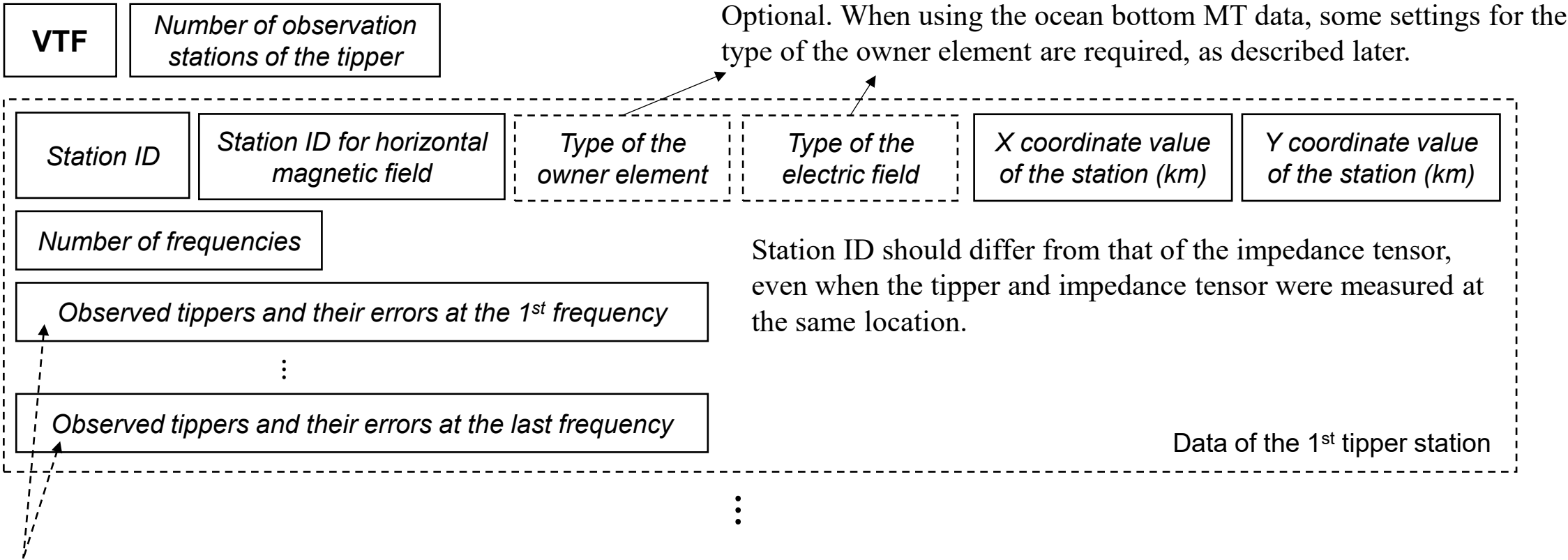


The format of observed impedance tensors and their errors

<i>Frequency</i> (Hz)	<i>Re(Z_{xx})</i> (V/A)	<i>Im(Z_{xx})</i> (V/A)	<i>Re(Z_{xy})</i> (V/A)	<i>Im(Z_{xy})</i> (V/A)	<i>Re(Z_{yx})</i> (V/A)	<i>Im(Z_{yx})</i> (V/A)	<i>Re(Z_{yy})</i> (V/A)	<i>Im(Z_{yy})</i> (V/A)
	<i>Re(Z_{xx}) error</i> (V/A)	<i>Im(Z_{xx}) error</i> (V/A)	<i>Re(Z_{xy}) error</i> (V/A)	<i>Im(Z_{xy}) error</i> (V/A)	<i>Re(Z_{yx}) error</i> (V/A)	<i>Im(Z_{yx}) error</i> (V/A)	<i>Re(Z_{yy}) error</i> (V/A)	<i>Im(Z_{yy}) error</i> (V/A)

Negative error excludes the corresponding component from the inversion.

observe.dat (4) Data format for the vertical magnetic transfer function (tipper)



The format of tippers and their errors

<i>Frequency (Hz)</i>	<i>Re(T_{zx})</i>	<i>Im(T_{zx})</i>	<i>Re(T_{zy})</i>	<i>Im(T_{zy})</i>	<i>Re(T_{zx}) error</i>	<i>Im(T_{zx}) error</i>	<i>Re(T_{zy}) error</i>	<i>Im(T_{zy}) error</i>
---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	-------------------------------------	-------------------------------------	-------------------------------------	-------------------------------------

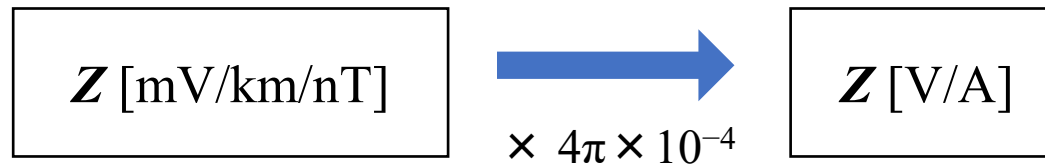
Negative error excludes the corresponding component from the inversion.

observe.dat (5) Unit conversion

- The unit of the impedance tensor should be V/A in this input file..

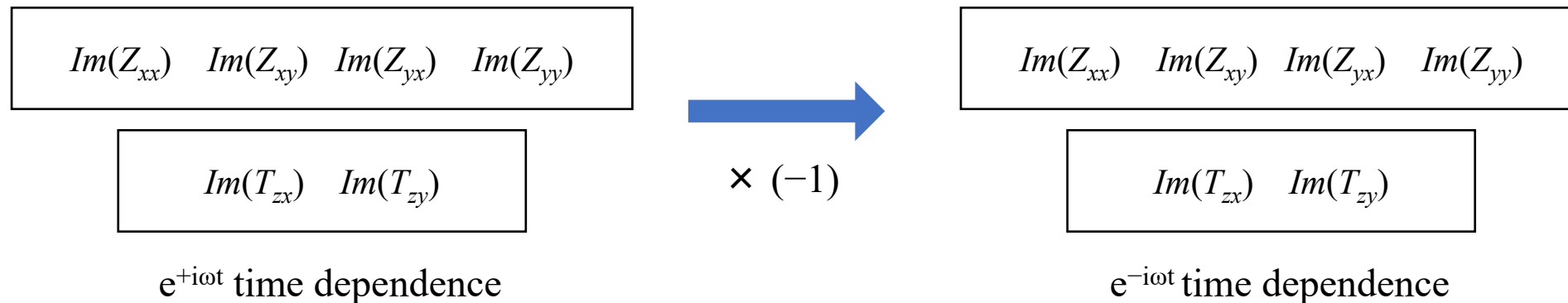
→ If another unit is used, unit conversion is necessary.

For example, the unit of the original impedance tensor is mV/km/nT, the impedance tensor components and their errors should be multiplied by $4\pi \times 10^{-4}$.



- Response functions must be based on the $e^{-i\omega t}$ time dependence.

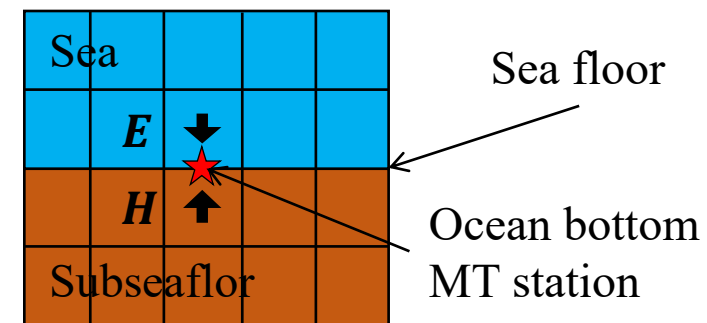
→ If the $e^{+i\omega t}$ time dependence is used, the sign of the imaginary parts of the response functions should be inverted.



observe.dat (6) Special treatment for ocean bottom MT data

For ocean bottom MT stations, it is better to interpolate the electric field and magnetic field from the upper-side and down-side elements, respectively (Usui et al. 2018).

This treatment needs a somewhat tricky setting.



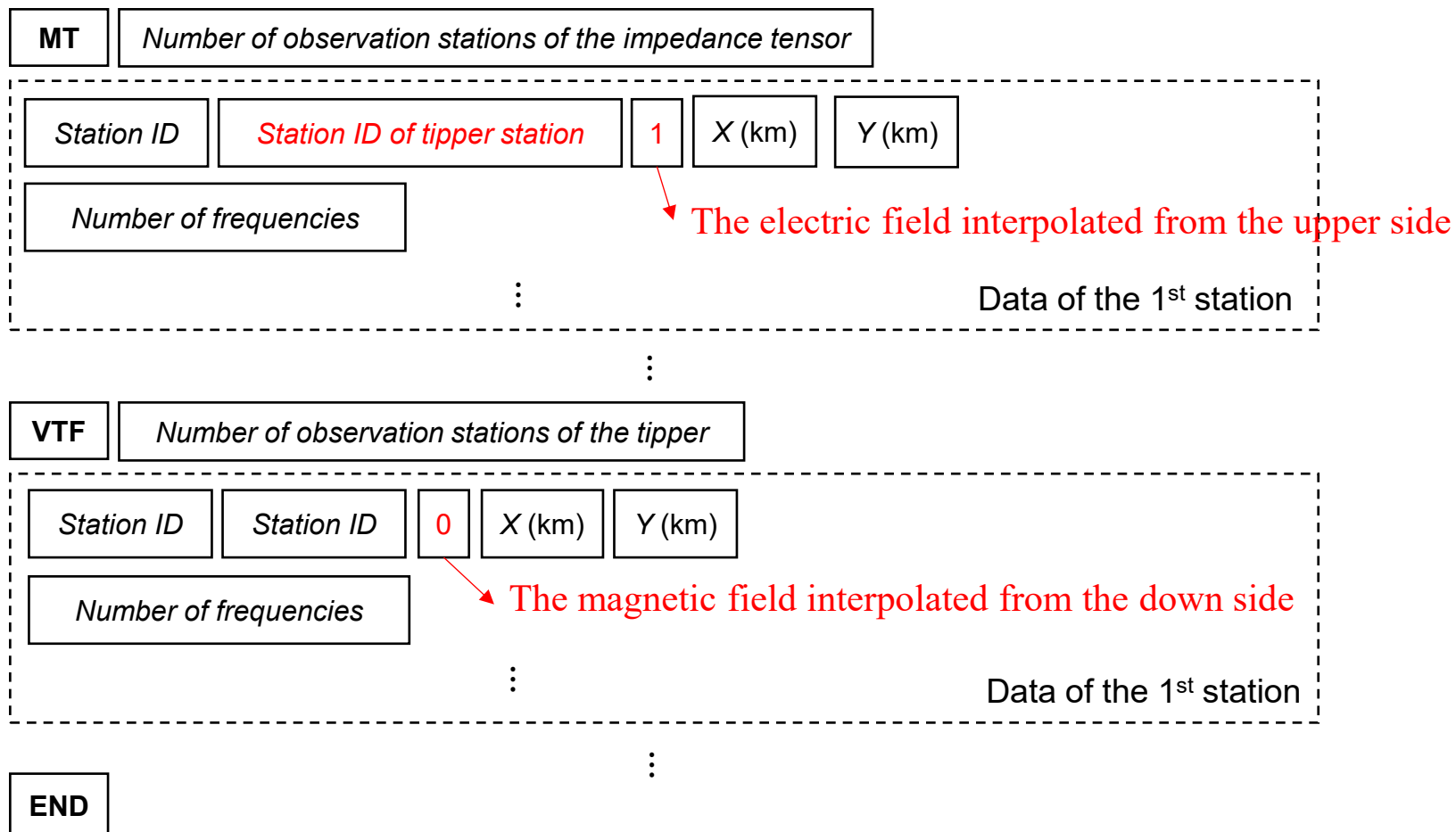
control.dat

Interpolation direction
of each station can be
individually selected

```

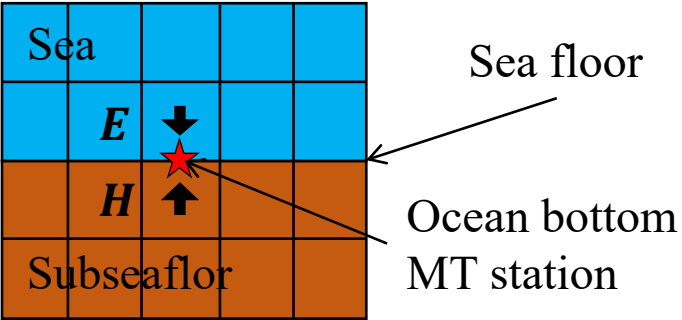
MESH_TYPE
2
OWNER_ELEMENT
-1
OUTPUT_PARAM
1
0
DIV_NUM_RHS_FWD
10
INV_METHOD
1
SMALL_VALUE
1.0e-4
DISTORTION
0
TRADE_OFF_PARAM
10.0
ITERATION
0 20
STEP_LENGTH
0.5 0.1 0.8
2
0.6 1.2
END
  
```

observe.dat



observe.dat (7) Special treatment for ocean bottom MT data

For ocean bottom MT stations, it is better to interpolate the electric field and magnetic field from the upper-side and lower-side elements, respectively (Usui et al. 2018).

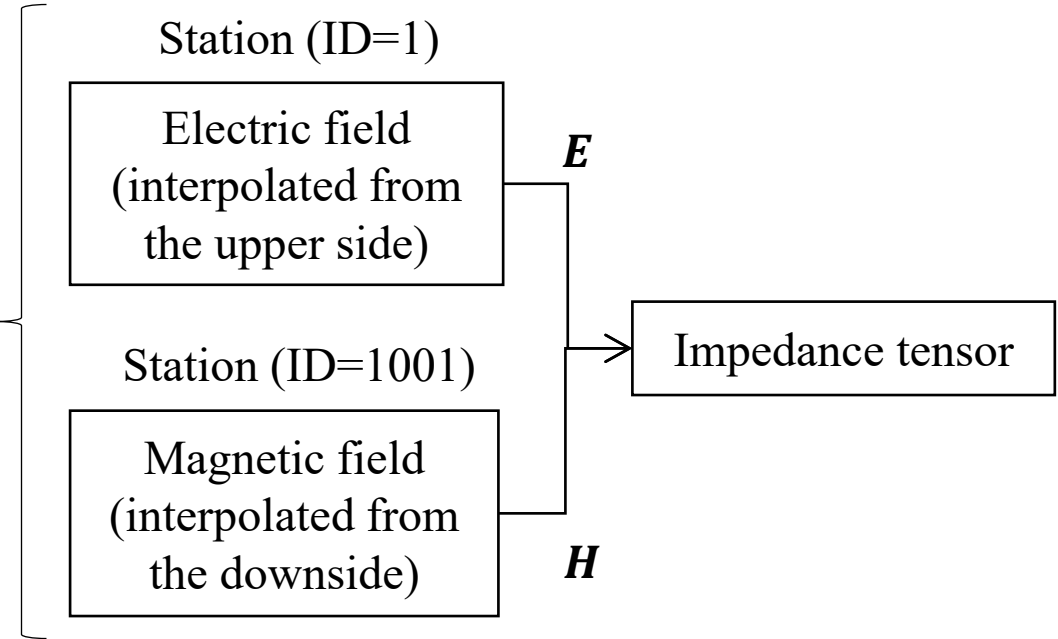


Example

```
MT 20
1 1001 1 (12.5 2.4)
12
[Observed impedance tensor and errors]
2 1002 1 23.1 -29.3
12
[Observed impedance tensor and errors]
:
VTF 20
1001 1001 0 (12.5 2.4)
12
[Observed tipper and errors]
1002 1002 0 23.1 -29.3
12
[Observed tipper and errors]
:
END
```

Same location

Same location



If only the impedance tensor is used in the inversion (no tipper station), please define dummy tipper stations whose number of frequencies is zero.

File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
distortion_iter0.dat	Initial values of galvanic distortion parameters
mesh.dat	Computational mesh
resistivity_block_iter0.dat	Parameter cells and initial resistivity values

control.dat

DISTORTION
0

→ Galvanic distortion parameters are NOT estimated → ‘distortion_iter0.dat’ is NOT required

DISTORTION
1

→ Distortion matrix components are estimated

DISTORTION
3

→ Gains (static shift) are estimated

The format of distortion_iter0.dat’ depends on the type of distortion parameters

Number of observation stations of the impedance tensor					
Station ID of the 1 st station	c'_{xx}	c'_{xy}	c'_{yx}	c'_{yy}	0: Distortion matrix is estimated, 1: Distortion matrix is fixed
Station ID of the 2 nd station	c'_{xx}	c'_{xy}	c'_{yx}	c'_{yy}	0: Distortion matrix is estimated, 1: Distortion matrix is fixed
⋮					
Station ID of the last station	c'_{xx}	c'_{xy}	c'_{yx}	c'_{yy}	0: Distortion matrix is estimated, 1: Distortion matrix is fixed

$Z_{dist} = CZ$

$C' = C - I = \begin{pmatrix} c'_{xx} & c'_{xy} \\ c'_{yx} & c'_{yy} \end{pmatrix}$ Model parameters

Zeros are recommended as initial values of the model parameters.

Example

36					
1	0.000000	0.000000	0.000000	0.000000	0
2	0.000000	0.000000	0.000000	0.000000	0
3	0.000000	0.000000	0.000000	0.000000	0
4	0.000000	0.000000	0.000000	0.000000	0
⋮					

Number of observation stations of the impedance tensor			
Station ID of the 1 st station	G_x	G_y	0: Gains are estimated, 1: Gains are fixed
Station ID of the 2 nd station	G_x	G_y	0: Gains are estimated, 1: Gains are fixed
⋮			
Station ID of the last station	G_x	G_y	0: Gains are estimated, 1: Gains are fixed

$$\mathbf{Z}_{dist} = \mathbf{C}\mathbf{Z}$$

$$\mathbf{C} = \begin{pmatrix} 10^{G_x} & 0 \\ 0 & 10^{G_y} \end{pmatrix} \quad \text{Unknowns: } G_x \text{ and } G_y$$

Zeros are recommended as initial values of the model parameters.

Example

36			
1	0.000000	0.000000	0
2	0.000000	0.000000	0
3	0.000000	0.000000	0
4	0.000000	0.000000	0
⋮			

File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
distortion_iter0.dat	Initial values of galvanic distortion parameters
mesh.dat	Computational mesh
resistivity_block_iter0.dat	Parameter cells and initial resistivity values

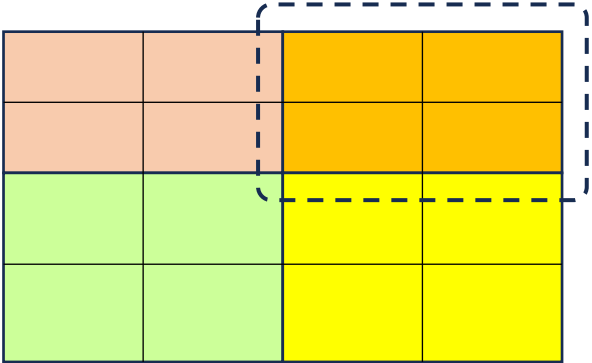
Structure of ‘mesh.dat’

<i>Node positions</i>
<i>Relationship between elements and nodes</i>
<i>Element faces on the outer boundaries and the Earth’s surface</i>

Structure of ‘resistivity_block_iter0.dat’

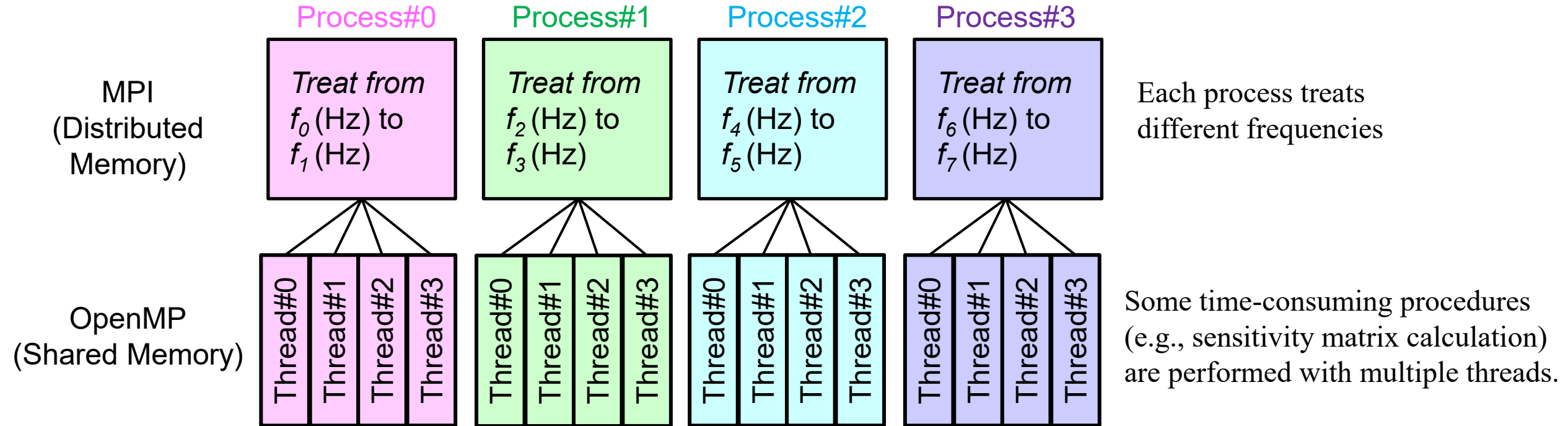
<i>Relationship between elements and parameter cells</i>
<i>Resistivity value of each parameter cell</i>

Parameter cell is defined as a group of adjacent elements that have the same resistivity.



It is nearly impossible to make those files manually.
→ Some mesh generation tools are shared on GitHub. I will introduce them later.

OpenMP-MPI Hybrid parallelization can be used to speed up the forward and inversion calculations.



- Total parallel number = Number of MPI processes $\times$ Number of threads
- MPI parallelization has higher parallelization efficiency than OpenMP.
- The required memory is proportional to the number of MPI processes.
- The number of threads has a minor influence on the consumed memory.

Execution commands depend on the computer system. What follows are simple examples.

Please check the usage of your computer system for MPI, OpenMP, and MPI-OpenMP hybrid programs.

1. Serial calculation

- 1) Write '1' below 'NUM_THREADS' in 'control.dat'.
- 2) Execute the following command in the work directory.
femtic

3. Parallel calculation only using OpenMP

- 1) Write number of threads below 'NUM_THREADS' in 'control.dat'.

Example (four threads)

NUM_THREADS

4

- 2) Execute the following command in the work directory.
femtic

2. Parallel calculation only using MPI

- 1) Write '1' below 'NUM_THREADS' in 'control.dat'.
- 2) Execute the following command in the work directory.
mpiexec -n [#Process] femtic

4. Hybrid MPI-OpenMP parallel calculation

- 1) Write number of threads below 'NUM_THREADS' in 'control.dat'.

Example (four threads)

NUM_THREADS

4

- 2) Execute the following command in the work directory.
mpiexec -n [#Process] femtic

[#Process] indicates the number of MPI processes.

Output files

File name	Content
femtic_#.log	Log information
femtic.cnv	Convergence of the Gauss-Newton iteration
result_#_iter*.csv	Calculated and observed response functions
resistivity_block_iter*.dat	Parameter cells and resistivity values obtained by the inversion
distortion_iter*.dat	Galvanic distortion parameters obtained by the inversion

‘#’ and ‘*’ indicate process number and iteration number, respectively.

Output files (1) Log and CSV files

File name	Content
femtic_#.log	Log information
femtic.cnv	Convergence of the Gauss-Newton iteration
result_#_iter*.csv	Calculated and observed response functions
resistivity_block_iter*.dat	Parameter cells and resistivity values obtained by the inversion
distortion_iter*.dat	Galvanic distortion parameters obtained by the inversion

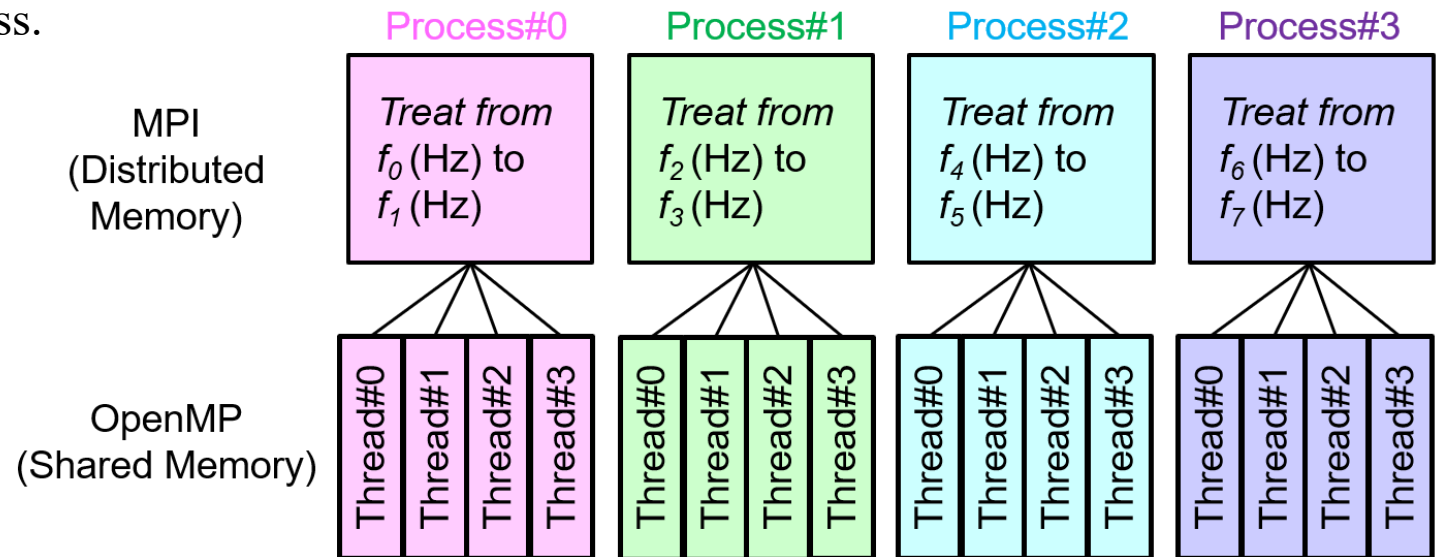
‘#’ and ‘*’ indicate process number and iteration number, respectively.

► These files are outputted for each MPI process.

CSV files contain the following values.

- Calculated response functions
- Observed response functions
- Errors of response functions
- Residuals of response functions

(Residuals may be redundant, because they can be calculated from the other three values)



File name	Content
femtic_#.log	Log information
femtic.cnv	Convergence of the Gauss-Newton iteration
result_#_iter*.csv	Calculated and observed response functions
resistivity_block_iter*.dat	Parameter cells and resistivity values obtained by the inversion
distortion_iter*.dat	Galvanic distortion parameters obtained by the inversion

‘#’ and ‘*’ indicate process number and iteration number, respectively.

Example

Iter#	Retrial#	Alpha	Damp	Roughness	Misfit	RMS	ObjFunc
0	0	1.000000e-01	5.000000e-01	0.000000e+00	2.118265e+03	1.715236e+00	2.118265e+03
1	0	1.000000e-01	5.000000e-01	6.550410e+03	2.134599e+03	1.721836e+00	2.200103e+03
1	1	1.000000e-01	2.500000e-01	1.637603e+03	1.919619e+03	1.632831e+00	1.935995e+03
2	0	1.000000e-01	2.500000e-01	1.888781e+03	1.523994e+03	1.454874e+00	1.542882e+03
3	0	1.000000e-01	2.500000e-01	5.282901e+03	1.082325e+03	1.226062e+00	1.135154e+03
4	0	1.000000e-01	3.125000e-01	1.076205e+04	7.350386e+02	1.010390e+00	8.426592e+02
5	0	1.000000e-01	3.906250e-01	1.755585e+04	4.993108e+02	8.327588e-01	6.748693e+02
6	0	1.000000e-01	4.882812e-01	2.412378e+04	3.590229e+02	7.061465e-01	6.002607e+02
7	0	1.000000e-01	5.000000e-01	2.805732e+04	2.993842e+02	6.448344e-01	5.799574e+02
8	0	1.000000e-01	5.000000e-01	3.014660e+04	2.732419e+02	6.160379e-01	5.747080e+02

File name	Content
femtic_#.log	Log information
femtic.cnv	Convergence of the Gauss-Newton iteration
result_#_iter*.csv	Calculated and observed response functions
resistivity_block_iter*.dat	Parameter cells and resistivity values obtained by the inversion
distortion_iter*.dat	Galvanic distortion parameters obtained by the inversion

‘#’ and ‘*’ indicate process number and iteration number, respectively.

Example

Iter#	Retrial#	Alpha	Damp	Roughness	Misfit	RMS	ObjFunc
0	0	1.000000e-01	5.000000e-01	0.000000e+00	2.118265e+03	1.715236e+00	2.118265e+03
1	0	1.000000e-01	5.000000e-01	6.550410e+03	2.134599e+03	1.721836e+00	2.200103e+03
1	1	1.000000e-01	2.500000e-01	1.637603e+03	1.919619e+03	1.632831e+00	1.935995e+03
2	0	1.000000e-01	2.500000e-01	1.888781e+03	1.523994e+03	1.454874e+00	1.542882e+03
3	0	1.000000e-01	2.500000e-01	5.282901e+03	1.082325e+03	1.226062e+00	1.135154e+03
4	0	1.000000e-01	3.125000e-01	1.076205e+04	7.350386e+02	1.010390e+00	8.426592e+02
5	0	1.000000e-01	3.906250e-01	1.755585e+04	4.993108e+02	8.327588e-01	6.748693e+02
6	0	1.000000e-01	4.882812e-01	2.412378e+04	3.590229e+02	7.061465e-01	6.002607e+02
7	0	1.000000e-01	5.000000e-01	2.805732e+04	2.993842e+02	6.448344e-01	5.799574e+02
8	0	1.000000e-01	5.000000e-01	3.014660e+04	2.732419e+02	6.160379e-01	5.747080e+02

Because the objective function increased at the first iteration, the damping factor was reduced from 0.5 to 0.25.

After that, the objective function decreased continuously, and then the damping factor was gradually increased.

Output files (3) Inverted resistivity values and distortion parameters

File name	Content
femtic_#.log	Log information
femtic.cnv	Convergence of the Gauss-Newton iteration
result_#_iter*.csv	Calculated and observed response functions
resistivity_block_iter*.dat	Parameter cells and resistivity values obtained by the inversion
distortion_iter*.dat	Galvanic distortion parameters obtained by the inversion

‘#’ and ‘*’ indicate process number and iteration number, respectively.

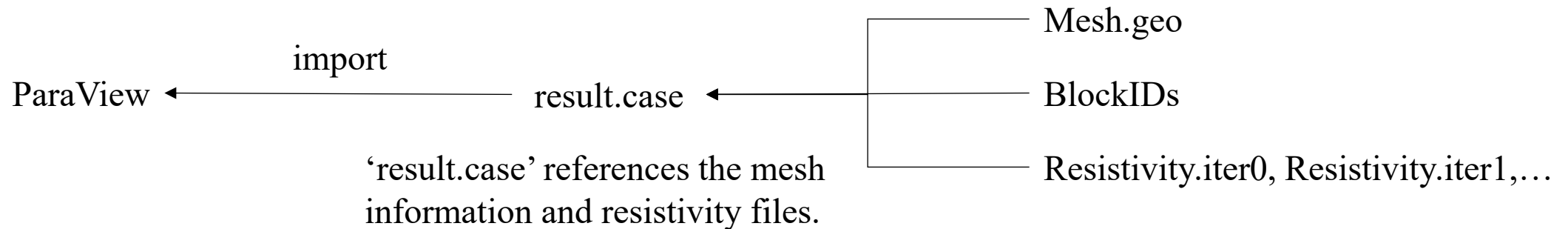
► File formats are the same as those of the input files, i.e., ‘resistivity_block_iter0.dat’ and ‘distortion_iter0.dat’

Output files (4) Files for the visualization with ParaView

The following output files can be imported into ParaView (<https://www.paraview.org/>).

File name	Content
result.case	EnSight Gold case file (ASCII)
Mesh.geo	EnSight Gold geometry file (Binary)
BlockIDs	Parameter cell of each element (EnSight Gold variable file format, Binary)
Resistivity.iter*	Resistivity values (EnSight Gold variable file format, Binary)
obs_loc.vtk	Locations of the observation stations used in the inversion (ASCII)
induction_arrow_#_iter*.vtk	Induction vector (Parkinson convention) at each tipper station (ASCII)

‘#’ and ‘*’ indicate process number and iteration number, respectively.



Useful functions of ParaView

Extract a part of the model using a threshold

Select item to be displayed

When binary files are outputted (on default), the iteration number can be selected here.

The model can be aligned with coordinate axes.

An arbitrary cross-section can be displayed

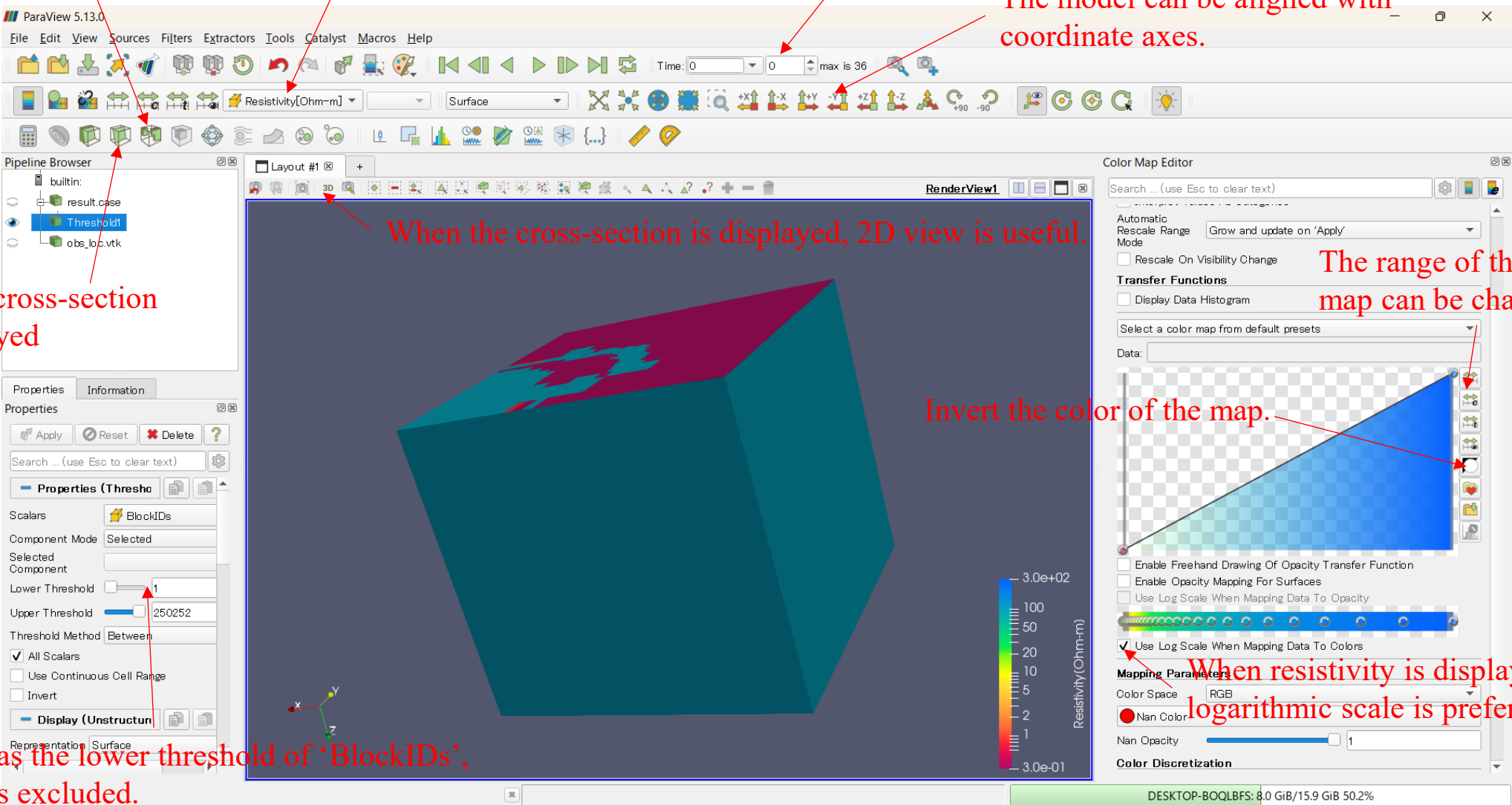
When the cross-section is displayed, 2D view is useful.

The range of the color map can be changed.

Invert the color of the map.

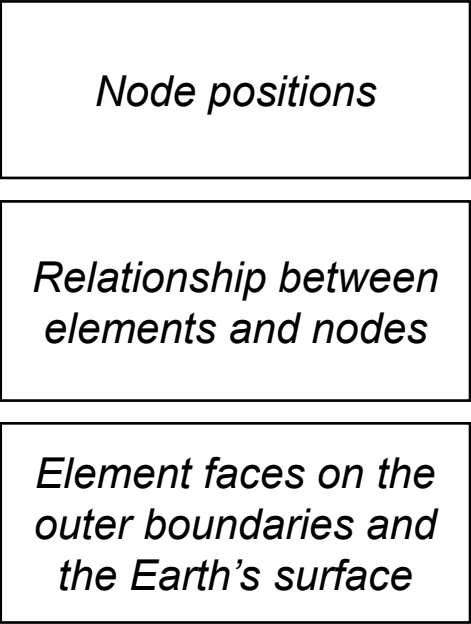
When resistivity is displayed, the logarithmic scale is preferable.

By writing 1 as the lower threshold of 'BlockIDs', the air layer is excluded.

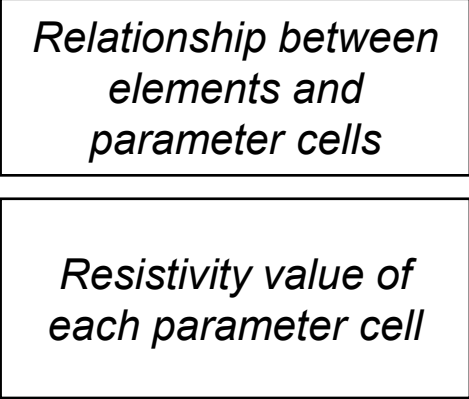


File name	Content
control.dat	Parameters controlling FEMTIC
observe.dat	Observation data
distortion_iter0.dat	Initial values of galvanic distortion parameters
mesh.dat	Computational mesh
resistivity_block_iter0.dat	Parameter cells and initial resistivity values

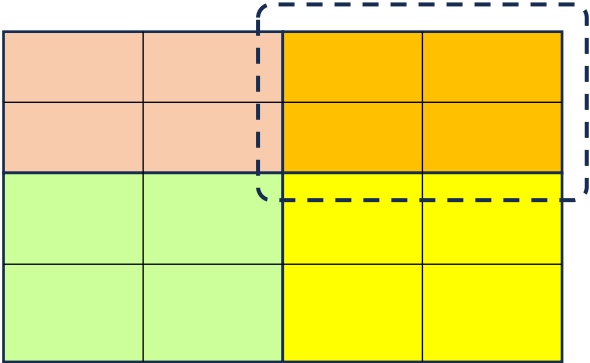
Structure of ‘mesh.dat’



Structure of ‘resistivity_block_iter0.dat’

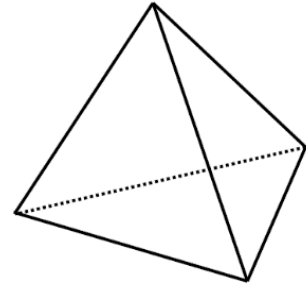
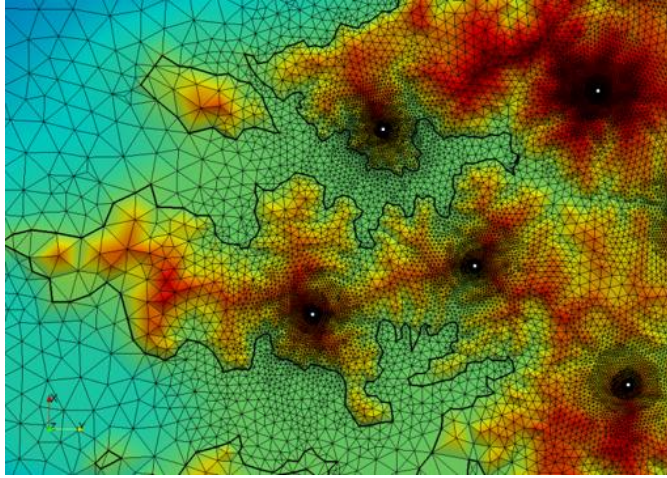


Parameter cell is defined as the group of adjacent elements that have the same resistivity.



It is nearly impossible to make those files manually.
→ Some mesh generation tools are shared on GitHub.

Mesh generation tools

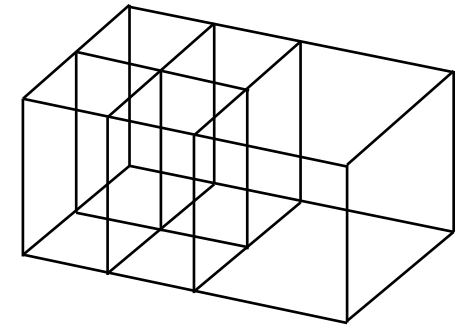
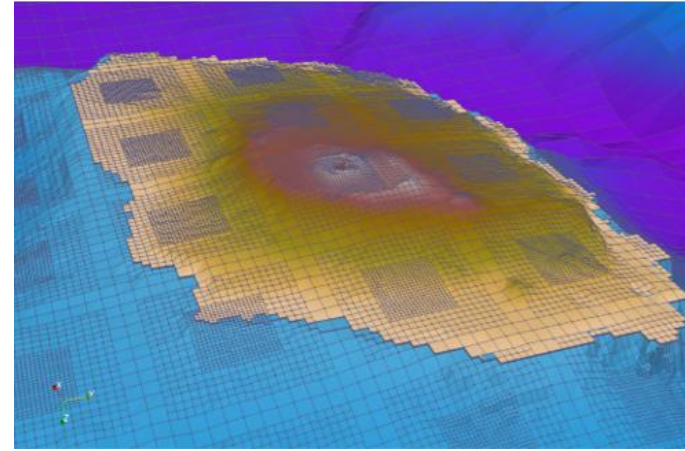


Tetrahedral mesh

- makeTetraMesh
<https://github.com/yoshiya-usui/makeTetraMesh>
- makeMtr
<https://github.com/yoshiya-usui/makeMtr>
- TetGen2Femtic
<https://github.com/yoshiya-usui/TetGen2Femtic>

In addition, TetGen (Si, 2015) is necessary.

<https://wias-berlin.de/software/tetgen/>

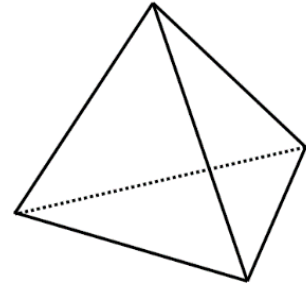
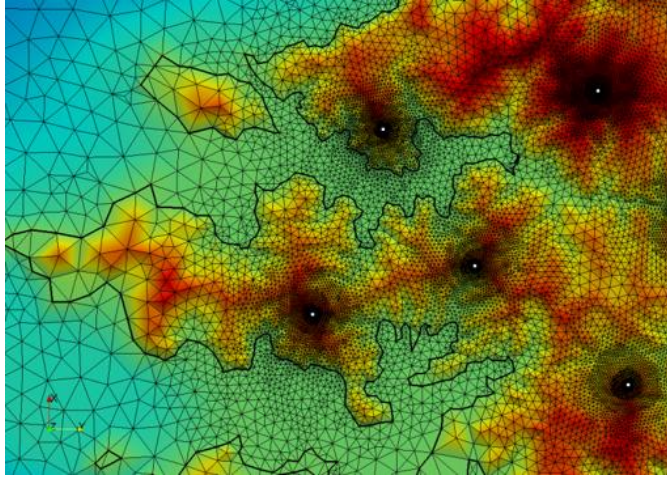


Non-conforming deformed hexahedral mesh

- makeDHexaMesh
<https://github.com/yoshiya-usui/makeDHexaMesh>

I recommend using a hexahedral mesh for the initial trial of FEMTIC because mesh generation can be completed using one tool.

Mesh generation tools

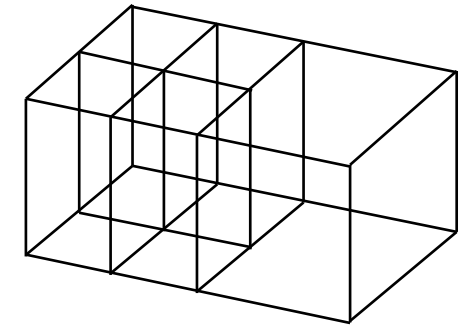
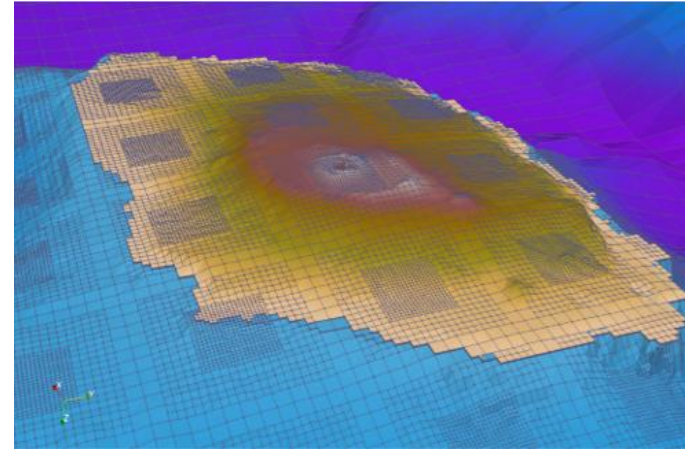


Tetrahedral mesh

- makeTetraMesh
<https://github.com/yoshiya-usui/makeTetraMesh>
- makeMtr
<https://github.com/yoshiya-usui/makeMtr>
- TetGen2Femtic
<https://github.com/yoshiya-usui/TetGen2Femtic>

In addition, TetGen (Si, 2015) is necessary.

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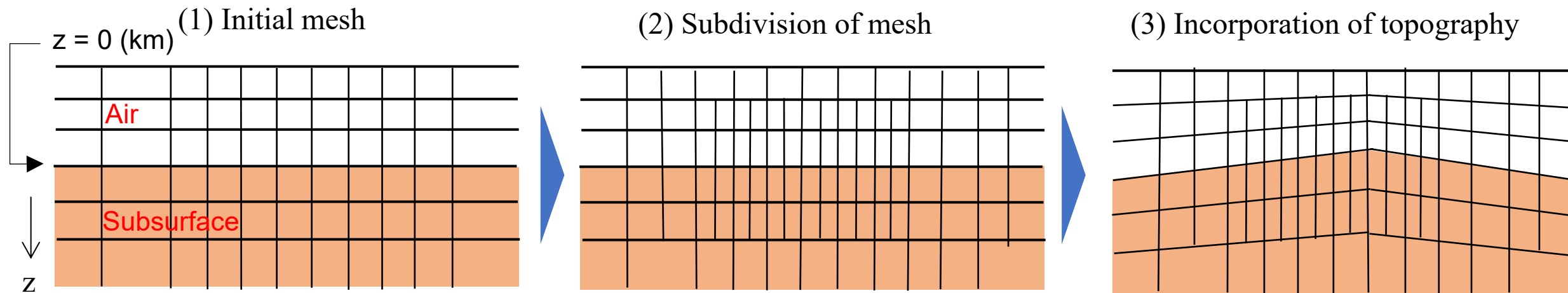
Non-conforming deformed hexahedral mesh

- makeDHexaMesh
<https://github.com/yoshiya-usui/makeDHexaMesh>

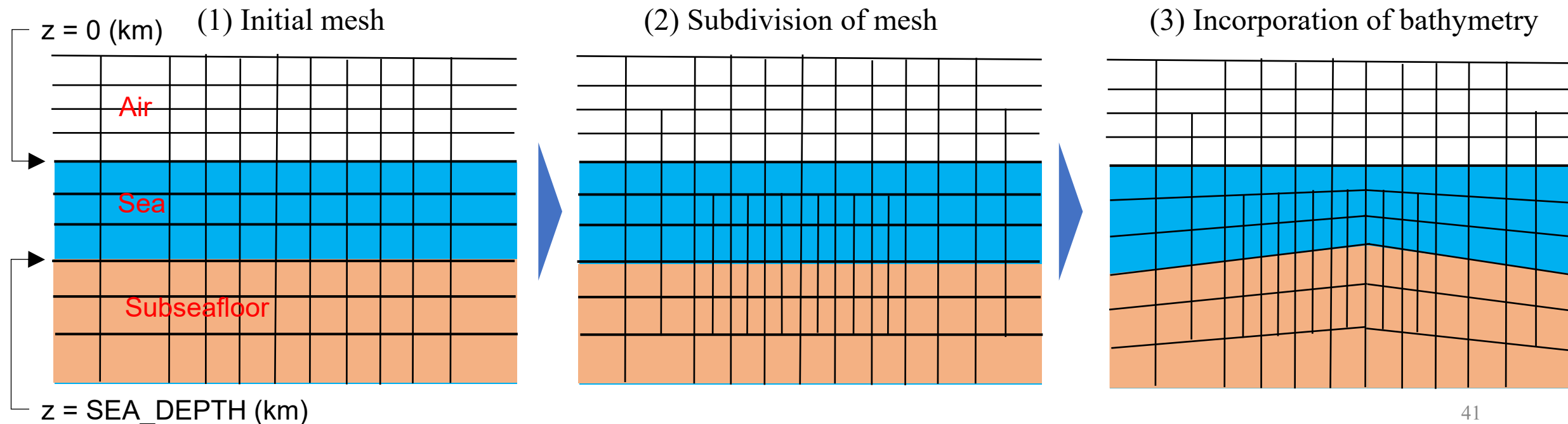
I recommend using a deformed hexahedral mesh for the initial trial of FEMTIC because mesh generation can be completed using one tool.

Mesh generation algorithm of 'makeDHexaMesh'

Land MT case



Ocean bottom MT case



Input files

File name	Contents
meshgen.inp	Controlling parameters
obs_site.dat	Desired edge lengths around observation sites
<i>Arbitrary name</i>	Topography/bathymetry (the name is defined in the keyword ‘TOPO’ of ‘meshgen.inp’)

Output files

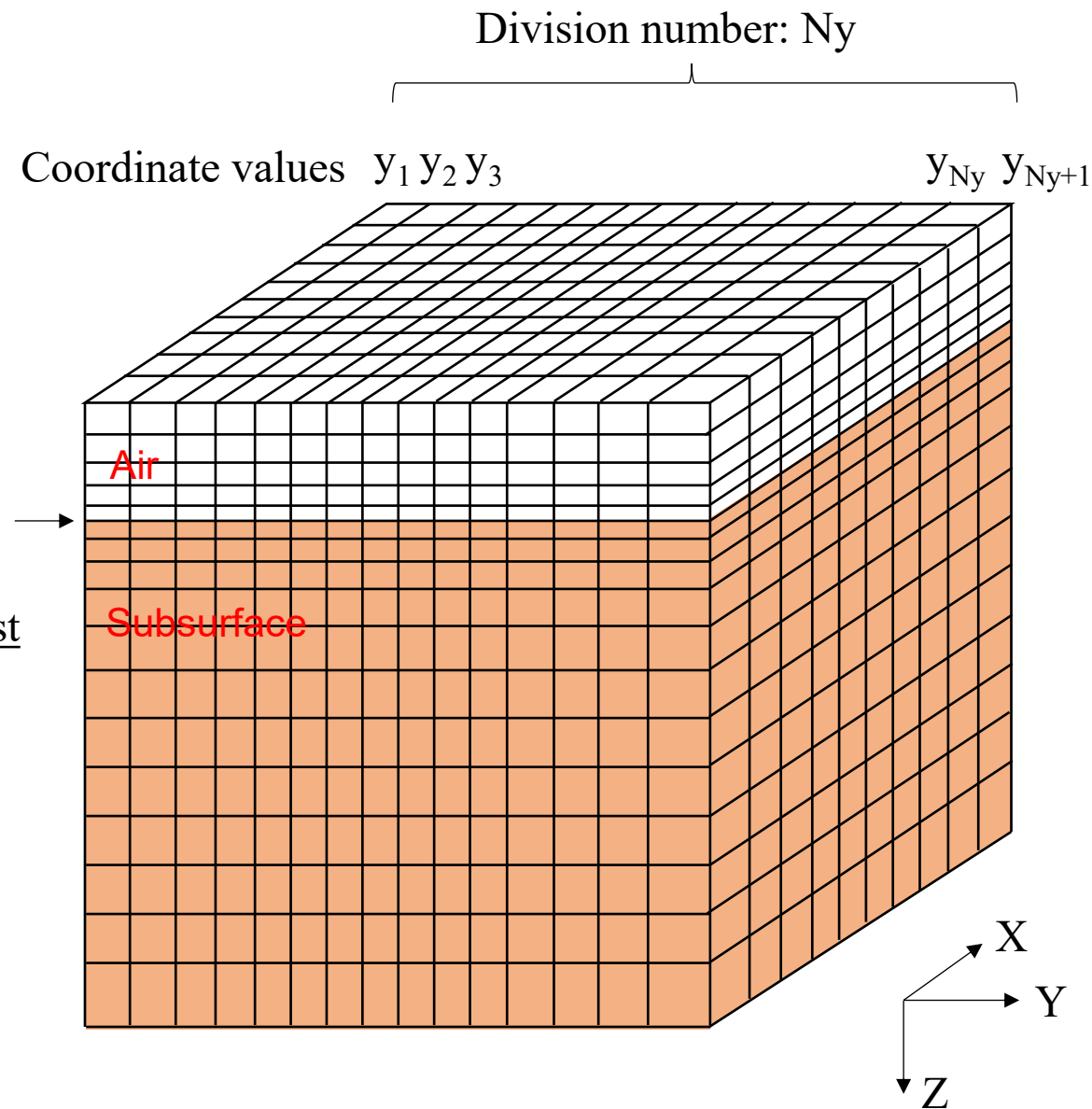
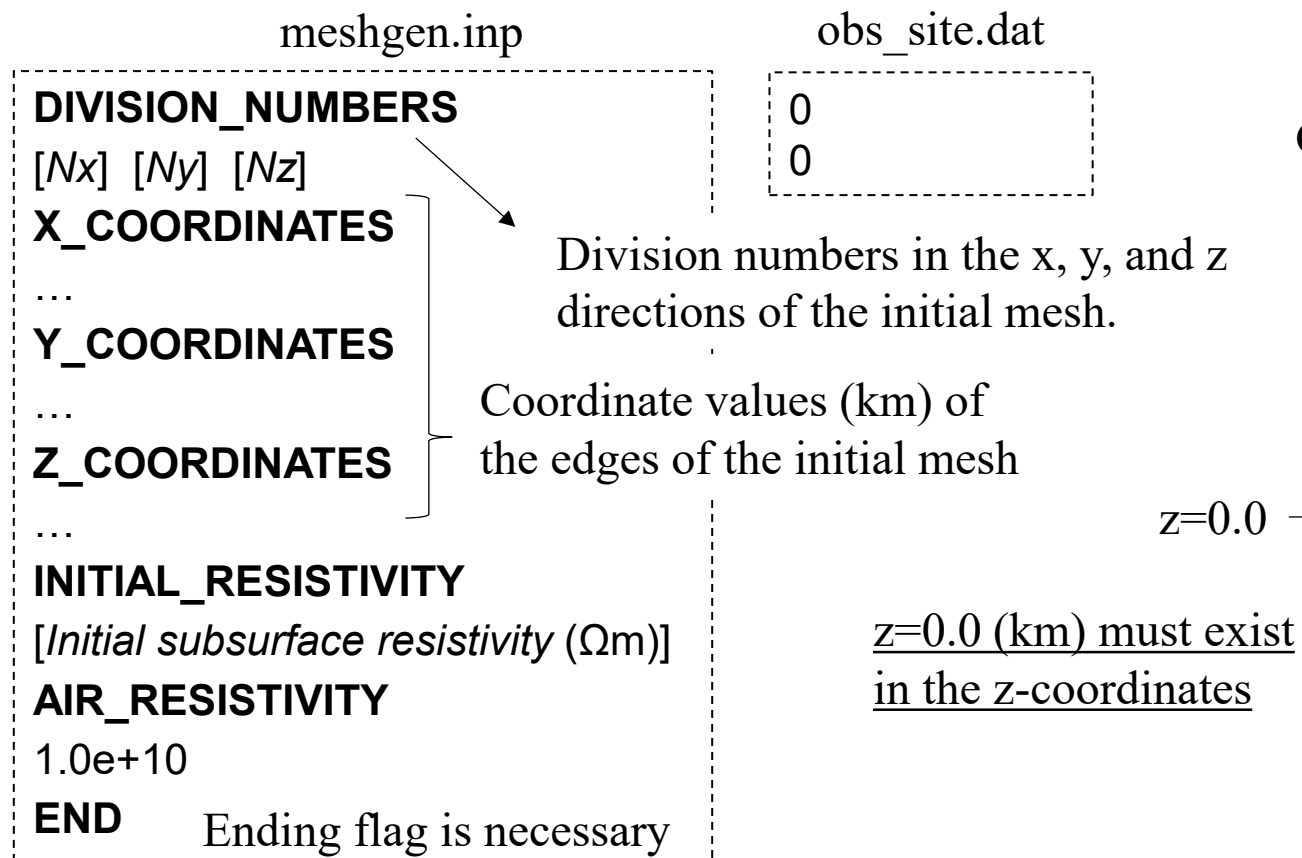
File name	Contents
mesh.dat	Data of computational mesh
resistivity_block_iter0.dat	Information about parameter cells and the initial resistivity values
MeshData.vtk	Data of computational mesh (for the visualization using ParaView)
depth.vtk	Data of the surface mesh of the computational mesh (for the visualization using ParaView)

Execution command

Please run the executable file ‘makeDHexaMesh’ in the directory where all the input files are stored.

Recommended setting for the initial trial (Land MT case)

7 October EMinars Season 4



This simplest setting does not incorporate topography and does not perform local refinement.

I recommend confirming that the response functions obtained from the simplest uniform half-space model are close to the analytical solutions before moving to elaborate meshing.

Recommended setting for the initial trial (Ocean bottom MT case)

meshgen.inp

obs_site.dat

DIVISION_NUMBERS

[Nx] [Ny] [Nz]

X_COORDINATES

...

Y_COORDINATES

...

Z_COORDINATES

...

INITIAL_RESISTIVITY

[Initial subsurface resistivity (Ωm)]

AIR_RESISTIVITY

1.0e+10

SEA_DEPTH

[Initial sea depth (km)]

SEA_RESISTIVITY

[Sea resistivity (Ωm)]

END

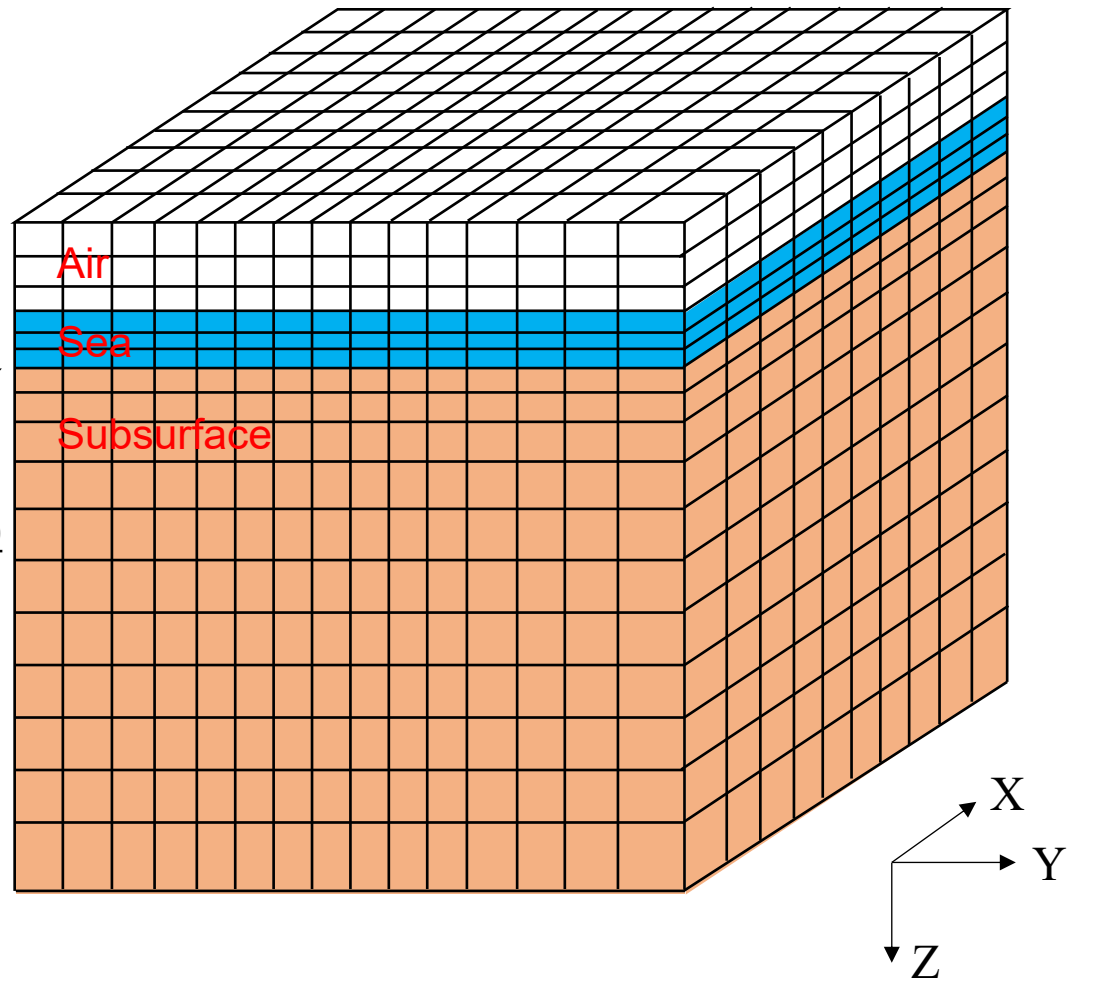
Ending flag is necessary

0

0

Division number: Ny
Coordinate values $y_1 y_2 y_3$ $y_{Ny} y_{Ny+1}$

$z=0.0$
 $z=\text{SEA_DEPTH}$
 $z=0.0$ and initial sea depth (km)
must exist in the z-coordinates



How to incorporate topography and bathymetry

7 October EMinars Season 4

meshgen.inp

DIVISION_NUMBERS

...

X_COORDINATES

...

Y_COORDINATES

...

Z_COORDINATES

...

INITIAL_RESISTIVITY

[Initial subsurface resistivity (Ωm)]

AIR_RESISTIVITY

1.0e+10

TOPO

[File name of topography/bathymetry data]

[Maximum point number used for interpolation]

[Search radius (km)]

[Small number to avoid zero divide (km)]

END

Example of topography data

-106.824031915	76.2046151595	2.100000e-003↓
-106.824276161	76.1651038245	6.000000e-003↓
-106.824459262	76.1354703228	8.500000e-003↓
-106.824642292	76.1058368206	7.400000e-003↓
-106.824886221	76.0663254836	9.000000e-003↓
-106.825069084	76.0366919804	9.300000e-003↓
-106.825251877	76.0070584767	8.600000e-003↓
-106.825495489	75.9675471378	8.600000e-003↓
-106.825678115	75.9379136331	9.400000e-003↓
-106.82586067	75.908280128	1.040000e-002↓
-106.826103966	75.8687687872	1.070000e-002↓
-106.826286354	75.839135281	9.300000e-003↓
-106.826468671	75.8095017744	1.110000e-002↓
-106.82671165	75.7699904316	1.390000e-002↓

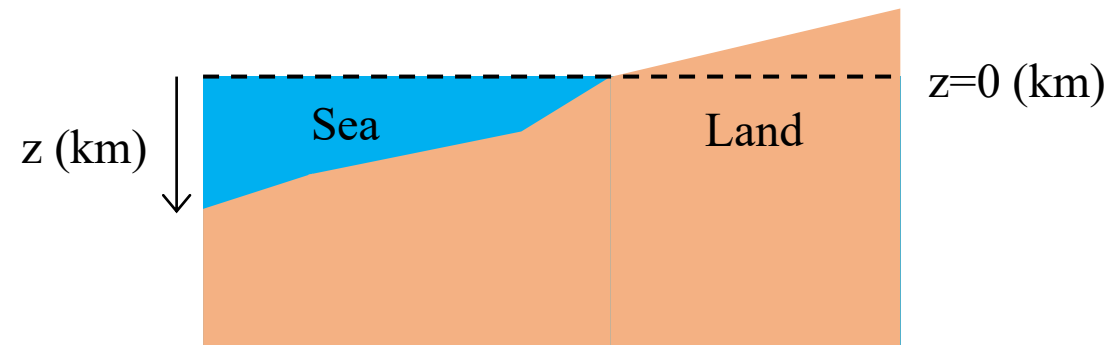
X (km)

Y (km)

Z (km)

X, Y, and Z coordinates
(km) of topography data
points, from left to right

Z-coordinates should be positive in the downward direction.



How to incorporate topography and bathymetry

7 October EMinars Season 4

meshgen.inp

DIVISION_NUMBERS

...

X_COORDINATES

...

Y_COORDINATES

...

Z_COORDINATES

...

INITIAL_RESISTIVITY

[Initial subsurface resistivity (Ωm)]

AIR_RESISTIVITY

1.0e+10

TOPO

[File name of topography/bathymetry data]

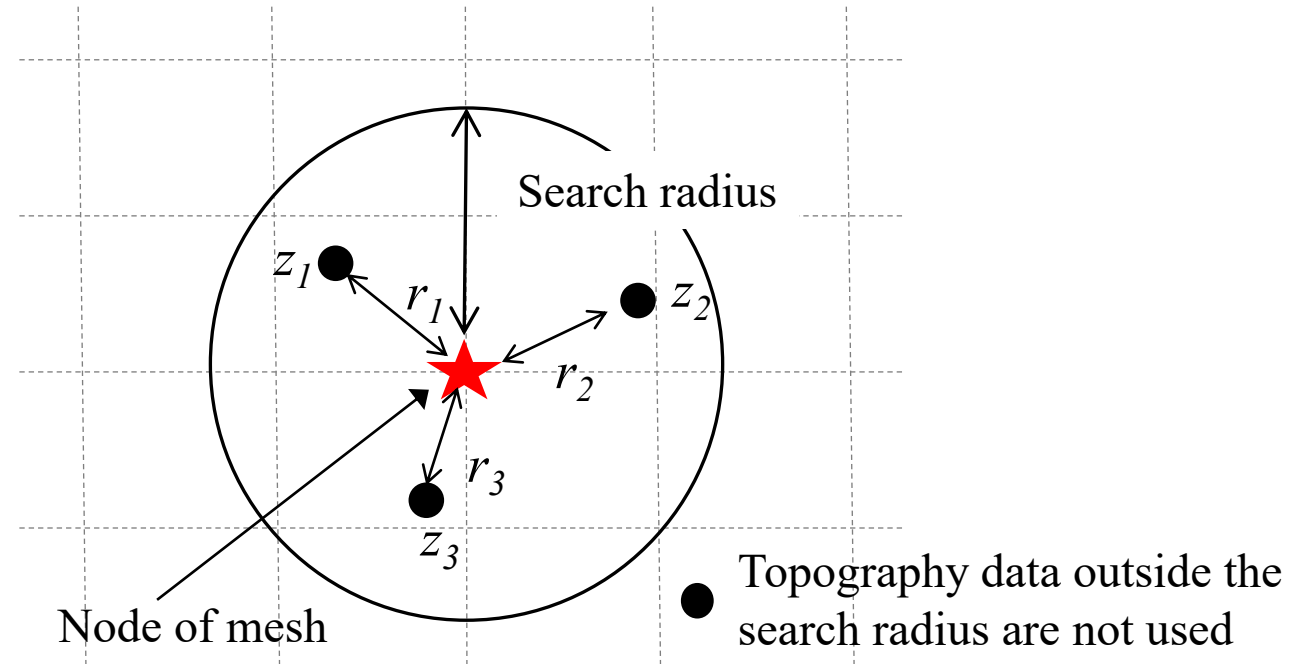
[Maximum point number used for interpolation]

[Search radius (km)]

[Small number to avoid zero divide ε (km)]

END

The inverse distance weighting method is used for interpolation of z coordinates on the Earth's surface



Interpolated z-coordinate
of mesh node

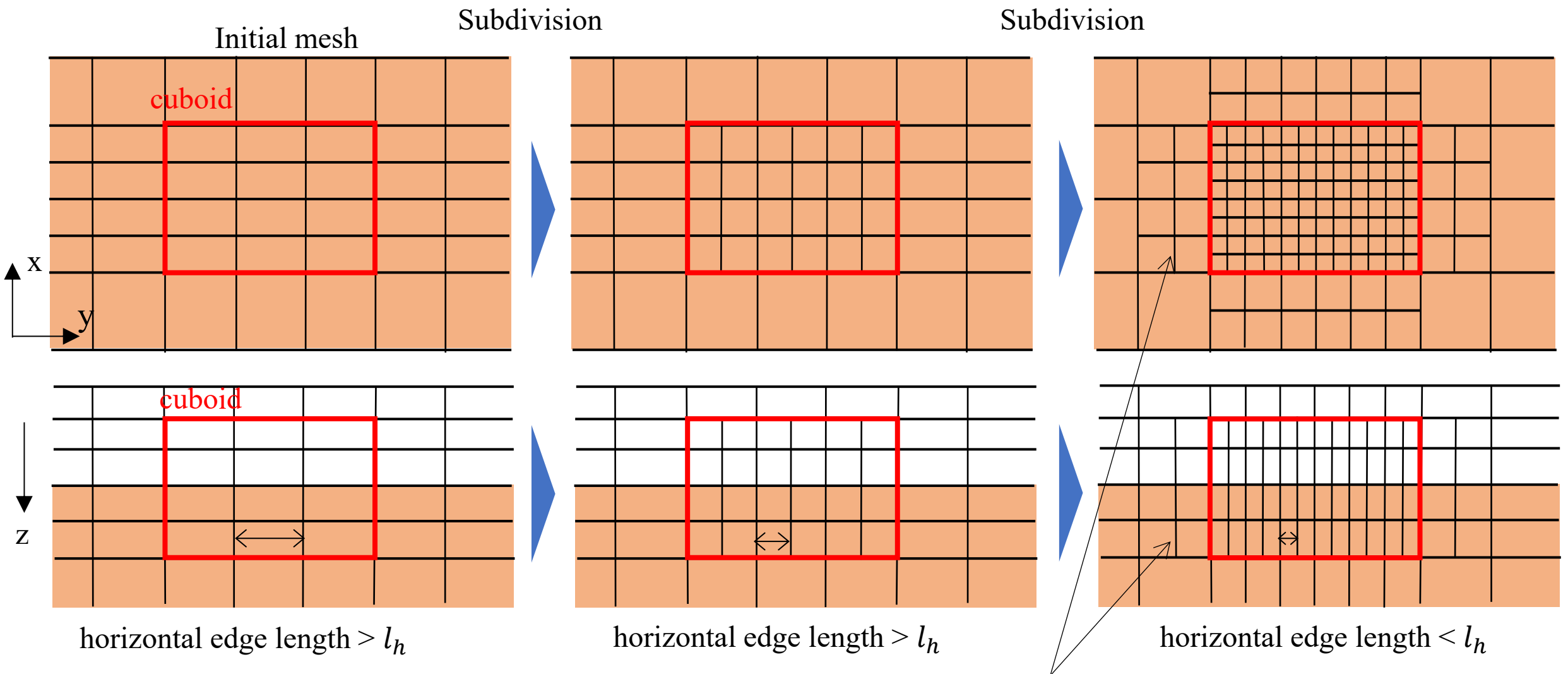
$$z = \frac{\sum_{i=1}^N w_i z_i}{\sum_{i=1}^N w_i}$$

Interpolation weight

$$w_i = \frac{1}{(r_i + \varepsilon)}$$

How to do subdivision

Subsivision is performed by defining cuboids and the upper limits (l_h) of the horizontal edge lengths within them.



The elements outside of cuboids can be subdivided because only bisection is allowed between adjacent elements.

How to do subdivision

meshgen.inp

DIVISION_NUMBERS

...

X_COORDINATES

...

Y_COORDINATES

...

Z_COORDINATES

...

INITIAL_RESISTIVITY

[Initial subsurface resistivity (Ωm)]

AIR_RESISTIVITY

1.0e+10

CUBOIDS

[Center coordinate of cuboids (km)]

[Rotation angle of cuboids (deg.)]

[Number of cuboids]

[Parameters for the 1st cuboid]

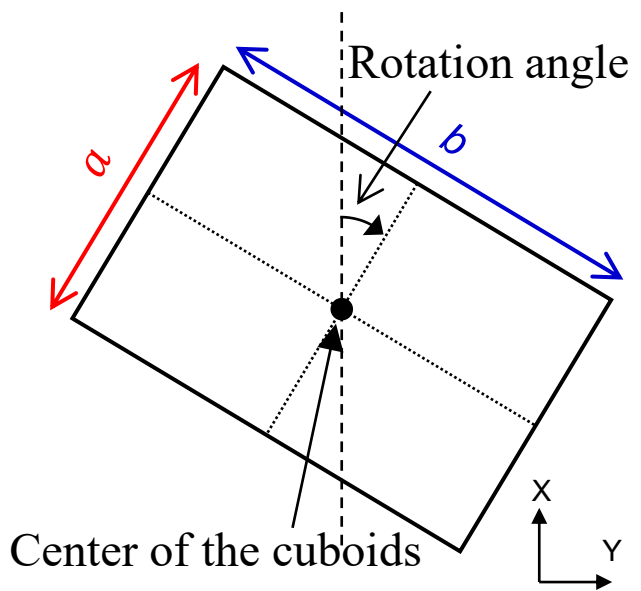
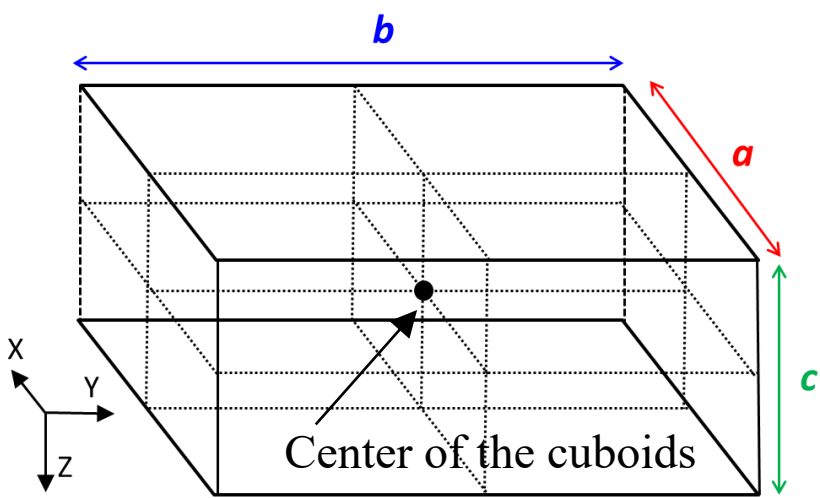
...

[Parameters for the last cuboid]

END

Subsequent cuboids must cover the formers (must be larger than the formers)

Cuboids are used to hierarchically change mesh fineness



Parameter for each cuboid

a	b	c	l_h
-----	-----	-----	-------

a : Length along x-axis (km)

b : Length along y-axis (km)

c : Length along z-axis (km)

l_h : Upper limit of the horizontal edge lengths within the cuboid (km)

How to do subdivision

meshgen.inp

DIVISION_NUMBERS

...

X_COORDINATES

...

Y_COORDINATES

...

Z_COORDINATES

...

INITIAL_RESISTIVITY

[Initial subsurface resistivity (Ωm)]

AIR_RESISTIVITY

1.0e+10

CUBOIDS

[Center coordinate of cuboids (km)]

[Rotation angle of cuboids (deg.)]

[Number of cuboids]

[Parameters for the 1st cuboid]

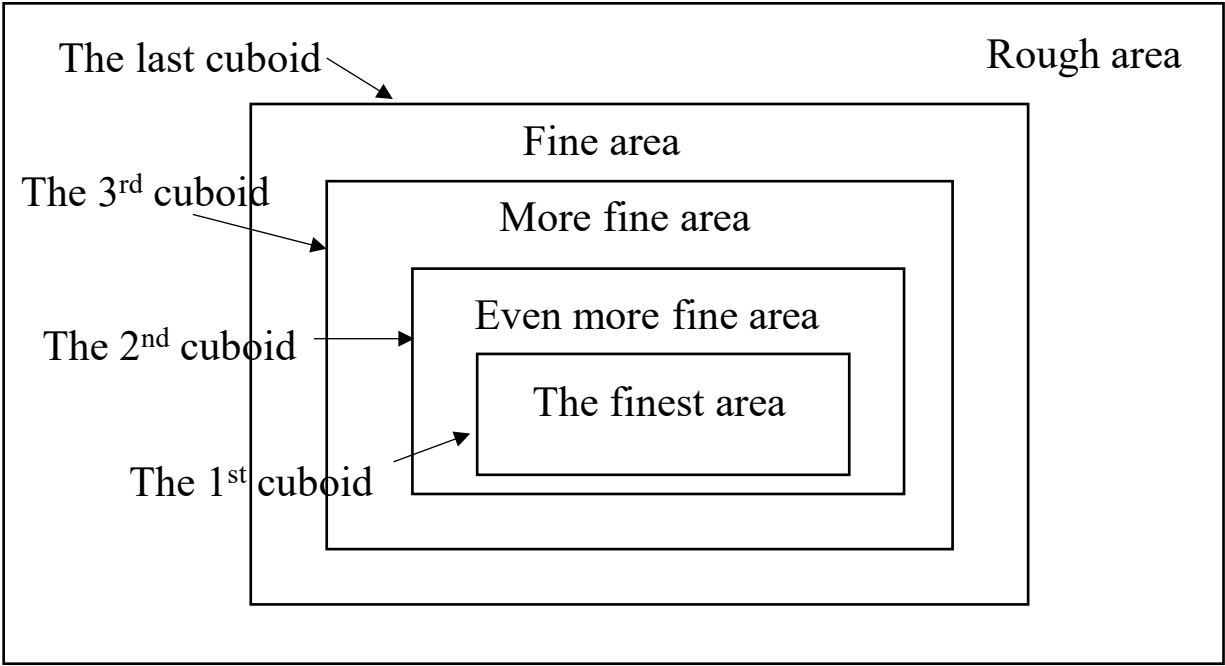
...

[Parameters for the last cuboid]

...

END

Cuboids are used to hierarchically change mesh fineness



Subsequent cuboids must cover the formers (must be larger than the formers)

a : Length along x-axis (km)

b : Length along y-axis (km)

c : Length along z-axis (km)

l_h : Upper limit of the horizontal edge lengths within the cuboid (km)

Parameter for each cuboid

a b c l_h

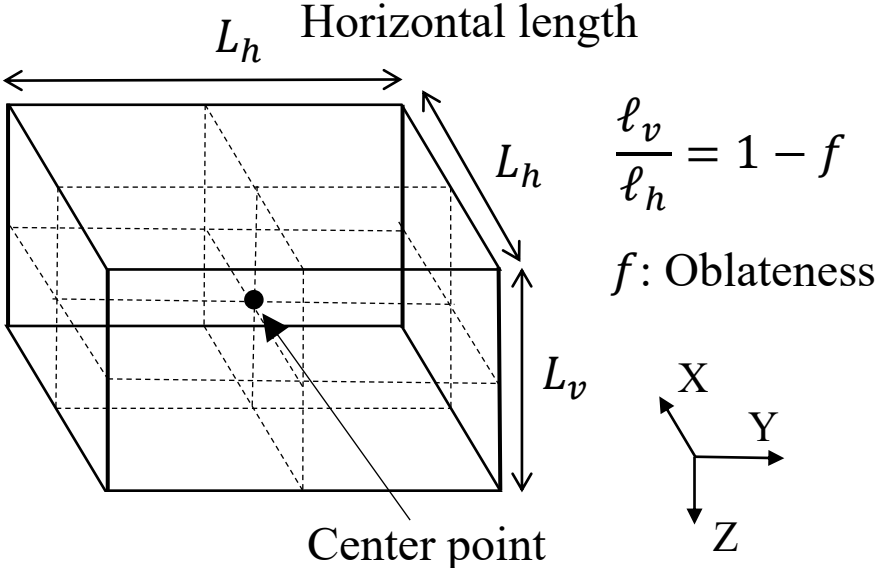
Trial and error are required to determine the cuboid settings

Local refinement around observation points

‘obs_site.dat’ is used to refine the areas around observation points.

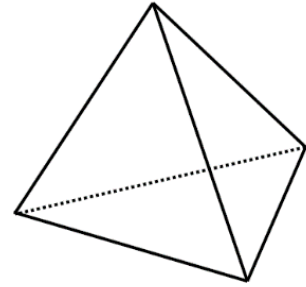
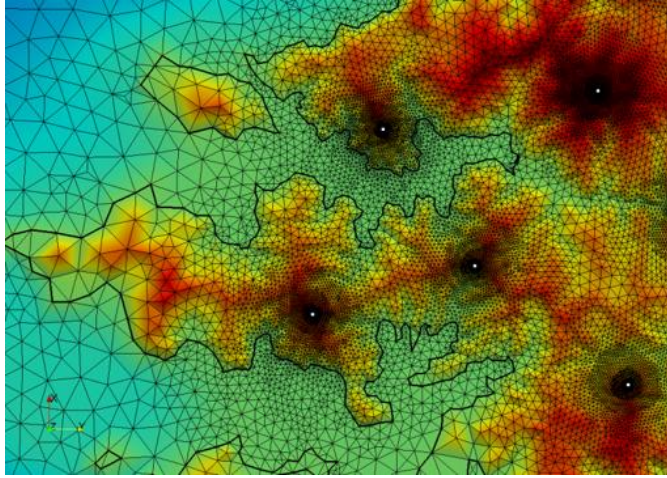
Format of ‘obs_site.dat’

Number of observation points			Data for the 1 st point		
X (km)	Y (km)	Z (km)	Location of the point (or the cuboid center)		
Number of cuboids					
Horizontal length of cuboid L_h (km)	The upper limit of horizontal edge lengths l_h (km)		Oblateness f	The 1 st cuboid	
⋮					
Horizontal length of cuboid L_h (km)	The upper limit of horizontal edge lengths l_h (km)		Oblateness f	The last cuboid	
⋮					
			Data for the last point		



Subsequent cuboids must cover the formers
(must be larger than the formers)

Mesh generation tools

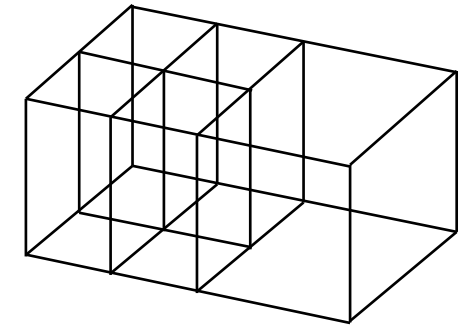
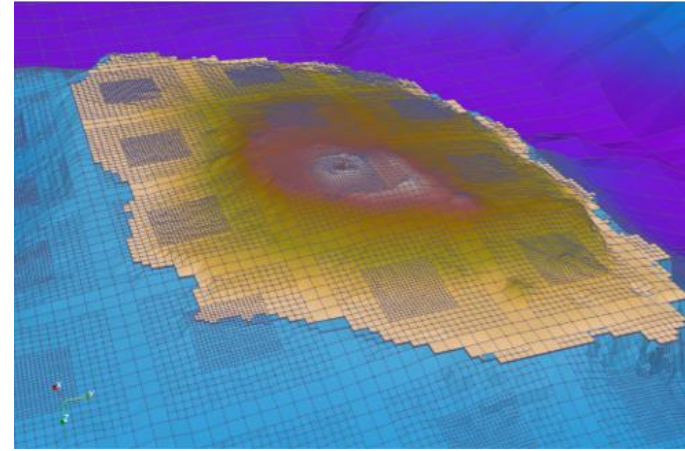


Tetrahedral mesh

- makeTetraMesh
<https://github.com/yoshiya-usui/makeTetraMesh>
- makeMtr
<https://github.com/yoshiya-usui/makeMtr>
- TetGen2Femtic
<https://github.com/yoshiya-usui/TetGen2Femtic>

In addition, TetGen (Si, 2015) is necessary.

<https://wias-berlin.de/software/tetgen/>

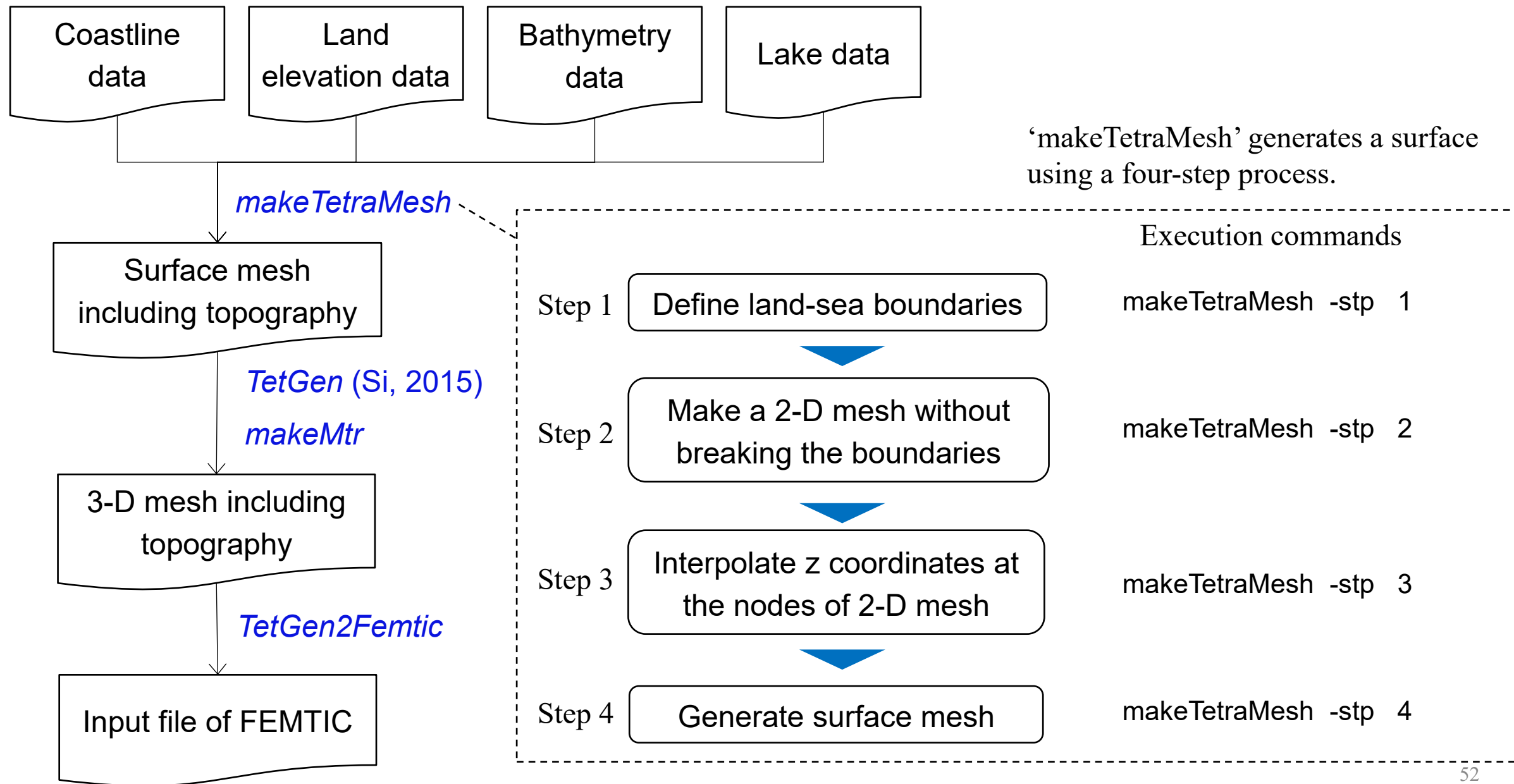


Non-conforming deformed hexahedral mesh

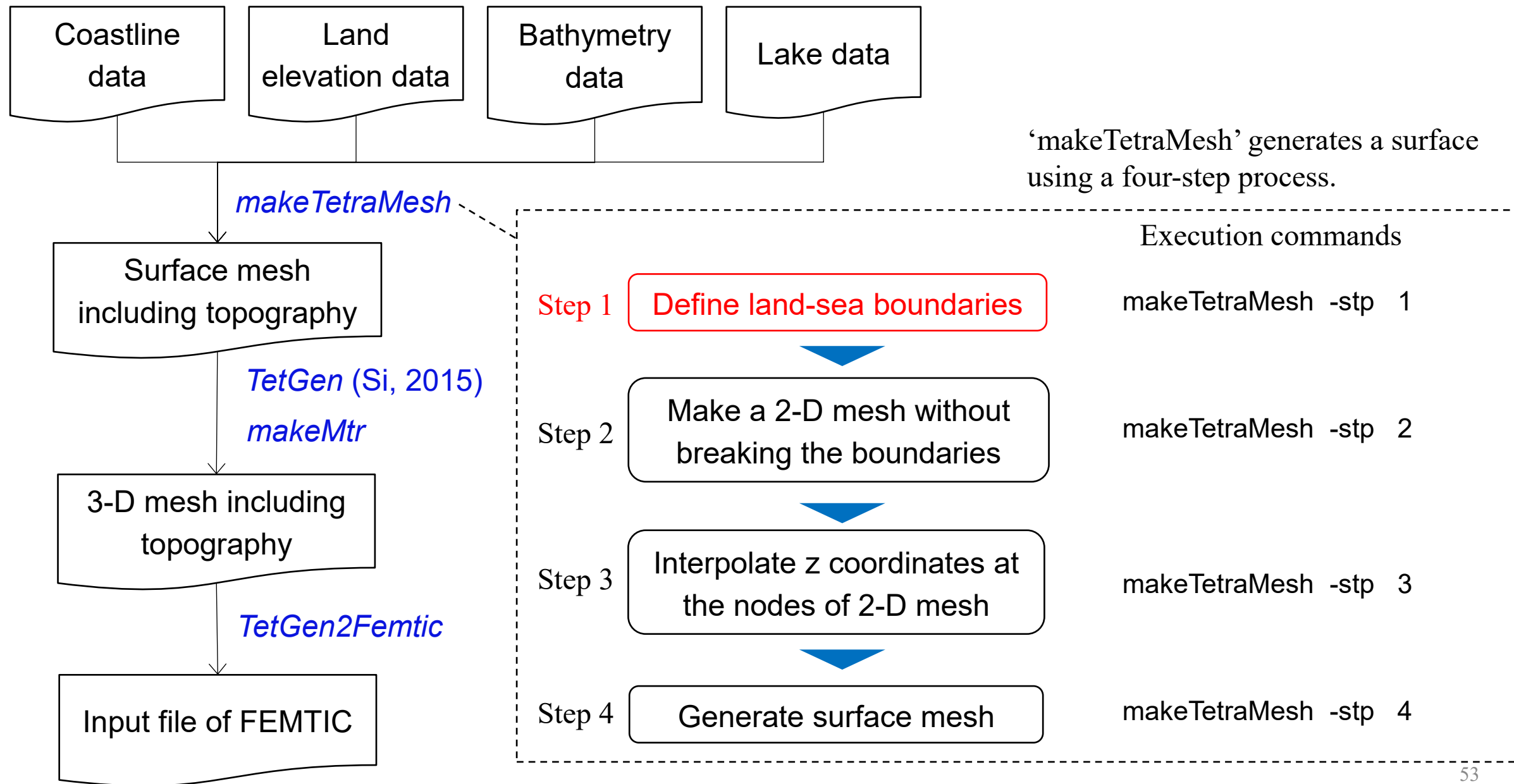
- makeDHexaMesh
<https://github.com/yoshiya-usui/makeDHexaMesh>

I recommend using a deformed hexahedral mesh for the initial trial of FEMTIC because mesh generation can be completed using one tool.

Procedure flow of tetrahedral mesh generation



Procedure flow of tetrahedral mesh generation



Input files for Step 1

File name	Contents
analysis_domain.dat	Size of computational domain
coast_line.dat	Information about land-sea/land-lake boundaries
control.dat	Parameters controlling mesh generation
observing_site.dat	Mesh size around observation sites

File format of 'analysis_domain.dat'

analysis_domain.dat

*The minimum X coordinate value
of the computational mesh (km)*

*The maximum X coordinate value
of the computational mesh (km)*

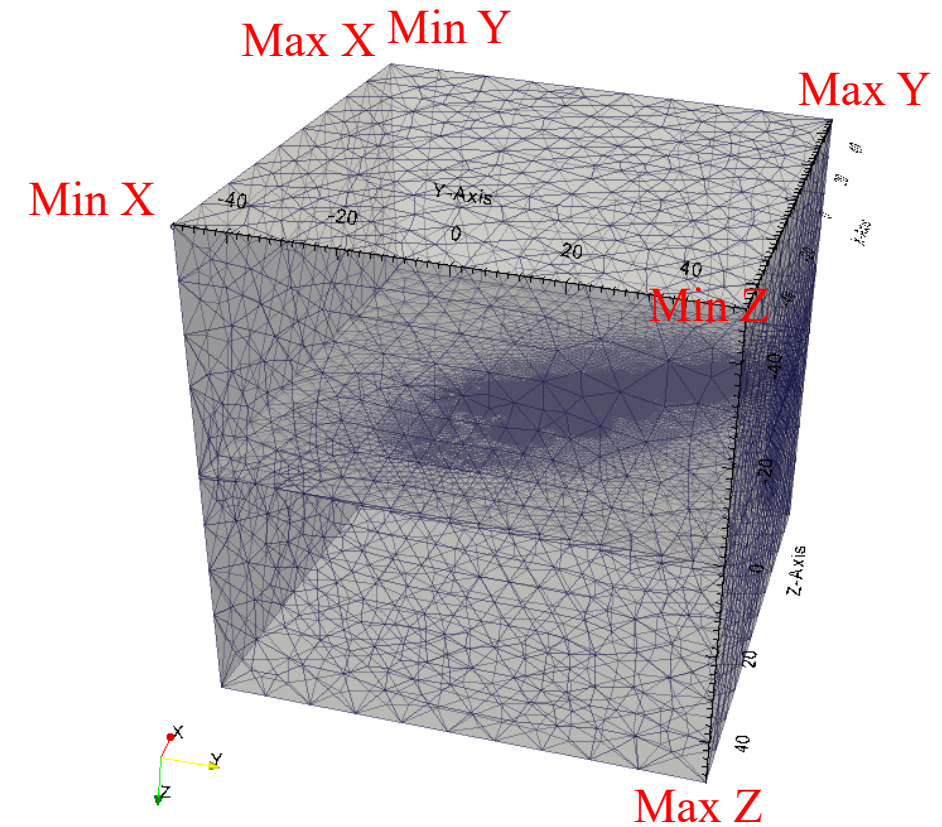
*The minimum X coordinate value
of the computational mesh (km)*

*The maximum Y coordinate value
of the computational mesh (km)*

*The minimum X coordinate value
of the computational mesh (km)*

*The maximum Z coordinate value
of the computational mesh (km)*

- The minimum coordinate value should be negative.
- The maximum coordinate value should be positive.



Example

```
-50.0  50.0 ↓
-50.0  50.0 ↓
-50.0  50.0 ↓
```

File format of 'coast_line.dat'

Number of boundaries			
X coordinate value (km)	Y coordinate value (km)	End flag	0
⋮			

End flag = 0

→ NOT the endpoint of a boundary

End flag = 1

→ End point of a closed boundary

End flag = -1

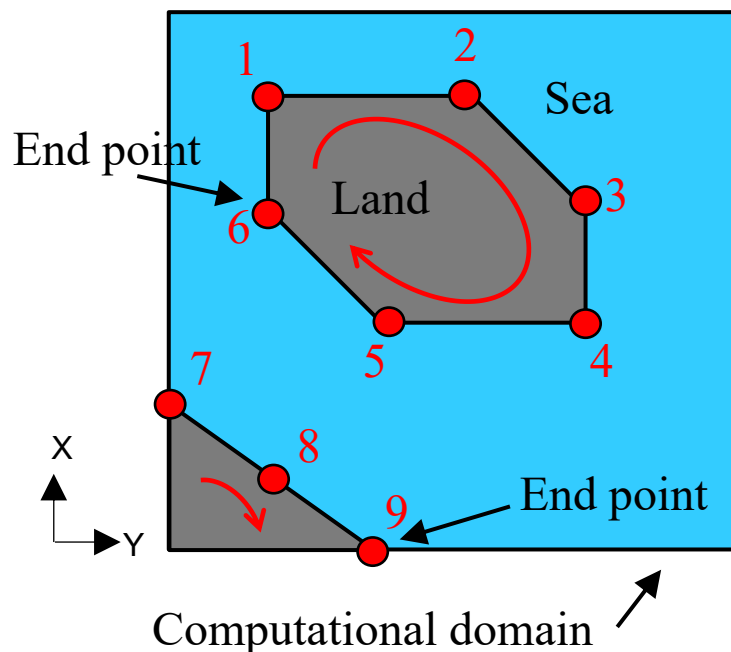
→ End point of an unclosed boundary

Example

```

2
X1 Y1 0 0
X2 Y2 0 0
X3 Y3 0 0
X4 Y4 0 0
X5 Y5 0 0
X6 Y6 1 0
X7 Y7 0 0
X8 Y8 0 0
X9 Y9 -1 0

```



- The first line should be the number of boundaries, not the number of points.
- Points should be written in the order that the land lies on the right-hand side.
- When there is no sea in the computational domain, please define a boundary that covers the whole of the computational domain.
- If there is no land area in the computational domain, please set 'Number of boundaries' to zero.

File format of 'control.dat' for Step 1

control.dat

CENTER

X (km)

Y (km)

Z (km)

Center coordinate of ellipsoids

ROTATION

Rotation angle (deg.) of the ellipsoids in the X-Y plane

ELLIPSOIDS

Number of ellipsoids

a l_h f_h f_v^+ f_v^-

The 1st ellipsoid

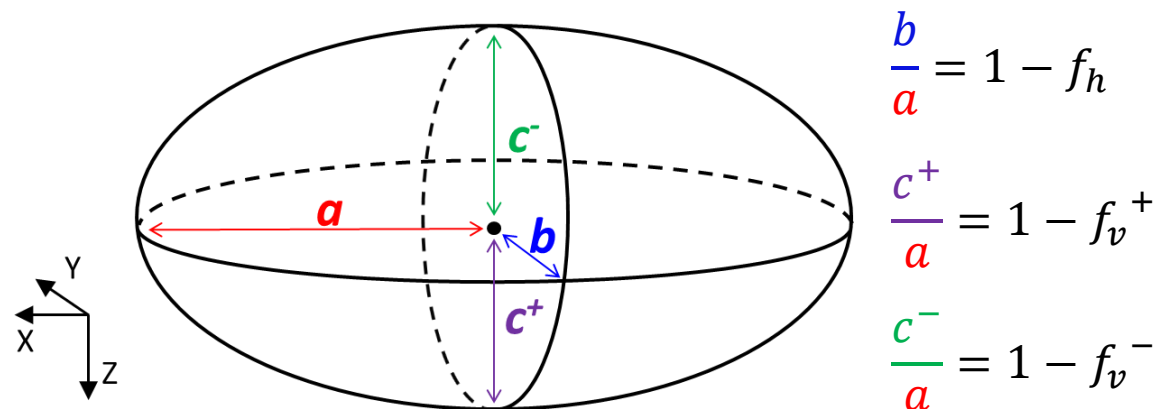
⋮

a l_h f_h f_v^+ f_v^-

The last ellipsoid

END

Ending flag is necessary



a : Length along the X axis (km)

l_h : Upper limit of the edge lengths within the ellipsoid (km)

f_h : Oblateness on the X-Y plane

f_v^+ : Oblateness on the Z-X plane (Upper side)

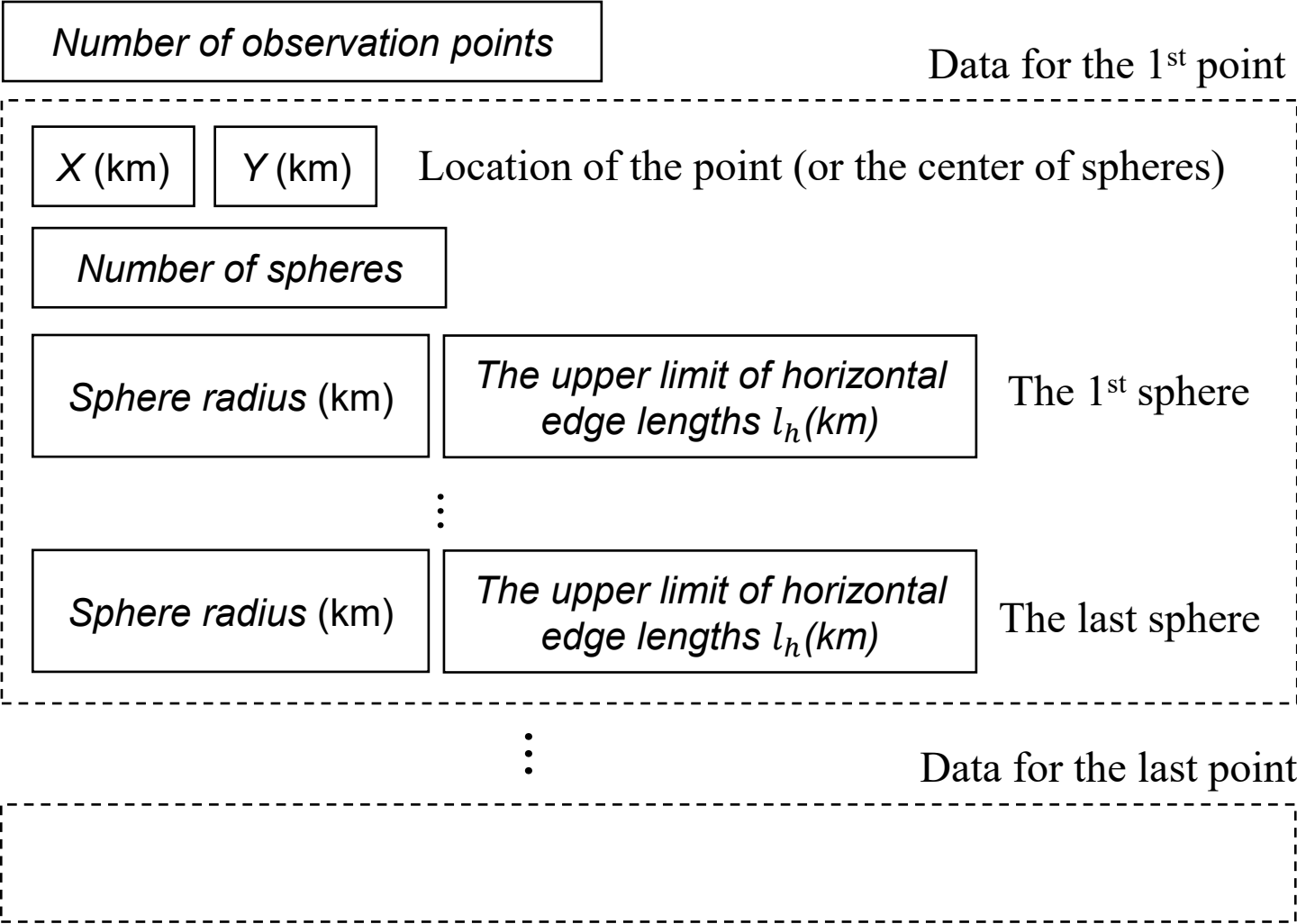
f_v^- : Oblateness on the Z-X plane (Lower side)

Subsequent ellipsoids must cover the formers
(must be larger than the formers)

File format of ‘observing_site.dat’

‘observing_site.dat’ is used to refine the areas around observation points.

observing_site.dat



Example

```
28↓
 0.0000000000 0.0000000000↓
5↓
0.1 0.02↓
0.3 0.05↓
1.0 0.10↓
3.0 0.30↓
5.0 0.50↓
```

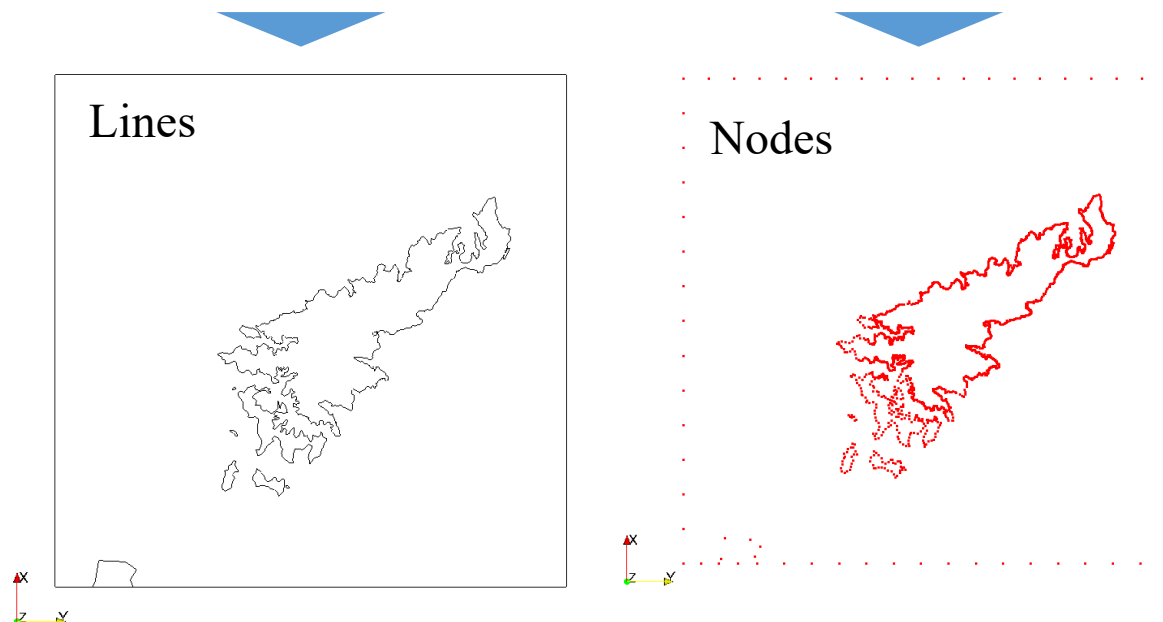
Subsequent spheres must cover the formers
(must be larger than the formers)

Bad boundary curve

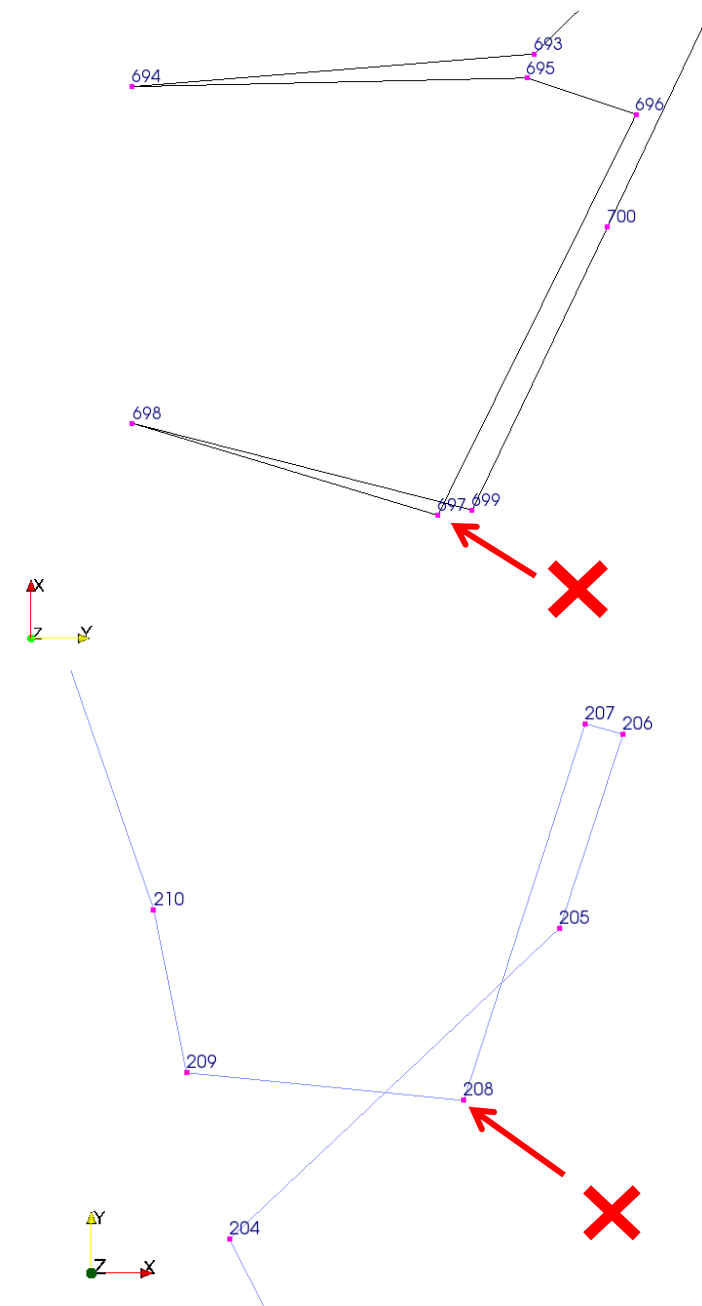
The result of Step 1, the boundary curves, can be visualized by importing two VTK files into ParaView.

boundary_curve_stp1.vtk

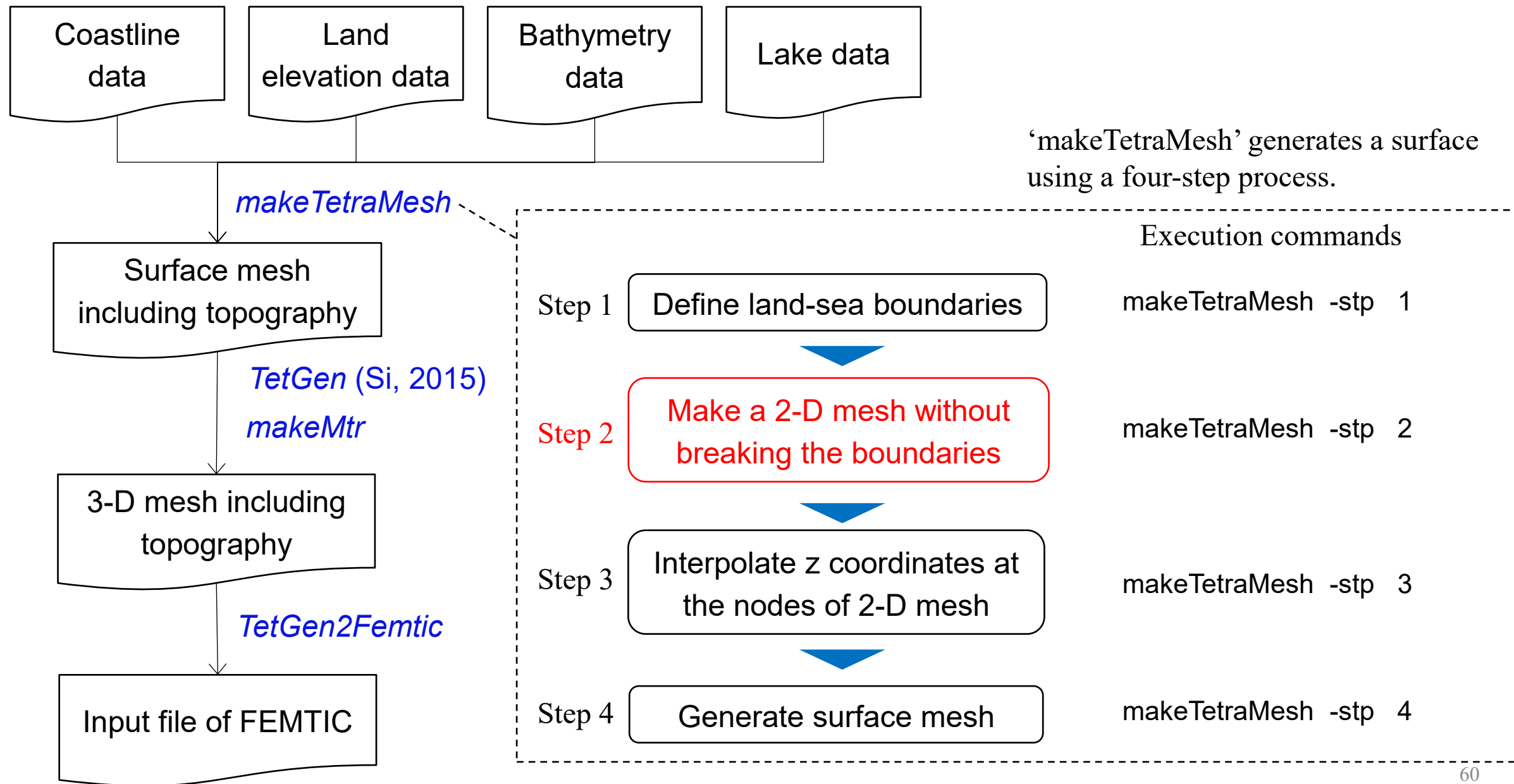
boundary_curve_node_stp1.vtk



- Boundary curves must NOT be crossed.
- It causes error at Step 2.
- If there are such boundaries, please correct them by executing Step 1 again by different parameters (or editing 'node_bound_curve.stp1' directly).



Procedure flow of tetrahedral mesh generation



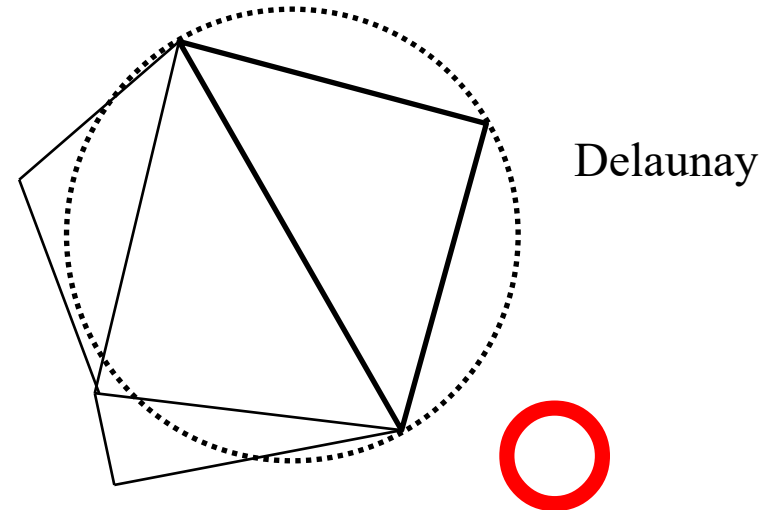
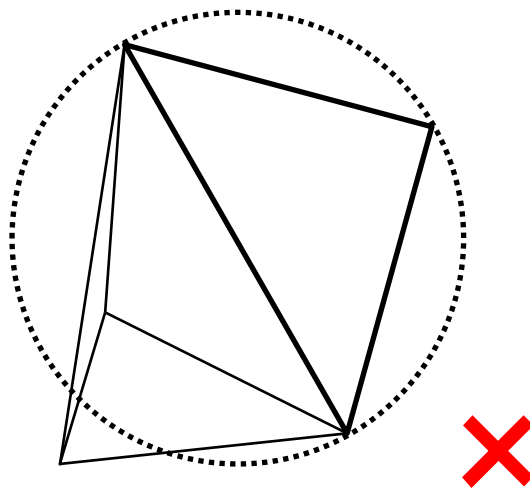
File name	Contents
analysis_domain.dat	Size of computational domain
control.dat	Parameters controlling mesh generation
observing_site.dat	Mesh size around observation sites
node_bound_curve.stp1	Coordinate values of the points of inner/outer boundaries (output file of step 1)
boundary_curves.stp1	Connections of the points of inner/outer boundaries (output file of step 1)
boundary_curves_relation.stp1	Relationship between inner/outer boundaries (output file of step 1)

The same files as for Step 1 and output files of Step 1 can be used for Step 2.

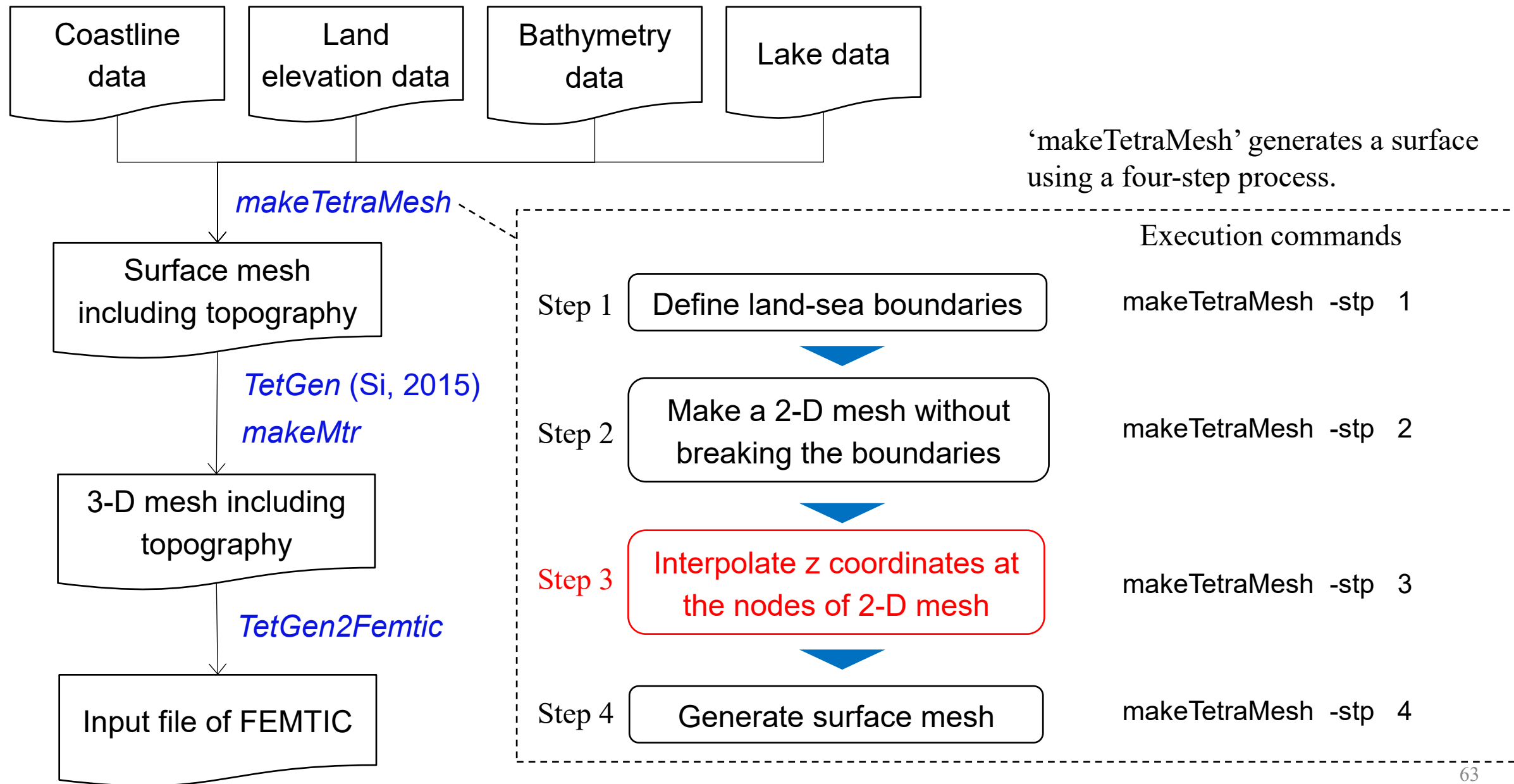
The Delaunay triangulation method with boundary constraints is used in Step 2.

Delaunay triangulation method

- In the context of a finite point set S , a triangle is '*Delaunay*' if its vertices are in S and its open circumdisk is empty.
- A *Delaunay triangulation* of S is a triangulation in which every triangle is *Delaunay*.



Procedure flow of tetrahedral mesh generation



Input files for Step 3

File name	Contents
analysis_domain.dat	Size of computational domain
control.dat	Parameters controlling mesh generation
observing_site.dat	Mesh size around observation sites
node_bound_curve.stp2	Coordinate values of the points of inner/outer boundaries (output file of Step2)
boundary_curves.stp2	Connections of the points of inner/outer boundaries (output file of Step2)
boundary_curves_relation.stp2	Relationship between inner/outer boundaries (output file of Step2)
node_mesh.stp2	Coordinate values of the points of 2-D mesh (output file of Step2)
triangle_list.stp2	Connections of the points of 2-D mesh (output file of Step2)
node_curves2mesh.stp2	Relation of the points of boundaries and the ones of 2-D mesh (output file of Step2)
<i>Defined in 'control.dat'</i>	Land topography data (land altitudes)
<i>Defined in 'control.dat'</i>	Bathymetry data (sea floor depths)

The same files as for Step 2 and output files of Steps 1 and 2 can be used, except for 'control.dat', the topography and bathymetry files.

File format of 'control.dat' for Step 3

control.dat

CENTER

X (km)

Y (km)

Z (km)

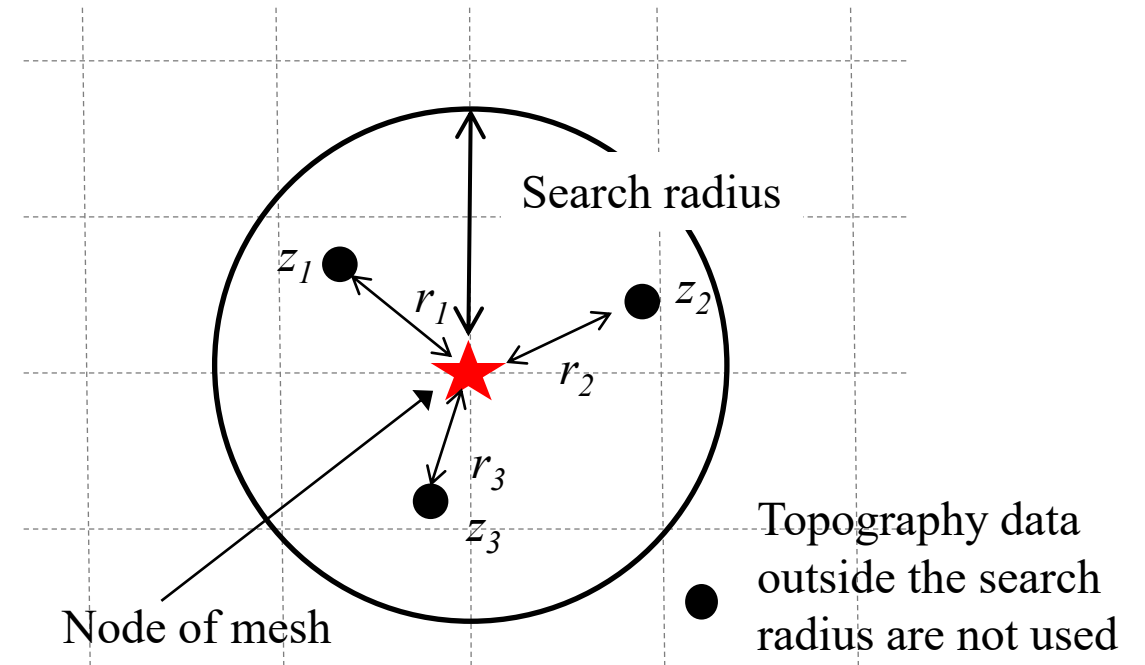
⋮

INTERPOLATE*Search radius (km)**Maximum point number used for interpolation**Small number to avoid zero divide ε (km)***ALTITUDE***Land altitude data file**Lower limit of altitude (km)**Upper limit of altitude (km)***SEA_DEPTH***Sea floor depth data file**Lower limit of depth (km)**Upper limit of depth (km)***END**

Ending flag is necessary

Three new keywords (INTERPOLATE, ALTITUDE, and SEA_DEPTH) are additionally required for Step 3.

The inverse distance weighting method is used for interpolation of z coordinates



Interpolated z-coordinate of mesh node

$$z = \frac{\sum_{i=1}^N w_i z_i}{\sum_{i=1}^N w_i} \quad w_i = \frac{1}{(r_i + \varepsilon)}$$

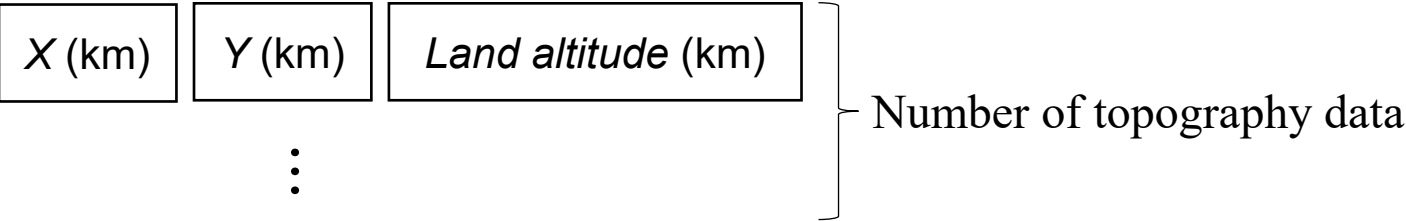
Interpolated z-coordinates are bounded within the input lower and upper values.

If $z > z_{UL} \rightarrow z = z_{UL}$

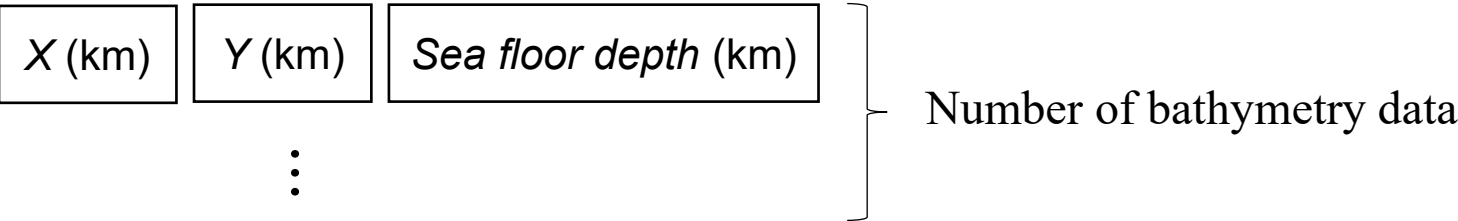
If $z < z_{LL} \rightarrow z = z_{LL}$

File format of topography and bathymetry data

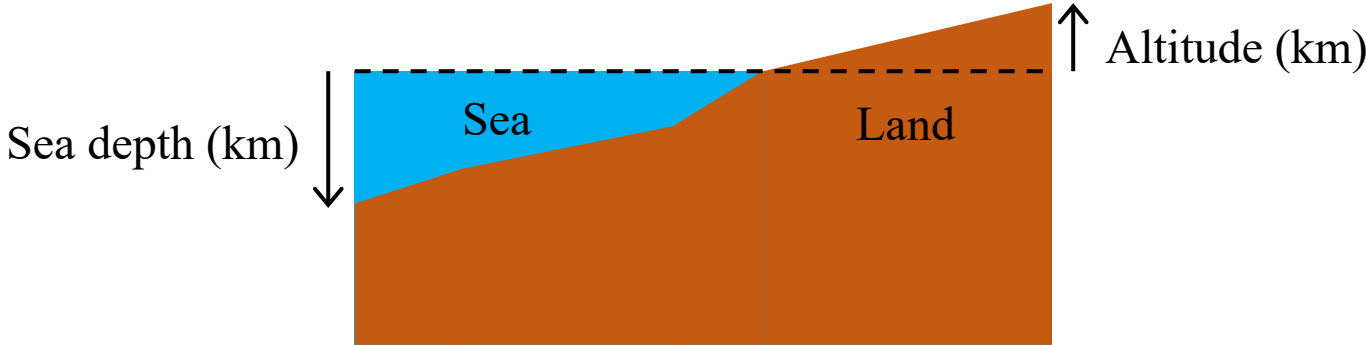
Land topography data (land altitudes)



Bathymetry data (sea floor depths)



Altitudes are positive in the upward direction, whereas sea depths are positive in the downward direction.



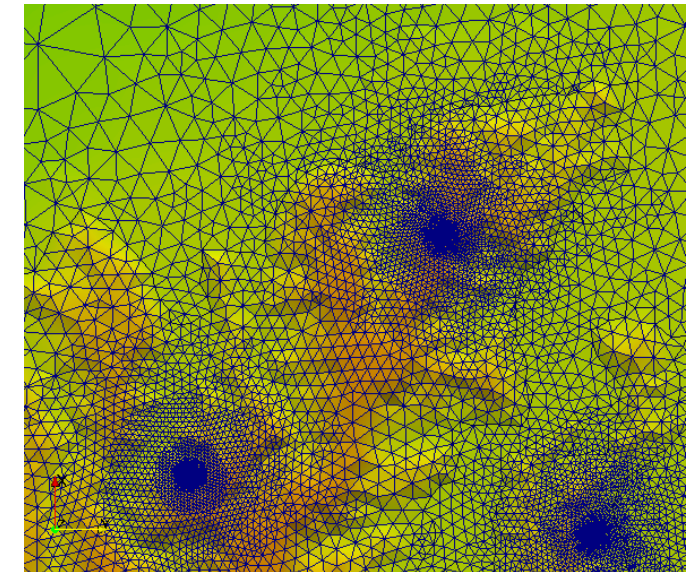
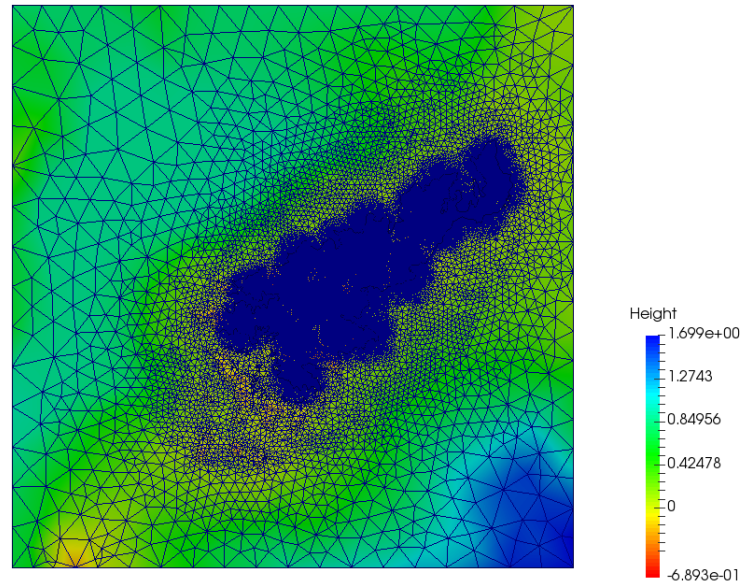
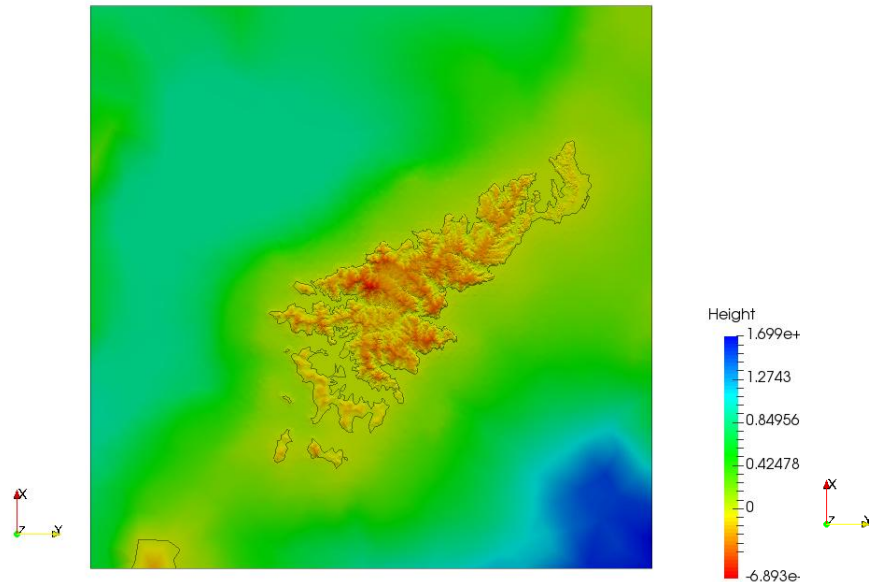
Example

-106.824031915	76.2046151595	2.100000e-003↓
-106.824276161	76.1651038245	6.000000e-003↓
-106.824459262	76.1354703228	8.500000e-003↓
-106.824642292	76.1058368206	7.400000e-003↓
-106.824886221	76.0663254836	9.000000e-003↓
-106.825069084	76.0366919804	9.300000e-003↓
-106.825251877	76.0070584767	8.600000e-003↓
-106.825495489	75.9675471378	8.600000e-003↓
-106.825678115	75.9379136331	9.400000e-003↓
-106.82586067	75.908280128	1.040000e-002↓
-106.826103966	75.8687687872	1.070000e-002↓
-106.826286354	75.839135281	9.300000e-003↓
-106.826468671	75.8095017744	1.110000e-002↓
-106.82671165	75.7699904316	1.390000e-002↓
...	...	...

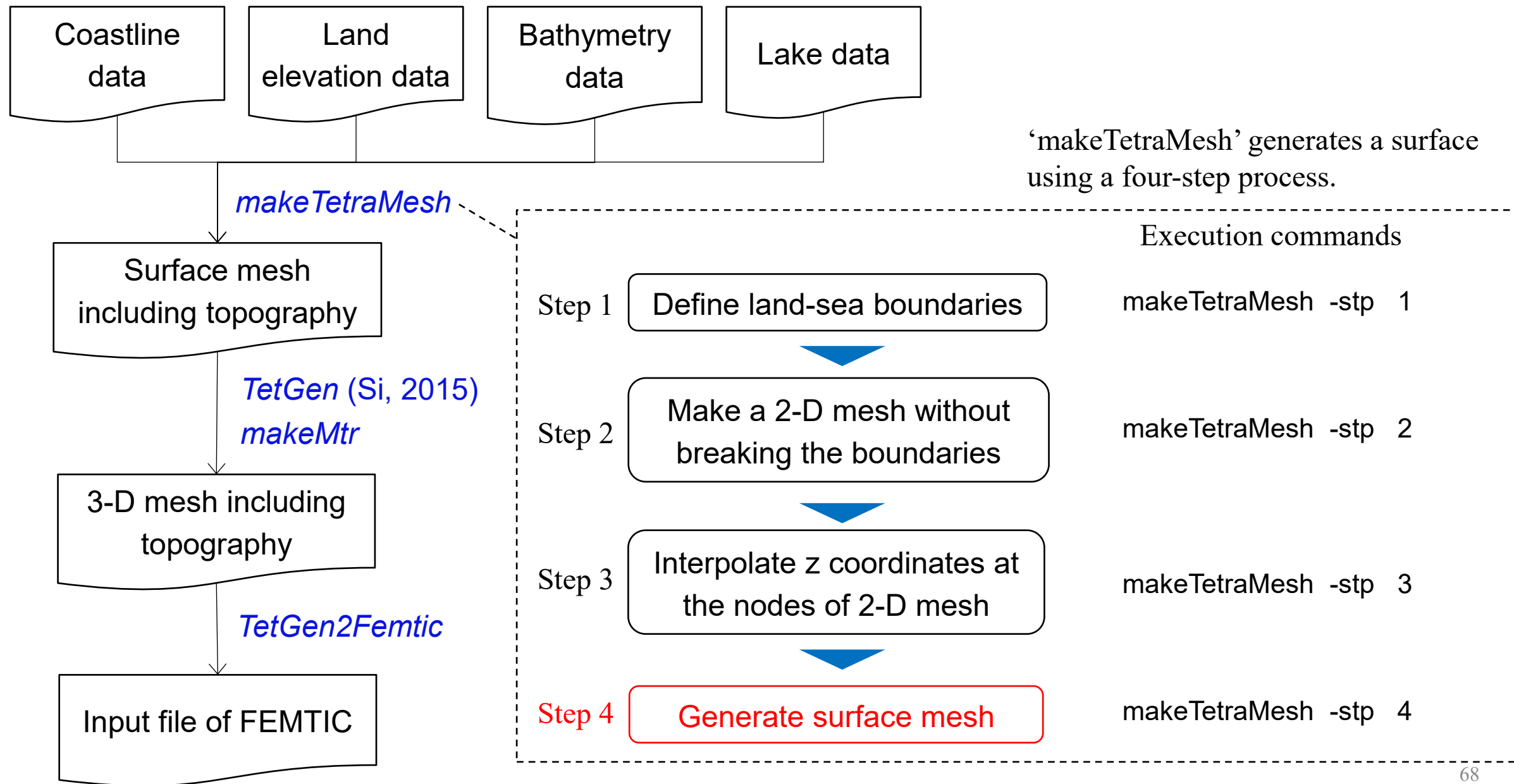
Visualization of Step 3 result with ParaView

The result of Step 3 can be visualized by importing the output vtk file 'triangles_with_height.vtk' into ParaView.

triangles_with_height.vtk



Procedure flow of tetrahedral mesh generation



Input files for Step 3

File name	Contents
analysis_domain.dat	Size of computational domain
control.dat	Parameters controlling mesh generation
observing_site.dat	Mesh size around observation sites
node_bound_curve.stp3	Coordinates of the points of inner/outer boundaries (Output file of Step3)
boundary_curves.stp3	Connections of the points of inner/outer boundaries (Output file of Step3)
boundary_curves_relation.stp3	Relationship between inner/outer boundaries (Output file of Step3)
node_mesh.stp3	Coordinates of the points of 2D mesh (Output file of Step3)
triangle_list.stp3	Connections of the points of 2D mesh (Output file of Step3)
node_curves2mesh.stp3	Relation of the points of boundaries and the ones of 2D mesh (Output file of Step3)
lake.dat	Information about lakes. This file is required only if you incorporate lakes in the model.

The same files as for Step 2 and output files of the previous steps can be used, except for ‘control.dat’.

File format of 'control.dat' for Step 3

control.dat

CENTER

X (km)

Y (km)

Z (km)

⋮

SURF_MESH

⋮

END

Ending flag is necessary

A new keyword 'SURF_MESH' is additionally required for Step 4.

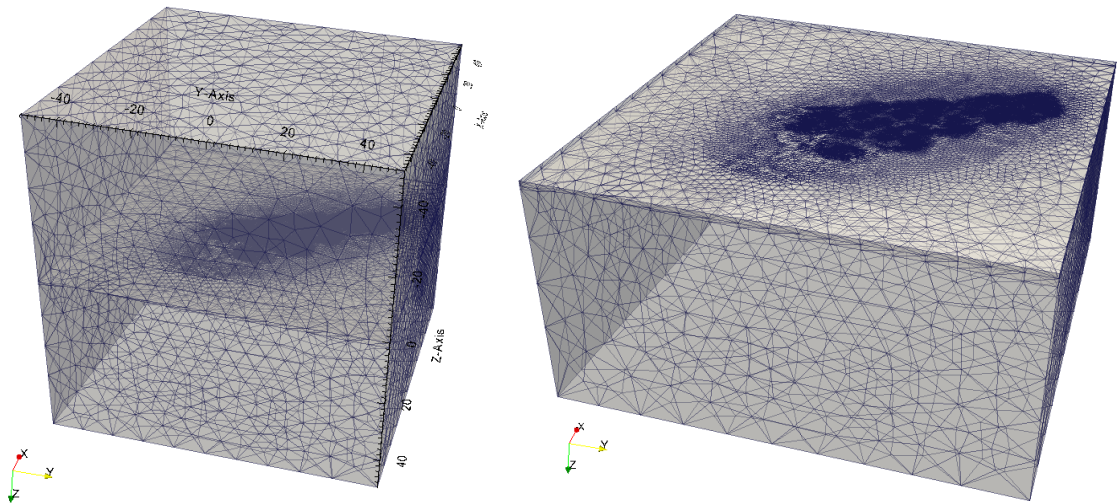
(You can write this keyword in Steps 1, 2, and 3. The existence or non-existence of this keyword does not influence the results of Steps 1, 2, and 3)

File name	Contents
makeTetraMesh.log	Log information
output.poly	Surface mesh covering the computational domain (input file of TetGen)
plc.vtk	Surface mesh covering the computational domain (input file of ParaView)

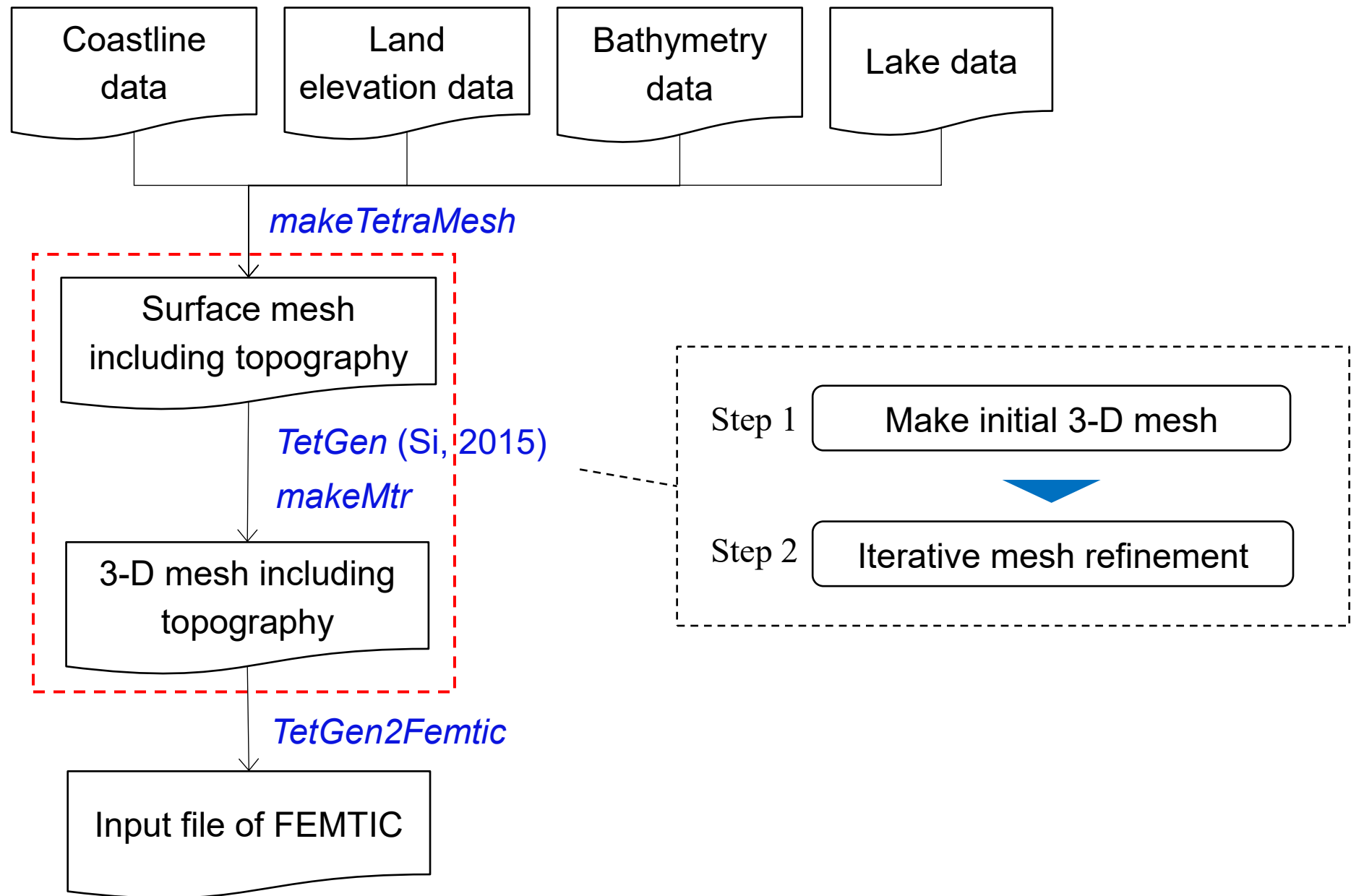
In addition to them, some intermediate file is outputted.

➔ Input file of TetGen (Si, 2015)

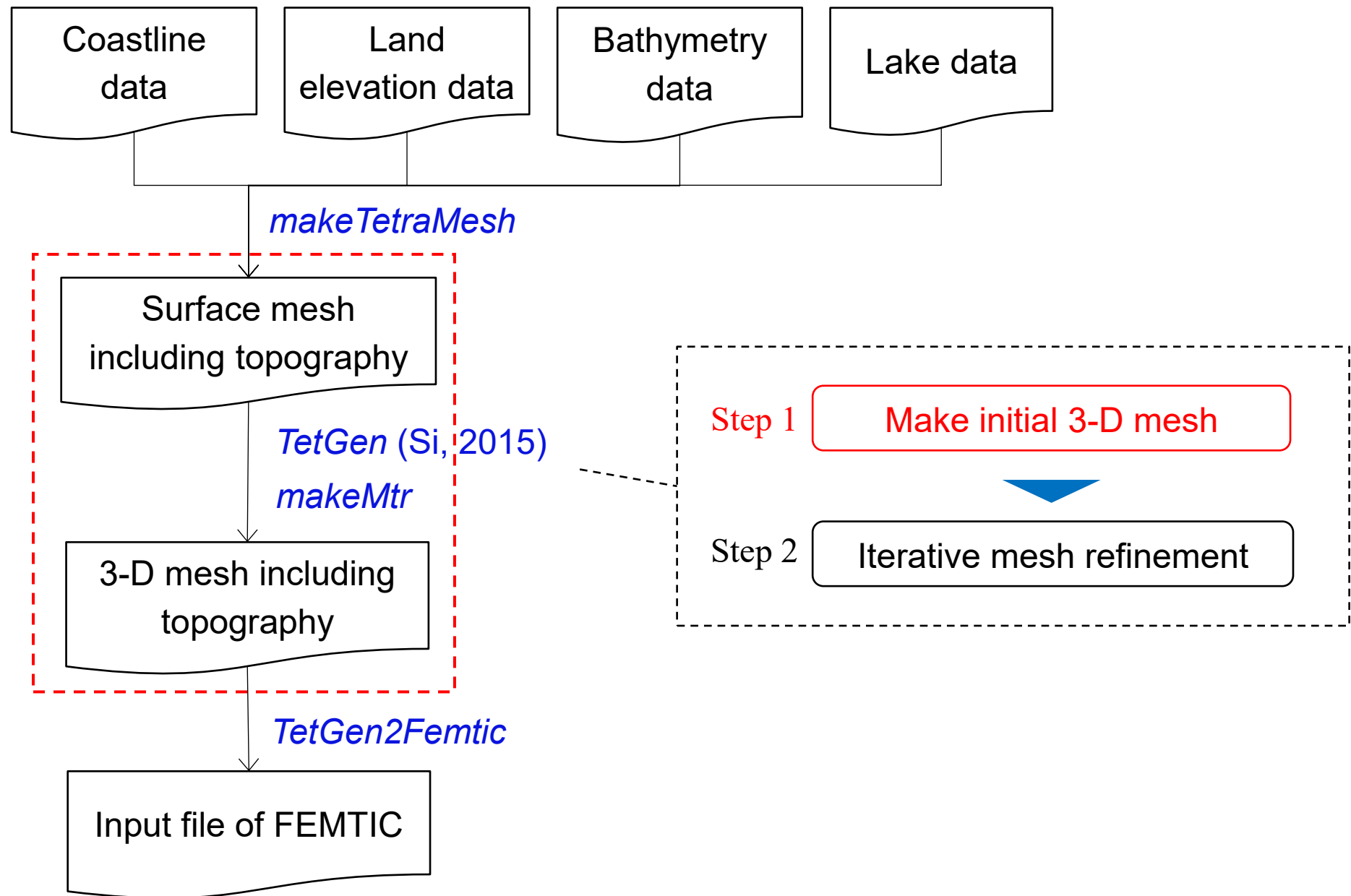
Surface mesh can be visualized by importing it into ParaView.



Procedure flow of tetrahedral mesh generation



Procedure flow of tetrahedral mesh generation



Add region attributes to 'output.poly'

- Edit 'output.poly' and add region attributes in its 'Part 4' section.
- The detailed format of poly file is written in the manual of TetGen.
- Unique region attributes (unique integers) should be given to the air, sea, and subsurface, respectively.

The last parts in 'output.poly'

The 'Part 4' section should be manually added.

```
# Part 3 ↓
0 ↓
# Part 4 ↓
3 ↓
1  0.0 0.0 -40.0 10 1e9 ↓
2  40.0 0.0  0.01 20 1e9 ↓
3  0.0 0.0  40.0 30 1e9 ↓
```

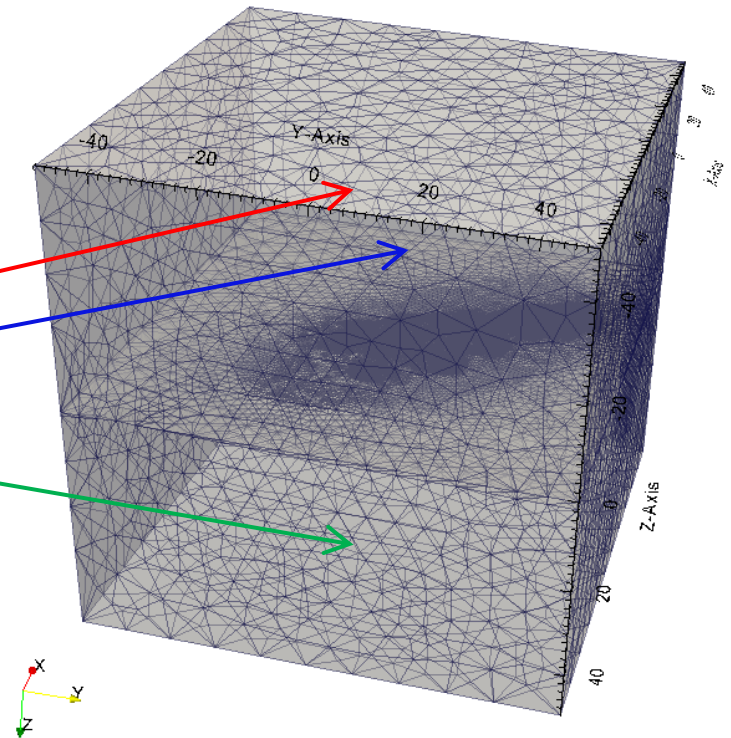
Position (X, Y, and Z in km) of a point locating within each region

Region attribute

Subsurface

Air

Sea



How to make initial 3-D mesh from a poly file

Execution command:

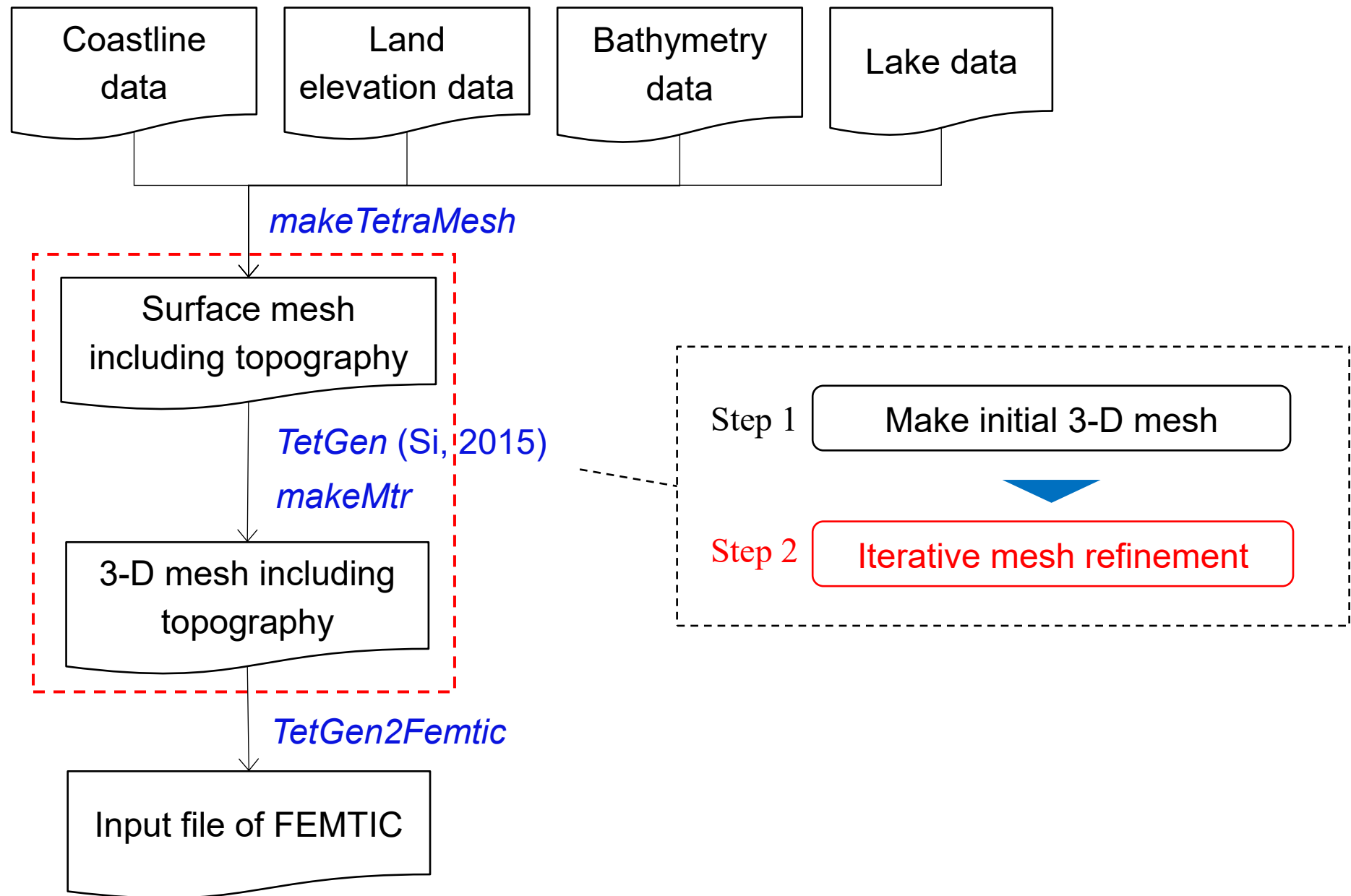
```
tetgen -nVpYAakq3.0/0 output.poly
```

Output files

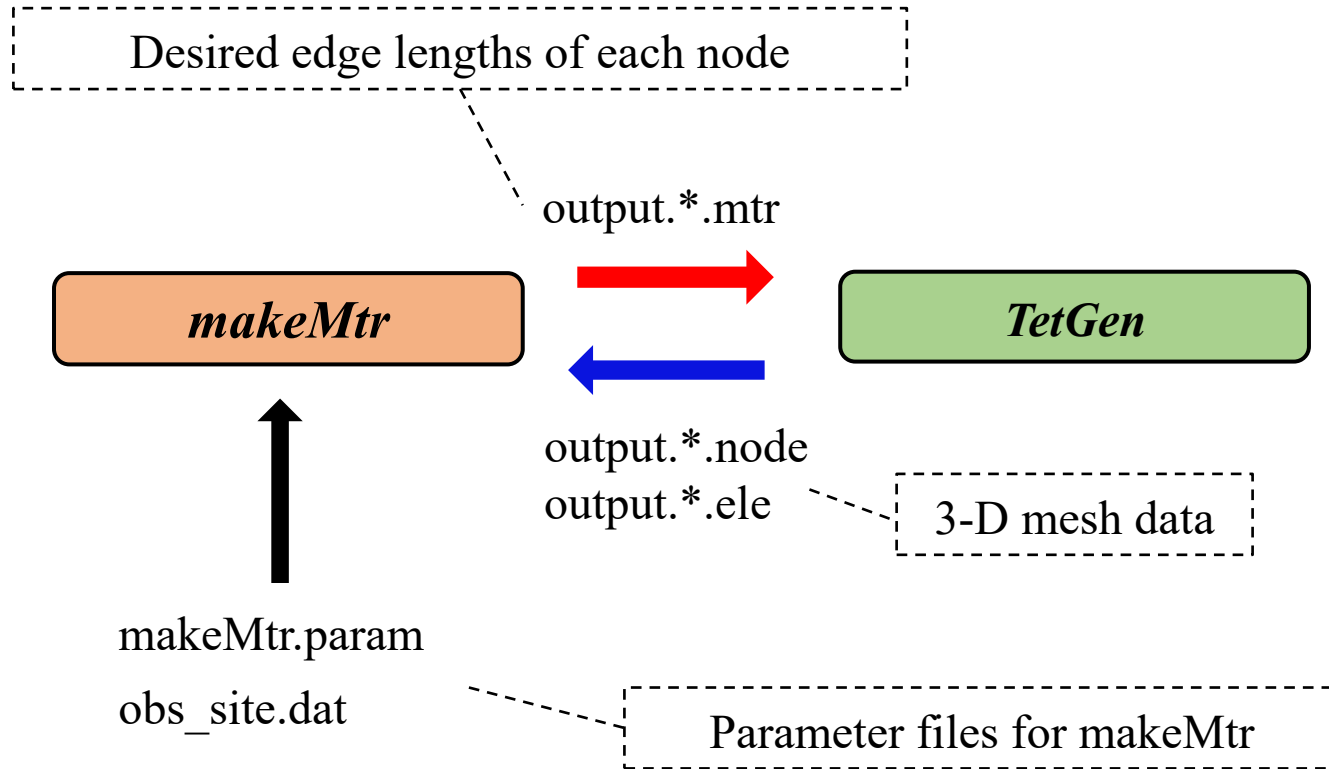
File name	Contents
output.1.node	Information of nodes (input file of iterative refinement)
output.1.ele	Information of elements (input file of iterative refinement)
output.1.face	Information of element faces
output.1.edge	Information of element edges
output.1.neigh	Information of adjacency relationship between elements (input file of iterative refinement)
output.1.vtk	3-D mesh data (input file of ParaView)

More detailed information about the command and the file formats are written in the manual of TetGen.

Procedure flow of tetrahedral mesh generation



The 3-D mesh should be refined iteratively to make element sizes sufficiently small.



Example of execution commands

```
for i in 1 2 3 4 5
do
  makeMtr output.$i
  tetgen -nmpYVrAakq3.0/0 output.$i
done
```

'*' indicates the interaction number

Input files

File name	Contents
output.*.ele	Element information of 3-D mesh (output file of TetGen).
output.*.node	Node information of 3-D mesh (output file of TetGen).
makeMtr.param	Desired edge lengths for the entire region of the model
obs_site.dat	Desired edge lengths around observation sites.

Output files

File name	Contents
output.*.mtr	Desired edge lengths of each node of 3-D mesh (input file of TetGen)

‘*’ indicates the interaction number

File format of 'makeMtr.param'

makeMtr.param

X (km)	Y (km)	Z (km)	Center coordinates of ellipsoids
--------	--------	--------	----------------------------------

Rotation angle (deg.) of the ellipsoids in the X-Y plane
--

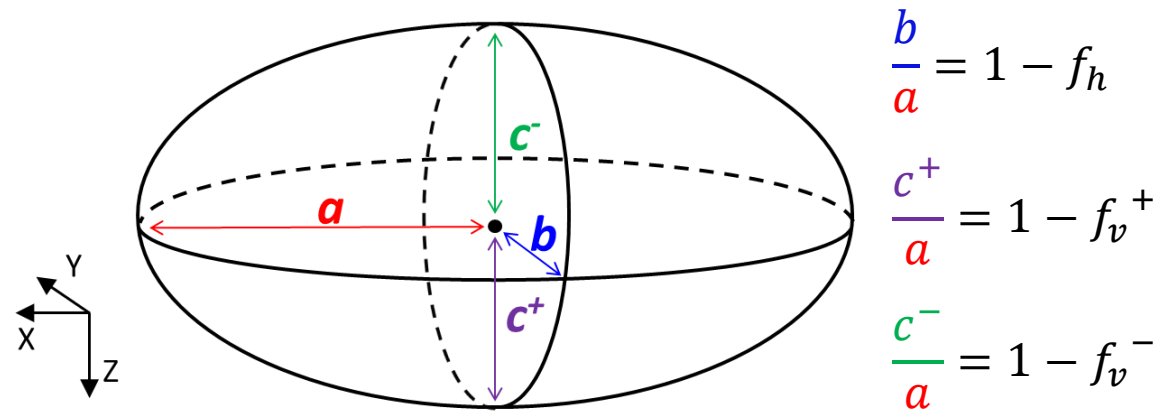
Number of the ellipsoids

a	l_h	f_h	f_v^+	f_v^-	The 1 st ellipsoid
-----	-------	-------	---------	---------	-------------------------------

⋮

a	l_h	f_h	f_v^+	f_v^-	The last ellipsoid
-----	-------	-------	---------	---------	--------------------

Subsequent ellipsoids must cover the formers
(must be larger than the formers)



$$\frac{b}{a} = 1 - f_h$$

$$\frac{c^+}{a} = 1 - f_v^+$$

$$\frac{c^-}{a} = 1 - f_v^-$$

a : Length along the X axis (km)

l_h : Upper limit of the edge lengths within the ellipsoid (km)

f_h : Oblateness on the X-Y plane

f_v^+ : Oblateness on the Z-X plane (Upper side)

f_v^- : Oblateness on the Z-X plane (Lower side)

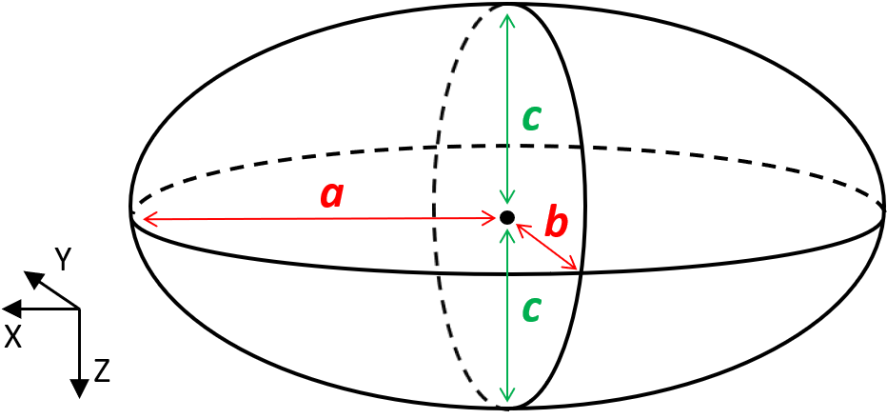
The required parameters are almost the same as in 'contron.dat' for 'makeTetraMesh'.

→ You can use the same parameters as for 'contron.dat'.

File format of ‘obs_site.dat’

makeMtr.param

<i>Number of observation points</i>			Data for the 1 st point
<i>X (km)</i>	<i>Y (km)</i>	<i>Z (km)</i>	Center coordinates of ellipsoids
<i>Number of the ellipsoids</i>			
<i>a</i>	<i>l_h</i>	<i>f_v</i>	The 1 st ellipsoid
⋮			Subsequent ellipsoids must cover the formers
<i>a</i>	<i>l_h</i>	<i>f_v</i>	
The last ellipsoid			
⋮			Data for the last point



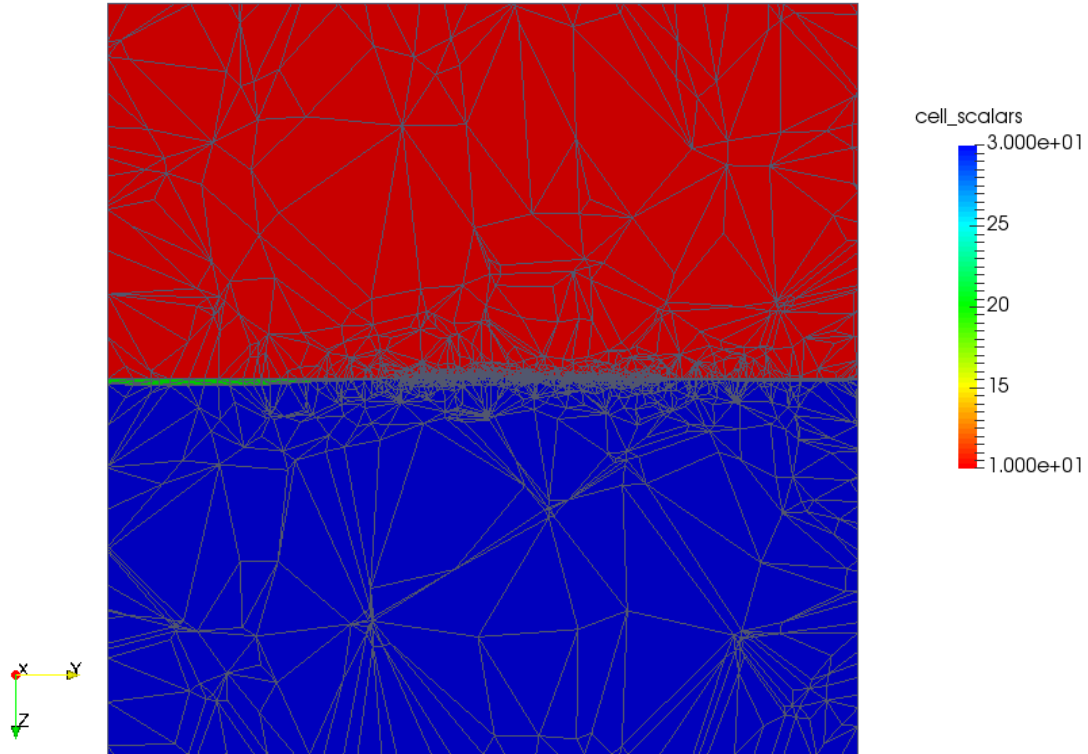
$$\frac{c}{a} = 1 - f_v$$

- a : Length along the X axis (km)
- l_h: Upper limit of the edge lengths within the ellipsoid (km)
- f_v: Oblateness on the Z-X plane

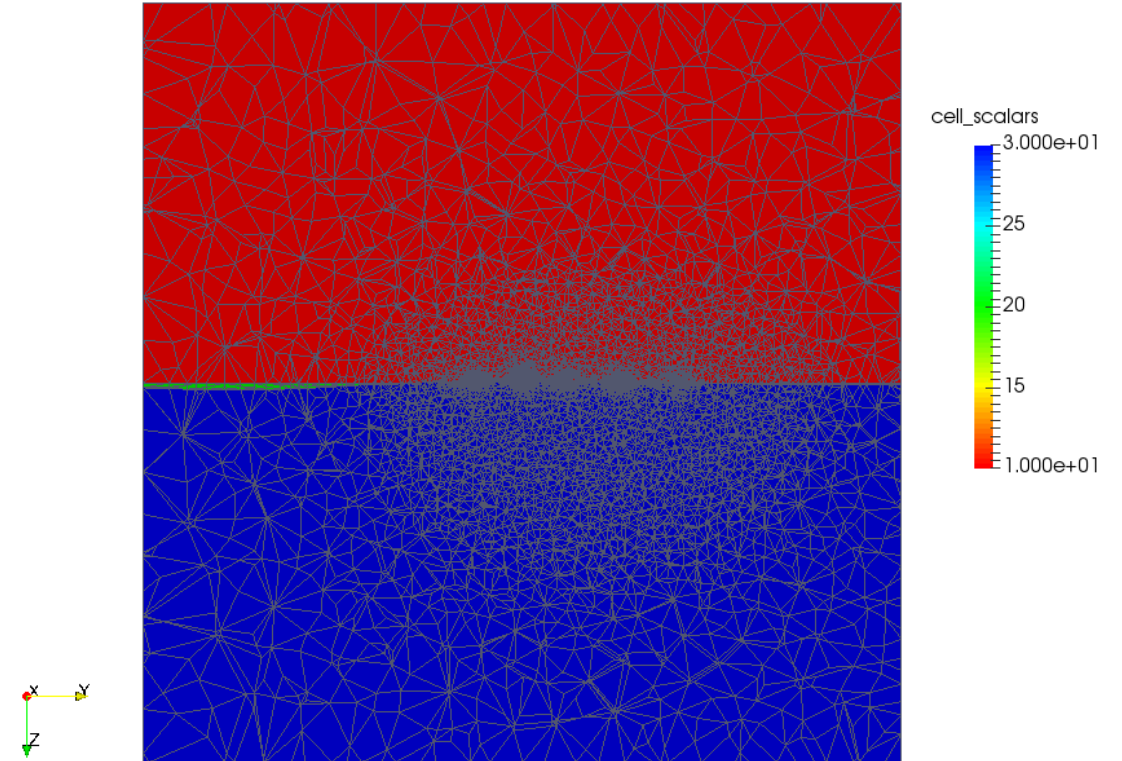
Visualization of the 3-D meshes with ParaView

The resulting 3-D mesh can be visualized by importing 'output.*.vtk' into ParaView.

output.1.vtk

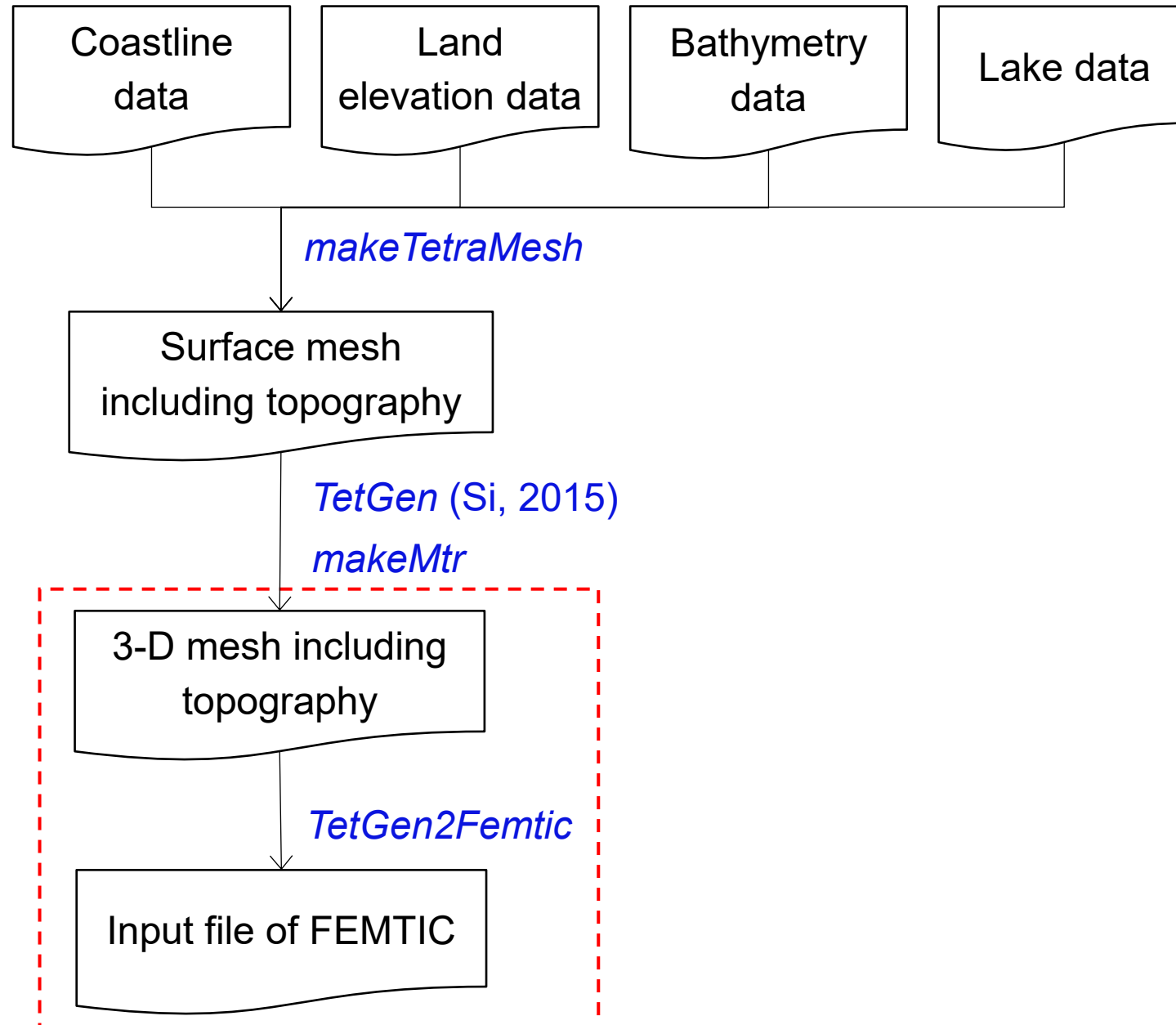


output.6.vtk



The color is based on the region attribute.

Procedure flow of tetrahedral mesh generation



Execution command:

```
TetGen2Femtic output.* -div_all
```

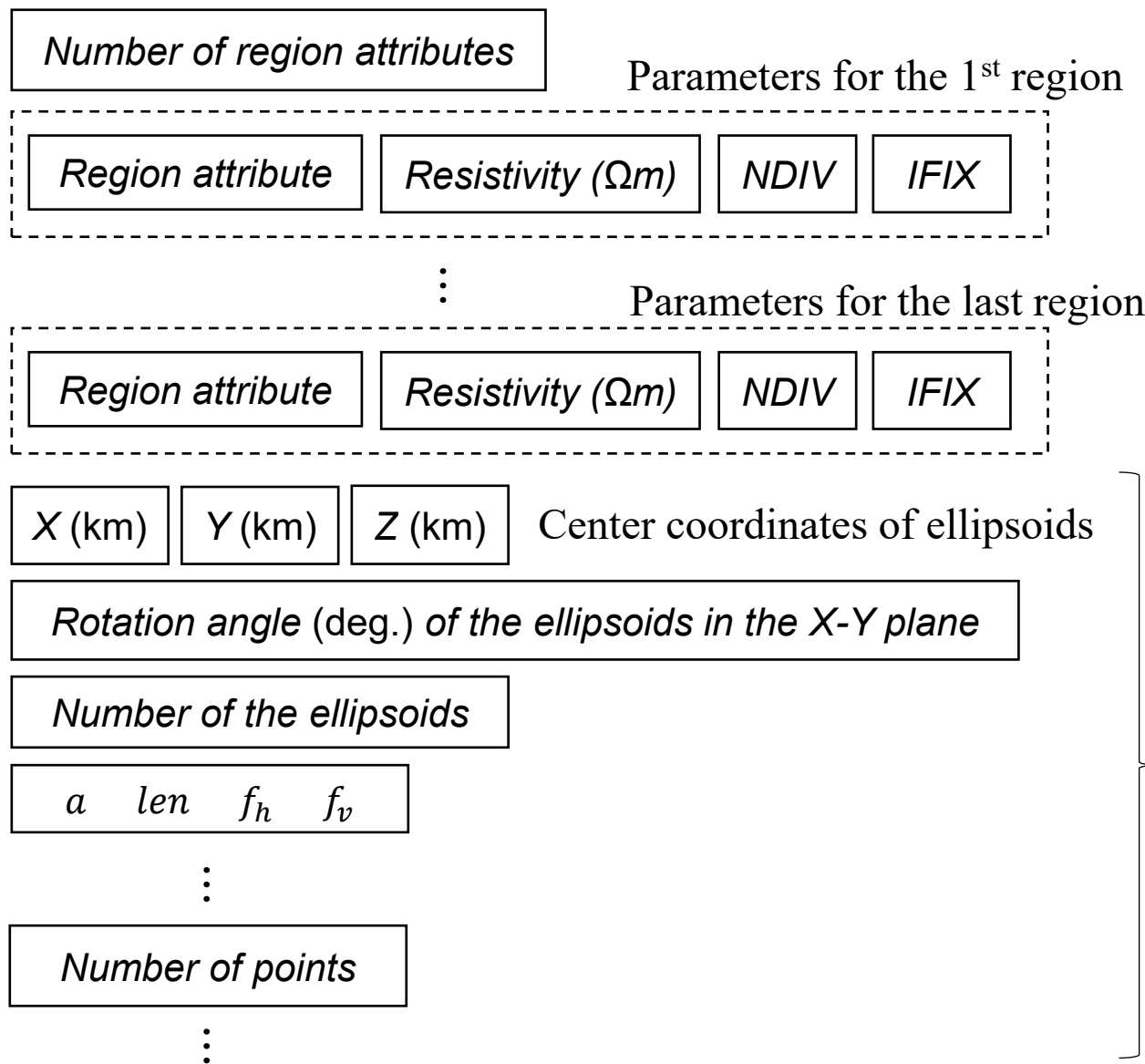
- ‘*’ indicates the final iteration number of the iterative refinement.
- I recommend using ‘-div_all’ option to make every individual subsurface element become a different parameter cell.

Input files:

File name	Contents
output.*.node	Node information (output file of TetGen)
output.*.ele	Element information (output file of TetGen)
output.*.face	Element faces (output file of TetGen)
output.*.neigh	Adjacency relationship between elements (output file of TetGen)
resistivity_attr.dat	Controlling parameters

File format of 'resistivity_attr.dat'

resistivity_attr.dat



NDIV: Repeat number of the partitioning of the region.
NDIV should be -1 for the air and sea regions.

IFIX:

1 → the resistivity value is fixed during the inversion.

0 → the resistivity value is modifiable during the inversion.

These parameters are used for partitioning the parameter cells.

When the option '-div_all' is used, these parameters are meaningless.

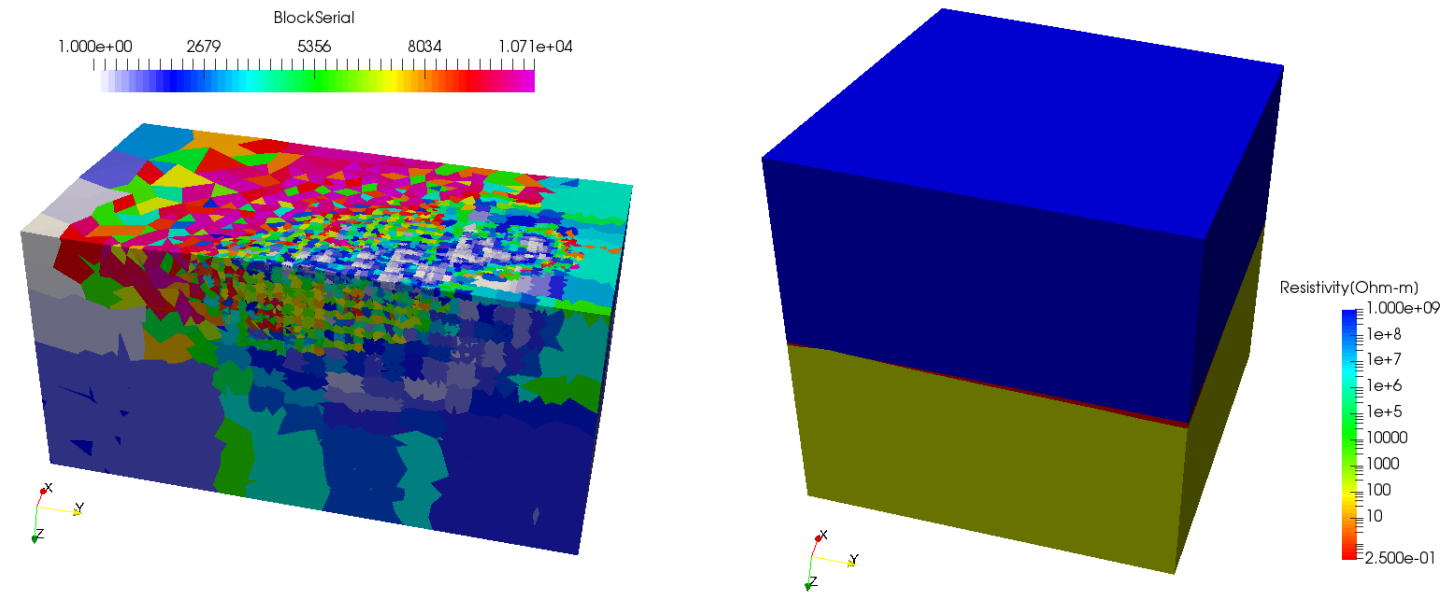
In this case, you can use the following to define neither ellipsoids nor points.

0.0 0.0 0.0	Dummy coordinates
0	No ellipsoid
0	No point

Output files

File name	Contents
mesh.dat	Computational mesh (input file of FEMTIC)
resistivity_block_iter0.dat	Initial resistivity values (input file of FEMTIC)
output.*.femtic.vtk	Data of computational mesh (input file of ParaView)

‘*’ indicates the final iteration number of the iterative refinement.



I developed some tools for post-processing FEMTIC results.

mergeResultOfFEMTIC

<https://github.com/yoshiya-usui/mergeResultOfFEMTIC.git>

This tool merges output CSV files containing calculated and observed responses into a single file.

changeResistivityForFemtic

<https://github.com/yoshiya-usui/changeResistivityForFemtic.git>

This tool can change resistivity values of a selected area for sensitivity tests.

makeCutawayForGMT

<https://github.com/yoshiya-usui/makeCutawayForGMT.git>

This tool extracts the resistivity values on horizontal or vertical planes. From the output file, a slice of the resistivity structure can be drawn using ‘psxy’ command of Generic Mapping Tools (Wessel et al. 2013).

Some researchers developed and opened helpful tools for FEMTIC inversion.

femticPy

<https://github.com/hseille/femticPy>

Author: Hoël Seille (CSIRO, Australia)

Python module to create inputs / read outputs of FEMTIC, a 3-D magnetotelluric inversion code.

MeshTran

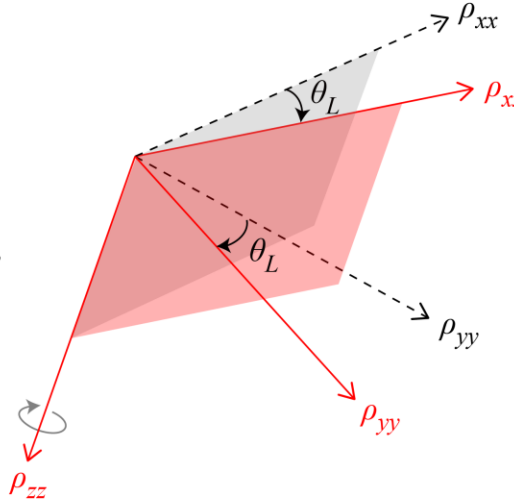
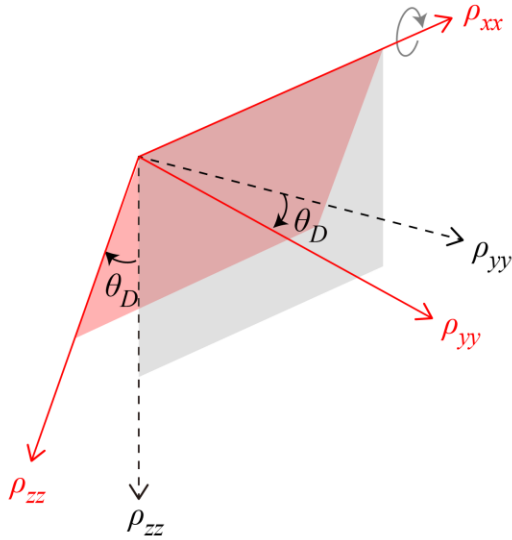
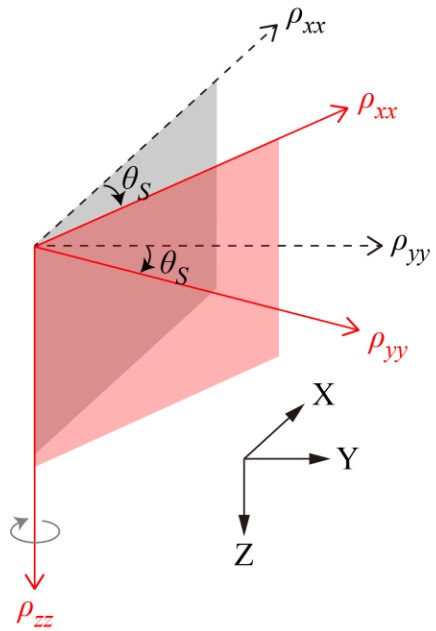
<https://github.com/molivag/MeshTran.git>

Author: Marco Oliva Gutierrez (Universitat de Barcelona, Spain)

Preprocessing toolkit to prepare input data for FEMTIC.

Anisotropic inversion

The latest version (v5.0) supports the anisotropic resistivity structure inversion.

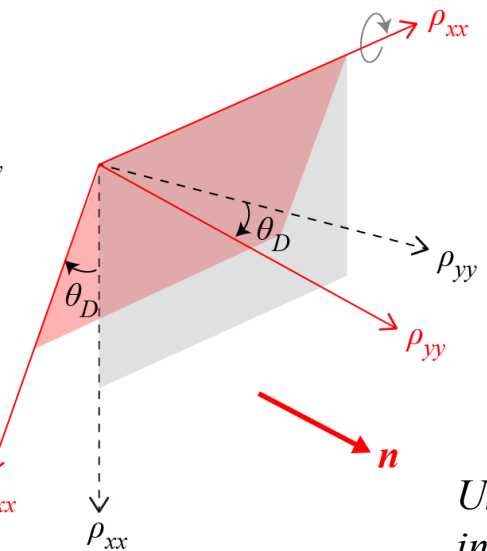
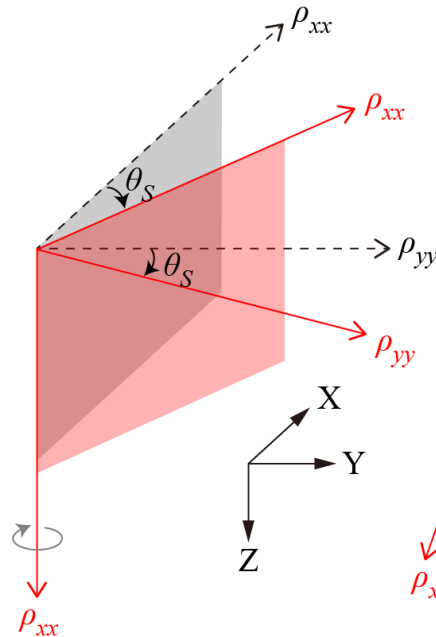


General anisotropy

Three principal resistivity: ρ_{xx} , ρ_{yy} , ρ_{zz}

Three rotation angles: θ_S , θ_D , θ_L

→ Usable only for the forward computation



Transverse isotropy (uniaxial anisotropy)

Two principal resistivity: ρ_{xx} , ρ_{yy}

Two rotation angles: θ_S , θ_D

$$\sigma = \mathbf{R}_Z(\theta_S) \mathbf{R}_X(\theta_D) \begin{pmatrix} 1/\rho_{xx} & 0 & 0 \\ 0 & 1/\rho_{yy} & 0 \\ 0 & 0 & 1/\rho_{xx} \end{pmatrix} \mathbf{R}_X^T(\theta_D) \mathbf{R}_Z^T(\theta_S)$$

Please read our preprint for the detailed methodology of the anisotropic inversion.

Usui, Y., Uyeshima, M., Sakanaka, S. Y., & Yamaya, Y. Three-dimensional magnetotelluric inversion code for transverse isotropic electrical resistivity structure. SSRN 7218262.

- (1) Previous studies (Yan et al. 2020; Usui 2024, 2026) have proposed mixing models that enable the quantitative interpretation of transversely isotropic conductivity.

Electrical conductivity of rocks with interconnected cracks/tubes

Electrical conductivity of rocks with interconnected cracks/tubes is calculated based on the mixing model proposed by Yoshiya Usui, Electrical conductivity model for transversely isotropic rocks with interconnected cracks, *Physics of the Earth and Planetary Interiors*, 354, 2024, <https://doi.org/10.1016/j.pepi.2024.107226>
The Python scripts performing the same calculation are available from GitHub <https://github.com/yoshiya-usui/TIECMIX.git>

Forward Calculation Inverse Calculation

Calculates transversely isotropic electrical conductivity given the Watson distribution parameter κ , volume fraction ϕ , and conductivity of conductive phase σ_1 . Select the pore geometry model (Crack or Tube).

Model: ☒ Crack ☐ Tube

$$\sigma_x = \sigma_y = \sigma_1 \phi \int f(\theta) (1 - \sin^2\theta \cos^2\phi)^{3/2} d\Omega$$
$$\sigma_z = \sigma_1 \phi \int f(\theta) \sin^2\theta d\Omega$$
$$f(\theta) = \exp(\kappa \cos^2\theta) / (4\pi {}_1F_1(1/2, 3/2; \kappa))$$

κ (kappa) ϕ — volume fraction (0–1) σ_1 — conductivity of conductive phase (S/m)

Calculate

<https://calctiec-web.vercel.app/>

- (2) Under the assumption of transverse isotropy, a unique symmetric axis (vector $\mathbf{n}$ in the previous slide) can be obtained for each conductivity tensor. This allows the use of a simple regularization for the spatial variation of the principal directions, as mentioned in a later slide.
- (3) Full anisotropy may be redundant in most cases, especially because MT data have low sensitivity to resistivity in the vertical direction

Input files

File name	Content
control.dat	Parameters controlling FEMTIC
mesh.dat	Computational mesh
resistivity_block_iter0.dat	Parameter cells and initial resistivity values
observe.dat	Observation data
distortion_iter0.dat	Galvanic distortion parameters

Only the formats of ‘control.dat’ and ‘resistivity_block_iter0.dat’ are different from those for the isotropic inversion

Output files

The principal resistivity and rotation angles are output, instead of a single resistivity.

OUTPUT_PARAM			
1		RhoXX.iter*	Strike.iter*
0		RhoYY.iter*	Dip.iter*
		RhoZZ.iter*	Slant.iter*
			AnisotropyTypes (0: isotropy, 1: transverse isotropy, 2: general anisotropy)
			Anisotropy.iter* (log(ρ_{xx}) – log(ρ_{yy}))
			‘*’ indicates the iteration number of the inversion.

What should be different from the isotropic inversion (1) 'control.dat'

Example

```

MESH_TYPE
2
OUTPUT_PARAM
1
0
DIV_NUM_RHS_FWD
10
INV_METHOD
1
SMALL_VALUE
0.0001
ANISOTROPY
DISTORTION
0
TRADE_OFF_PARAM
10.0 1.0 100.0
ITERATION
0 20
STEP_LENGTH
0.5 0.1 0.8
2
0.6 1.2
END

```

→ Keyword for activating the anisotropic inversion

→ Trade-off parameters ($\alpha_1, \alpha_2, \alpha_3$)

(When galvanic distortion parameters are also estimated, trade-off parameters for them should be described as well)

Objective function: $\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha_1^2 \phi_r(\mathbf{m}) + \alpha_2^2 \phi_a(\mathbf{m}) + \alpha_3^2 \phi_c(\mathbf{m})$

$\phi_d(\mathbf{m})$: Data misfit, $\phi_r(\mathbf{m})$: Roughness, $\phi_a(\mathbf{m})$: Anisotropy, $\phi_c(\mathbf{m})$: Cross product

$$\log(\rho_{xx}^A / \rho_{xx}^B)^2 \rightarrow \phi_r$$

$$\log(\rho_{xx}^A / \rho_{yy}^A)^2 \rightarrow \phi_a$$

$$\log(\rho_{xx}^B / \rho_{yy}^B)^2 \rightarrow \phi_c$$

$$\|\mathbf{n}^A \times \mathbf{n}^B\|^2 = \left\| \begin{pmatrix} -\cos(\theta_D^A) \sin(\theta_S^A) \\ \cos(\theta_D^A) \cos(\theta_S^A) \\ \sin(\theta_D^A) \end{pmatrix} \times \begin{pmatrix} -\cos(\theta_D^B) \sin(\theta_S^B) \\ \cos(\theta_D^B) \cos(\theta_S^B) \\ \sin(\theta_D^B) \end{pmatrix} \right\|^2 \rightarrow \phi_c$$

7. Geophysical Inversion

Season 4

What should be different from the isotropic inversion (2) ‘resistivity_block_iter0.dat’

- The format of the upper half of ‘resistivity_block_iter0.dat’, the relationship between elements and parameter cells, is the same as that for the isotropic inversion.
- The format of the lower half, the resistivity parameters of each parameter cell, is different from that of the isotropic inversion.
- The format of the resistivity parameter depends on whether it is isotropic or anisotropic.
- Isotropic and anisotropic parameter cells can be mixed in the same file.

Structure of
‘resistivity_block_iter0.dat’

*Relationship between
elements and parameter cells*

*Resistivity value of each
parameter cell*
→ Only this part is changed
from the isotropic inversion

Format for a parameter cell with **isotropic** resistivity:

<i>Parameter cell index</i>	0	<i>Resistivity (Ωm)</i>	<i>Lower limit of resistivity (Ωm)</i>	<i>Upper limit of resistivity (Ωm)</i>	<i>Fixing resistivity or not</i>
-----------------------------	----------	--	---	---	----------------------------------

Example

0: Resistivity is NOT fixed
1: Resistivity is fixed

0	0	1.000000e+10	1.000000e-20	1.000000e+20	1	← Air layer				
1	0	1.000000e+02	1.000000e-20	1.000000e+20	0					
2	0	1.000000e+02	1.000000e-20	1.000000e+20	0					
3	1	1.000000e+02	1.000000e+02	0.000000e+00	0.000000e+00	1.000000e-20	1.000000e+20	0	0	0
4	1	1.000000e+02	1.000000e+02	0.000000e+00	0.000000e+00	1.000000e-20	1.000000e+20	0	0	0
5	1	1.000000e+02	1.000000e+02	0.000000e+00	0.000000e+00	1.000000e-20	1.000000e+20	0	0	0

What should be different from the isotropic inversion (2) ‘resistivity_block_iter0.dat’

Format for a parameter cell with **transverse isotropic** resistivity:

Parameter cell index	1	ρ_{xx} (Ωm)	ρ_{yy} (Ωm)	θ_S (deg.)	θ_D (deg.)	Lower limit of resistivity (Ωm)	Upper limit of resistivity (Ωm)
Fixing ρ_{xx} or not		Fixing ρ_{yy} or not		Fixing θ_S or not		Fixing θ_D or not	

Example

0: Principal resistivity or rotation angle is NOT fixed
1: Principal resistivity or rotation angle is fixed

0	0	1.000000e+10	1.000000e-20	1.000000e+20	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												</
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Format for a parameter cell with **generally anisotropic** resistivity:

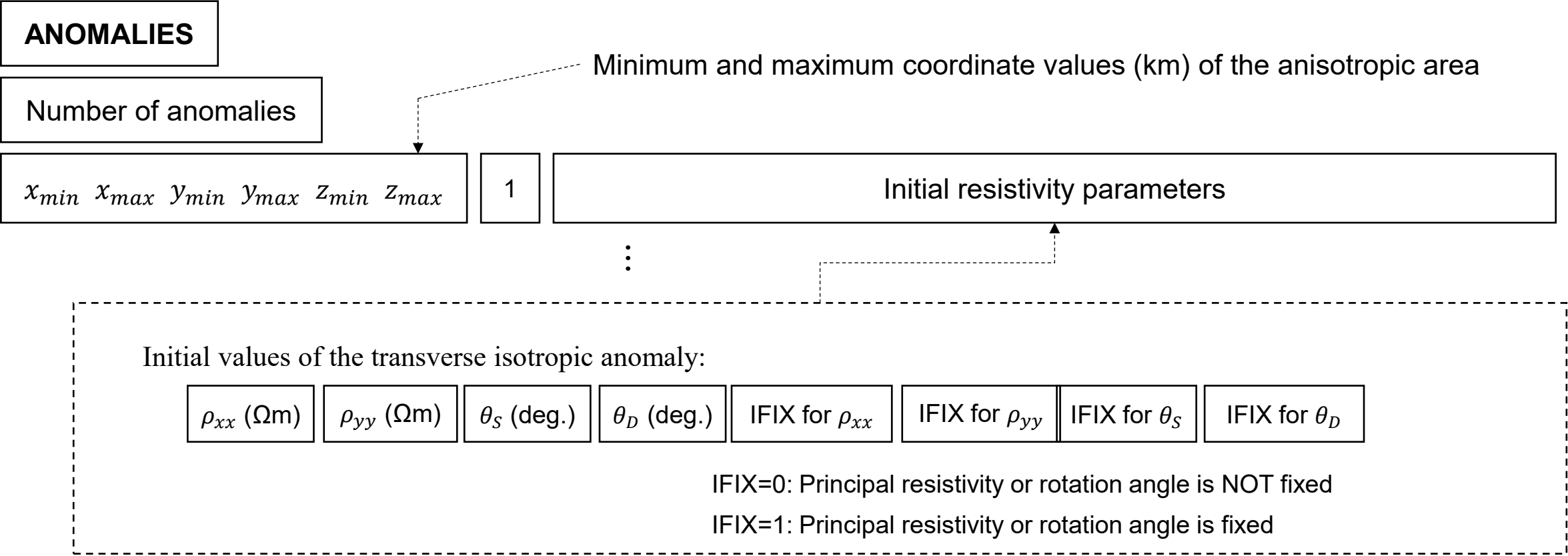
General anisotropic resistivity can be used only for the forward computation

Parameter cell index	2	ρ_{xx} (Ωm)	ρ_{yy} (Ωm)	ρ_{zz} (Ωm)	θ_S (deg.)	θ_D (deg.)	θ_L (deg.)
Lower limit of resistivity (Ωm)		Upper limit of resistivity (Ωm)		Fixing ρ_{xx} or not		Fixing ρ_{yy} or not	Fixing ρ_{zz} or not
Fixing θ_S or not		Fixing θ_D or not		Fixing θ_L or not			

0: Principal resistivity or rotation angle is NOT fixed
1: Principal resistivity or rotation angle is fixed

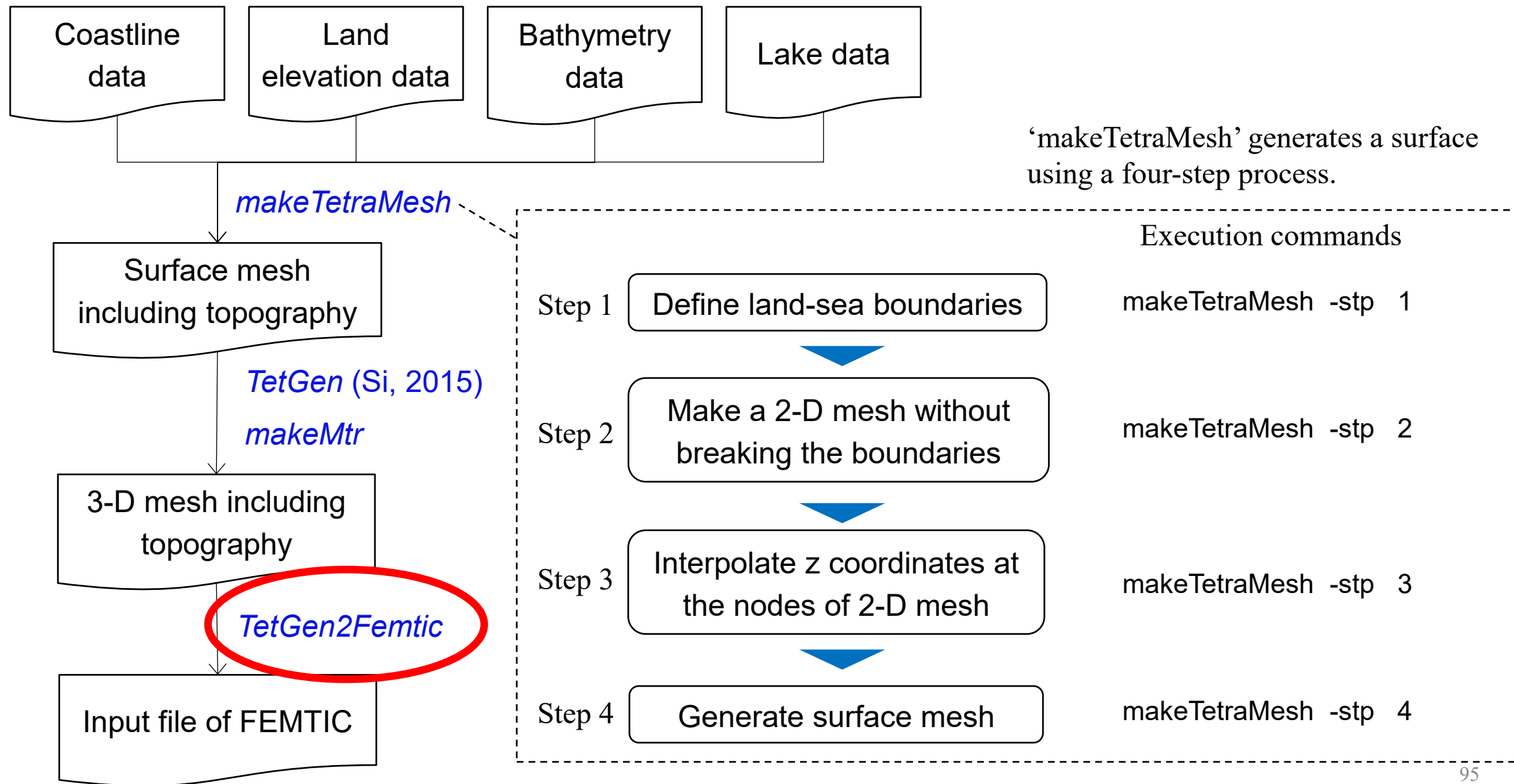
How to make 'resistivity_block_iter0.dat' for anisotropic inversion (1) Hexahedral mesh

- (1) Write the keyword 'ANISOTROPY' in 'meshgen.inp'.
- (2) Under the keyword 'ANOMALIES' in 'meshgen.inp', write anisotropic resistivity areas and initial resistivity for each area. Otherwise, the whole model has isotropic resistivity.



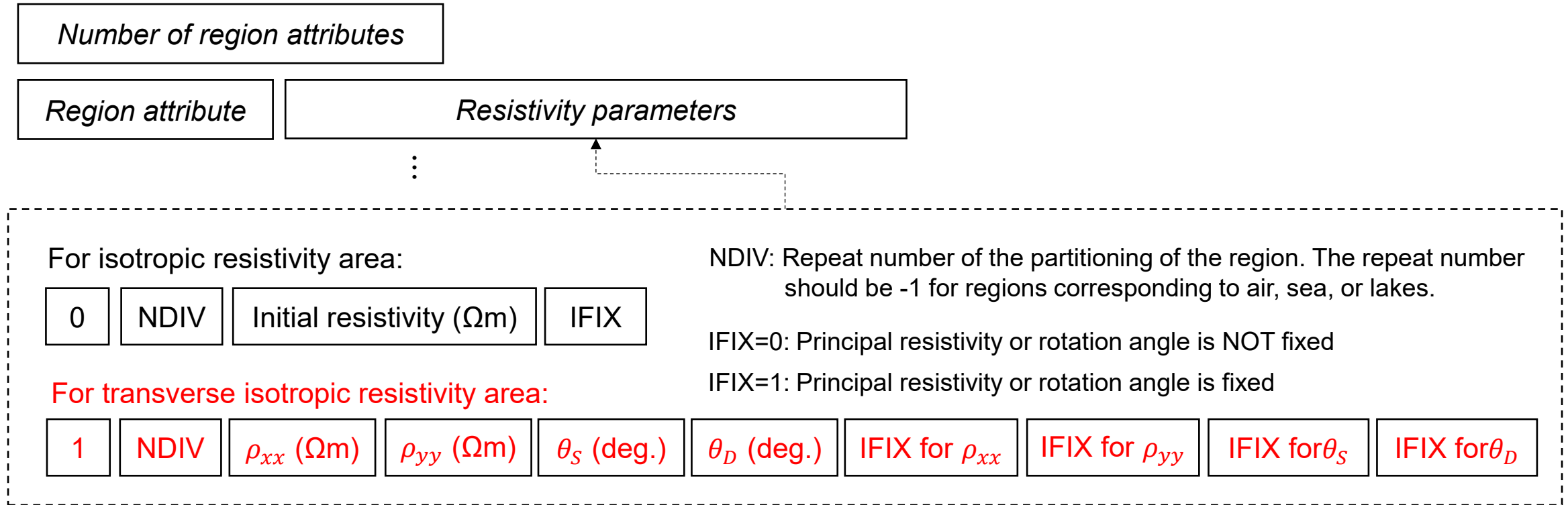
(3) Execute 'makeDHexaMesh' in the same way as in generating a usual isotropic model.

Procedure flow of tetrahedral mesh generation



How to make 'resistivity_block_iter0.dat' for anisotropic inversion (2) Tetrahedral mesh

- (1) Set initial values of anisotropic resistivity parameters in 'resistivity_attr.dat'. The format for the isotropic resistivity area is also different from the isotropic inversion mode.



- (2) Add '-aniso' in execution arguments. I recommend using '-div_all' as well to make every different subsurface element become a different model parameter.

TetGen2Femtic output.* **-aniso** -div_all

'*' in the left sample command indicates the iteration number of the iterative mesh refinement.

FIN

Thank you for listening

Please contact me if you have questions about FEMTIC and its tools.